招商斯里兰卡CICT40台轮胎吊

40 RUBBER-TYRED GANTRY CONTAINER CRANES FOR CICT

Project No. ZP12-1874

使用说明书及维护保养手册
（机械部分）

User Manual
(Mechanical)

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Chapter 1 Safety

Summary

Safety is a priority either in operation or in maintenance and inspection. This chapter defines the caution terms and definitions throughout the whole manual and informs you of the possible hazards and necessary precautions to be taken when operating or maintaining the crane.

All the maintenance personal should read, understand and obey the safety rules described in this chapter before operating or maintaining the crane.

1.1 WARNINGS AND SYMBOLS

The following signs and symbols are used in the manual to designate instructions of particular importance.

Symbol	Description	Symbol	Description
	Special information and/or orders and prohibitions directed towards preventing damage.		Hazardous voltage
	Warn of ELECTRIC VOLTAGE.		Keep hands and body clear
	Stay clear		No standing on it
	No access		Keep hands clear of the moving parts.
	Do not enter, Authorized personnel only.		Danger of falling
	Refer to specified sections in the manual.		Special information for increasing efficiency
	Crush hazard		Keep away from suspended loads
	Refers to information in other parts of the Manual		Additional Information
	Wear protective clothing		Danger of tripping

In this manual, as well as on the crane, warnings or warning signs are categorized into four levels according to the severity of danger (or seriousness of accident) as shown in Table1.1-1.

Warnings	Definitions
DANGER	Refers to imminently dangerous situations. Used when serious injury or even death will occur if specified procedures or instructions are not followed.
WARNING	Refers to potentially dangerous situations. Used when serious injury or even death might occur if specified procedures or instructions are not followed.
CAUTION	Refers to potentially dangerous situations. Used when medium or slight injury will occur or when the machine or equipment will be damaged if specified procedures or instructions are not followed.
NOTE	Used when attention must be paid to or when information must be stressed.

1.2 GENERAL PRECAUTIONS



All the safety warnings, safety signs and safety operating principles in this section are suitable for all the chapters in the manual, and need to be obeyed strictly.

Before operating or maintaining the crane, the operation and maintenance instructions have to be read carefully and understood by all maintenance and repair personnel. Safety should always be the priority throughout the whole process, no matter operating or maintenance. The maintenance and repair personnel should familiar with the crane's characteristic. Only the authorized and trained professional staff can be allowed to maintain and repair the crane.

This machine is built according to the latest standards of technology and recognized safety technical rules, and is equipped with safety devices. It has been checked for function and safety before delivery and hand-over. However, in case of faulty operation, misuse, or use not in accordance with the instructions, dangers and/or damages could still arise and threaten the health of the operation personnel or third persons, the container crane or other material, and/or the efficient operation of the crane itself.

Always use safety harnesses and/or other safety gear as required whenever in an elevated working place. It is always a good practice to work in a team of at least two persons in a place where potential dangers may arise.

It is need to pay attention while the crane is in the status of maintenance or repairmen, for some parts may have been separated or in unstable status. Above safety procedures must be observed strictly to secure the proper protection of staff whenever working on an equipment that may release any form of hazardous energy including, but not limited to, electrical, rotational, mechanical, chemical, hydraulic, or pneumatic energy, while the equipment is shut down.

In addition to the operation and maintenance instructions provided herein, legal（generally valid） and other obligatory regulations issued by the local authorities for accident prevention and environmental protection must also be observed.

It is our admonition that working cautiously and carefully is the best protection from accidents.

None of us has all the expertise or training needed to complete every repair that may be required. Do not try to undertake a task on which you are lack of training or qualifications.



IT'S ABOUT YOUR SAFETY!

The manufacturer, Shanghai Zhenhua Heavy Industry Co. (ZPMC), has tried his best to assemble and make available a complete set of instructions in the form of manuals and data sheets to provide all necessary information for crane operation and maintenance.

1.3 GENERAL RULES OF CONDUCT AND SAFETY REGULATIONS

The crane may only be operated in technically healthy condition and in compliance with the operating and maintenance instructions. Particular malfunctions that may impair the safety must be eliminated immediately. Before each commissioning, the crane must be checked for operational safety!

Uncompleted or illegible operation and maintenance must be replaced immediately.

Protection and safety devices (covers, safety barriers, etc.) must not be removed while operating the unit. The crane must not be operated without safety devices at any time.

Appoint personnel solely responsible for the crane operation and maintenance respectively. Also, do not permit unauthorized personnel to operate the crane.

The operating and maintenance staff has to be checked by the employer responsible for the operation. The person responsible for the operation is obliged to operate the crane only in proper condition and to instruct the operating and service staff to wear protective clothing, etc. whenever necessary or required by the regulations.

If there are any alterations concerning the safety of the machine or changes in the operational behavior, the machine has to be set to a standstill and the fault has to be reported to the relevant authority or person. Damages affecting the safety or malfunctions at the crane have to be eliminated immediately.

Arbitrary constructive modifications and the alteration of parameters beyond the operating and maintenance instructions are not permitted for the sake of safety technical.

All the maintenance and repair work should recheck and inspect to make sure its effectiveness.



IMPORTANT SAFETY PRACTICE!

The following safety rules must be observed strictly in practice:

(1) Verify that there is no danger to other related personnel before operating the machine.

(2) Do not touch the rotating parts or the moving parts of the machine during operation.

(3) Do not perform inspection and adjustment of the machine during operation.

(4) Do not touch the live parts inside the control cabinets or terminal boxes. The internal live parts carry high voltage current and pose the risk of electric shock when touched.

(5) Do not lubricate the components of the machine during operation.

(6) Immediately press the emergency stop pushbutton to stop the machine when an accident has occurred or is likely to occur.

(7) Always keep the machine and its surroundings clean. Especially keep the passageways at the elevated place clean so that personnel can walk safely.

(8) Warn all the service personnel of an unsafe intended action that may endanger them.

(9) Report any injury and make sure the wounded person can receive fast medical treatment.

1.4 SAFETY RULES FOR OPERATION

Pre-service Precautions:

(10) Before starting work at the crane, all persons have to be instructed in theoretic and practical use of emergency measures, the use of the emergency exits, and the use of rescue aids, in case of an accident.

(11) Unauthorized persons should not enter the crane.

(12) All safety devices and facilities with protection functions must always be connected and operable. They must not be rendered ineffectively or used improperly.

(13) The factory setting of the components must not be changed.

(14) Observe the inspection procedure stated in this manual for the inspection of each component.

Precautions during service:

(1) Do not step into an area below the cargo lifted when the crane is operating. It may result in a critical accident.

(2) The lifting load and the reach during the loading or unloading operation must be within the limit values defined by the specifications of the crane. Excess reach or excess load may result in a critical accident such as a fall-down, a break-down or a damage of the crane.

(3) Immediately stop operating the crane when an abnormal sound or vibration arises. Inform the responsible personnel immediately to eliminate the malfunctions or conduct maintenance.

(4) Do not leave the crane with open hatch covers, or other openings into which a person could step or tools and objects could fall. And do not leave lubricants, paints, or other materials uncovered or un-attended.

(5) Always operate the crane smoothly to prevent the shock of start-up or stoppage. Avoid operations such as abrupt loading or unloading at full speed with excess load or unnecessary repetition of

start-up or stoppage.

(6) When the crane encounters a strong wind over 25 m/s, conduct the crane stowage procedure against wind according to the related content in the manual.



DANGER ZONE MUST BE BLOCKED BY SUITABLE MEASURES. HAZARD AREAS MUST BE INDICATED PERMANENTLY BY CLEAR INDICATIONS, SUCH AS COLORED SIGNS OR WARNING SIGNS.

1.5 SAFETY RULES FOR MAINTENANCE AND REPAIR

The maintenance work is different from the ordinary operations. When the crane is under maintenance, some of its parts may be disassembled or unstable, thus special attentions should be paid to.



BEFORE COMMENCEMENT OF SERVICE WORK THE CONTAINER CRANE HAS TO BE MADE FREE OF LOAD, SET TO MAINTEN¬ANCE POSITION AND SWITCHED OFF, IF NO OTHER NOTE IS GIVEN.

Observe the following safety rules when conducting maintenance or inspection work on the crane:

(1) Personnel unauthorized by the person responsible for the crane should not perform maintenance or inspection work.

(2) Do not perform maintenance or inspection work when the crane is operating.

(3) Before conducting maintenance or inspection work, cut off the main power supply and all the power resources; then verify if the crane is stopped completely.

(4) Before conducting maintenance or inspection work, release all the energy stored in the related system. Spreader and headblock should be lowered down on the ground.

(5) Before conducting maintenance or inspection work, warn all the related workers of the work and prohibit entry of those unauthorized workers into the maintenance and inspection areas.

(6) The crane should be in maintenance mode when conducting maintenance or inspection work.

(7) Secure the repair area by closing a wider area. During the maintenance of the crane, be sure to close hatches on the machinery house and other places, or protect them with fences.

(8) When inspecting one of the cranes on the same gantry rails, attach the mechanical stoppers or station a person to prevent collision of the cranes.

(9) Never fail to close the door to the walkways and the chains at the gate of the walkways.

(10) When removing bolts, pins, wire ropes, and cables, warn workers other than the service personnel of the danger of attached parts falling.

(11) When relieving the residual pressure of the trim/list/skew and rope tensioning device, warn maintenance personnel of the device. The sheaves may move and each wire rope may loosen due to the relief of the pressure of the hydraulic cylinder. It may cause injury.

(12) Verify that there is no residual pressure left in the hydraulic equipment and the compressor. Residual pressure may cause injury.

(13) If hands will be on the rope for inspection, they must be away from the pinch point.

(14) The maintenance persons should pay more attention to the moving/rotating parts.

(15) Do not use the machine parts as climbing aids.

(16) Reinstall and test safety devices (which have been disassembled during the assembly or test and repair work) immediately after the end of the maintenance and the repair work.

(17) Correctly tighten the screw joints that were loosened during the test and the maintenance work.

(18) Replace the fastening elements and seals (e.g. self-securing nuts, washers, split pins, O-rings and seals) which cannot be used again.

(19) After the maintenance/repair work, update the fault log and the maintenance check list.

(20) Take away the tools, external matters and operation materials which left in the working area, and clean the repair area. The crane has been fixed with gratings walkway at its necessary place. All the passage to the cabin and E-room are higher than 15m, object falling down to ground at such high position will reach very fast speed which will make server damage, even threat to life, if the object hit the person on the ground. Picking up the foreign matters or debris on the crane is everyone's responsibility and obligation.

(21) Do test running and functional check of the components and safety devices before starting the operation and service again.

1.6 CONDUCT IN CASE OF ACCIDENTS AND EMERGENCY

(1) In the case of wire rope broken or trolley wheel broken during the operation, first try to release the load attached to the crane, if possible, then carry out the repair procedure described.

(2) All personnel are to be trained and familiar with the site specific emergency and accident response procedures. The employer shall assure that such procedures and training remain current.

(3) In the case of an accident with injury to persons, damages to devices or other materials, immediately inform the supervisor after providing the injured with first aid. For well-directed rescue efforts, clearly give the degree of severity of the injury to persons and the damage to property.

(4) In case of disasters (e.g. fire), all persons must leave the crane using the indicated emergency exits and/or other facilities.

(5) Rescue actions may only be carried out by instructed operating and rescue personnel. Use the intended protective equipment.

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Chapter 2 Brief introduction of RTG

Summary：

This chapter is a starting point. It aims at providing the fundamental knowledge about the construction, functions, performances and features of the crane for the operators and maintenance personnel in order that they can conduct the operating and maintenance in proper, safe and efficient way.

This chapter will describe the RTG’s the general features, including structure constructions and the compositions, layout, technical parameters, functions, characteristics of the mechanisms. The three main mechanisms, the Main Hoist, Trolley Drive and Gantry Travel, will be the primary part and detailed. Furthermore, the auxiliary components of the crane will be included as well.

2.1 Outline and main parameters for RTG

2.1.1 The types of RTG

It's shown in table 2-1 for the types of normal RTG

Table 2-1 the types of normal RTG

No.	Classification method	Form	Using in this project
1	The type of power supply	Diesel generator power supply, high voltage cable reel, Low voltage (400V~1000V) cable reel, Low voltage (400V~690V) contact bus	diesel generator + lithium battery system
2	The number of gantry wheels	8 wheels, 16 wheels	8 wheels
3	The type of hoist rope reeving	4 ropes, 8 ropes	8 ropes
4	The type of span	6+1，7+1	6+1
5	The hoist height	1 over 4 ， 1 over 5 ， 1 over 6 ， 1 over 7	1 over 6
6	The rated load	40t(Lt)， 45t(Lt)，50t(Lt)，60 t(Lt)，65t(Lt)	40.5t
7	The type of trolley driving	Rack and pinion, single drive (universal shaft drive), double drive (universal shaft drive), double drive (direct drive wheel)	Double drive (direct drive wheel)

Note: this project's form is shown in table 2-1. The main features are described in section 2.1.2. The main parameters are shown in section 2.2. In order to keep manual's universal property; this manual includes more information than normal project. In using, please check this project's as built drawings. As built drawings are special ready for this project. The other information's are not belonging to this project and for reference only.

2.1.2 The main features of this project

This equipment is specifically designed, manufactured and installated container handling mechineryaccording the contract for CICT, Sri Lanka, totally 40 rubber-tyre gantry cranes (hereinafter referred to as RTG)

The RTG is the latest model for handling container. The design and manufacture is based on the long-term experience of ZPMC in crane manufacturing field and the latest technology. Compared with traditional RTG, it has advanced performance of pratical and efficient operation.

The RTG's type is: trolley double driving (direct driving wheel), double box section girder, rated load 40.5 ton, span 23.47m, and hoist height 21.2m above ground.

In the following are the RTG's main features

◇ The RTG will be worked on the costumer' s yard. It meets the site' s infrastructure condition and the local weather conditions. Design, manufacture, erecting and acceptance are according to international and domestic standards and rules. It could be operated and maintenance 24 hours every day in normal weather conditions. The electrical, electronic and mechanical device shall meet the requirements of the weather conditions and the work cycle.

◇ The RTG' s main structure is composed of two groups of box section structures of door frame, supported by 8 inflatable rubber tires. The trolley is located on the girders' rail. It could move following the girder' s direction to handle the containers between the legs. The driver cabin is fixed under the trolley. The driver can control the whole movements of the RTG.

◇ The RTG has a electrical telescopic spreader to handle a container which weight is 40.5tons. It could lift a 9' 6" high container to pass six stacking 9' 6" containers' height. It also could cross width with six row 8' containers with the gap between the containers 400 mm, and a container truck lane. The trolley' s effective travel range is not less than 18.5m.

◇ The electric spreader is hanged under the trolley by wire rope system. The spreader is supplied by BROMMA. It could telescopic to 20' and 40' positions. It also could lift 45' or 48' container (with the corner fittings at the 40'position). The spreader system is composed by headblock and spreader body to achieve rapid replacement.

◇ There are three main mechanisms, main hoist, trolley travelling and gantry travelling. The main hoist, trolley travelling could be separated or simultaneously operated with rated load. As well as the gantry travelling, the trolley travelling could be separated or simultaneously operated with rated load too.

◇ The RTG could move along the normal lane as straight line. Each wheel can be rotated 90 degrees, to form a right angle with the original way, in order to change operation block. It also could be 360 ° turning around.

◇ Operation and control of the RTG are realized by the all digital AC variable frequency drives, PLC and CMS. The control function is reliable and effective. The main electrical control equipments are installed in the electrical room which is above the sill beam.

Electrical room is equipped with air condition, to ensure proper working temperature.

◇ Main hoist mechanism is mounted on the trolley. It works with constant power control and according to the lifting load automatically adjusts the lifting speed. It is equipped with overload protection device, over speed protection device, to ensure work safety and efficiency.

◇ Trolley travelling mechanism is mounted on the trolley. It works with AC frequency conversion control. The trolley is driven by the wheel directly driven method.

◇ Gantry travelling mechanism works with AC frequency conversion control. In order to avoid collision between crane to crane, or crane to container, the crane is provided with a manual and automatic steering system, and anti - collision device.

◇ In safety facilities, there is overload protection device, diesel engine over speed protection, high oil temperature and low oil pressure signal device, wind speed indicator, crane anti-collision device on four corners, emergency stop button and imit switch, signal in the mechanism.

◇ There is lighting and signal apparatus on the crane for night operations, and alarm when gantry travelling. The driver can communicate with others with the phone, enlarger and radio.

◇ Diesel generator set is arranged in the power room on sill beam. It’s consisted by CAT diesel engine and Zhejiang Zhongyuan generator. It is equipped with a lithium battery system, and the energy recovery system. The generator is brushless self-excitation type. The gen-set has the following characteristics.

a. Strong impact resistance, suitable for RTG frequent starting conditions. The frequency and voltage fluctuation is very small.
b. Reliable electronic speed governor in diesel engine, and has high water temperature, low oil pressure, over speed protection device.
c. Enough standby power, so the engine has enough power and keeps good work condition even after wearing.

◇ The hoist wire ropes are used to anti sway by eight rope reeving system.

◇ The spreader has skew function to reduce position time.

2.2 Main parameters for this project

2.2.1 Main technical parameters

It's shown in table 2-2 for main technical parameters of RTG

Table 2-2 main technical parameters of RTG

item			value	unit	note
Rated load capacity			40.5	t	Under spreader
Lift height			21.2	m	
speed	hoist	With rated load	23	m/min	
		Without load	52	m/min	
	Gantry travelling	With rated load	30	m/min	
		Without load	90	m/min	
	Trolley travelling		70	m/min	
Span			23.47	m	
base			7.9	m	
Total wheels/driving wheels			8/4	pcs	
Spreader skewing			±5	degree	
Diesel engine	type		C7		CAT
	power		205	Kw	
	speed		1800	rpm	
generator	type		TYFZ18000-280L-G		ZHONGYUAN
	power		180	KVA	380V，50HZ，3PH
	speed		1800	rpm	
Number of containers stacked by the crane			7	containers	
Number of containers the crane may pass over			6	containers	

2.2.2 Main hoist mechanism parameters

It's shown in table 2-3for main hoist mechanism parameters

Table 2-3 main hoist mechanism parameters

item	value	unit	Note
Number of ropes on drum	8	pcs	
Ratio of rope reeving	1		
Rate load(under spreader)	40.5	t	

Hoist speed	With rate load	23	m/min	
	Without load	52	m/min	
Drum	Nominal diameter of the drum	1200	mm	
	Number of groups of grooves on wire rope drum	8	Sets	
	Number of wraps of rope for fixing	3	pcs	
	Length of rope on the drum(one group of grooves)	44.9	m	
motor	Type	5RN315L08B9		YASKAWA
	Power	180	kw	
	Speed	778/1760	rpm	
	Voltage	380	V	
	Number	1	set	
reducer	Type	ML4PSF140E-1565		SEW
	Input power	180	kw	
	Ratio	127.5		
	Number	1	set	
Wire rope	Type	IWRC 6XWS36		
	diameter	∅28		mm
	length	43.5x2	m	
		42.3x4	m	
		44.9x2	m	

2.2.3 Trolley travelling mechanism parameters

It's shown in table 2-4 for trolley travelling mechanism parameters

Table 2-4 trolley travelling mechanism parameters

item		value	unit	Note
Trolley travelling speed		70	m/min	
Reducer building in motor and brake	Type	KH107DV160		SEW
	Ratio	37		
	Output speed	44.6	rpm	
	Output torque	3212	Nm	
	Motor power	15	kw	
	Motor speed	1740	rpm	
	Voltage	380	V	
	Brake torque	130	N.m	
	Number	2	Pcs	
wheel	Diameter	∅500	Mm	
	Number	4	Pcs	
Trolley rail		80x80	Square steel	

2.2.4 Gantry travelling mechanism parameters

It's shown in table 2-5for gantry travelling mechanism parameters

Table 2-5 Gantry travelling mechanism parameters

item			value	unit	Note
Gantry travelling speed		With rated load	30	m/min	
		Without load	90	m/min	
Motor with brake	Type		FEIK-C21 MOUV		YASKAWA
	Power		25	kw	
	Speed		1680	rpm	
	Voltage		380	V	
	Number		4	台	
	Type		Built in motor		
	Brake torque		214	N.m	
	Number		4	Pcs	
Ruder and chain drive	Type		FVS370.30.D1B		ZPMC
	Input power		25	Kw	
	Ratio		30.61		
	Number		4	Pcs	
	Chain type		160SW - 2×102		GB1243.1-83
	Pitch		50.80	Mm	
	Ratio		58/17=3.412		
	Totally ratio		104.43		
tyre	Type		21.00-35 40 层(无内胎)		SIEMX
	Working diameter		~1860	Mm	
	Air pressure		10	kg/cm²	
	number (driving/totally)		4/8	pcs	

2.3 General introduce for RTG' s configuration

RTG is one kind of port equipment used to handle container between yard block and truck. The below is RTG's main components. The general arrangement is as shown in Figure 2-1. The trolley arrangement is shown in Figure 2-2.

Ⓐ **Gantry frame**: The main bodies are box section members. It's consist by girder, leg and sill beam.

Ⓑ **Gantry travelling mechanism: T**he RTG is supported by 8 rubber tyres. Motor drive and power transferred by reducer. It's including the driving wheel system and driven wheel system.

Ⓒ **Trolley**: Trolley is the main component of the RTG. It's consist by four steel wheels and trolley frame. There is main hoist mechanism, trolley travelling mechanism, driver cabin (including control device) and kinds of auxiliary devices installed on the trolley. The reeving system is using to hang headblock and spreader. The driver in cabin can control RTG to handle stock and carry container.

Ⓓ **Main hoist mechanism:** The main hoist mechanism is located on the trolley. It's used to lift spreader to hang container.

Ⓔ **Trolley travelling mechanism**: The trolley travelling mechanism is located on the trolley. It's used to move the trolley to the required position. The trolley is self drive type.

Ⓕ **Main hoist wire rope reeving system:** The wire rope through sheave and connect the hoist drum and headblock. It could transfer relative movement between drum and headblock.

Ⓖ **Spreader skewing mechanism:** It is achieved by two sets of electric push rod fixed on the headblock.

Ⓗ **Spreader anti-sway system**: The ansiway function is achieved by 8 hoist rope system, no other auxiliary antisway system.

Ⓘ **Trolley's walkway and platform**: provided on trolley for maintenance equipments on the trolley.

Ⓙ **Cabin**: Driver can operate RTG and spreader in it.

Ⓚ **Trolley anchor device**: Two sets of manual operation pin are located on both sides of the trolley for locking the trolley, preventing accidentally moved

Ⓛ **Service crane**: None

Ⓜ **Cable chain system**: Cable chain system transport the power and control signal to main hoist mechanism, trolley travelling mechanism, spreader skewing mechanism, headblock and spreader.

Ⓝ **Power unit**: It provides power for RTG all motion mechanism and auxiliary equipment. It locates on the sill beam. In this project, it is consisted by diesel generator + lithium battery system.

Ⓞ **Walkway, ladder and platform**: Including all the others platforms, handrails, stairs, ladders except fixed on the trolley. It is used for boarding the crane, and providing the maintaince space.

Ⓟ **Electric house**: Electric house installs on the sill beam. There is control system, including the driver cabinet, control cabinet, computer and other related equipment in it. It's equipped with air conditioning, in order to keep temperature indoor.

Ⓠ **Headblock**: the bottom is connected to spreader. The top is hanged by wire ropes through 8 adjusting screw rods.

Ⓡ **Spreader**: spreader is under the headblock. It's used to lift load. Mainly used for ISO standard containers, also can use the auxiliary lug on the edge to lift special goods.

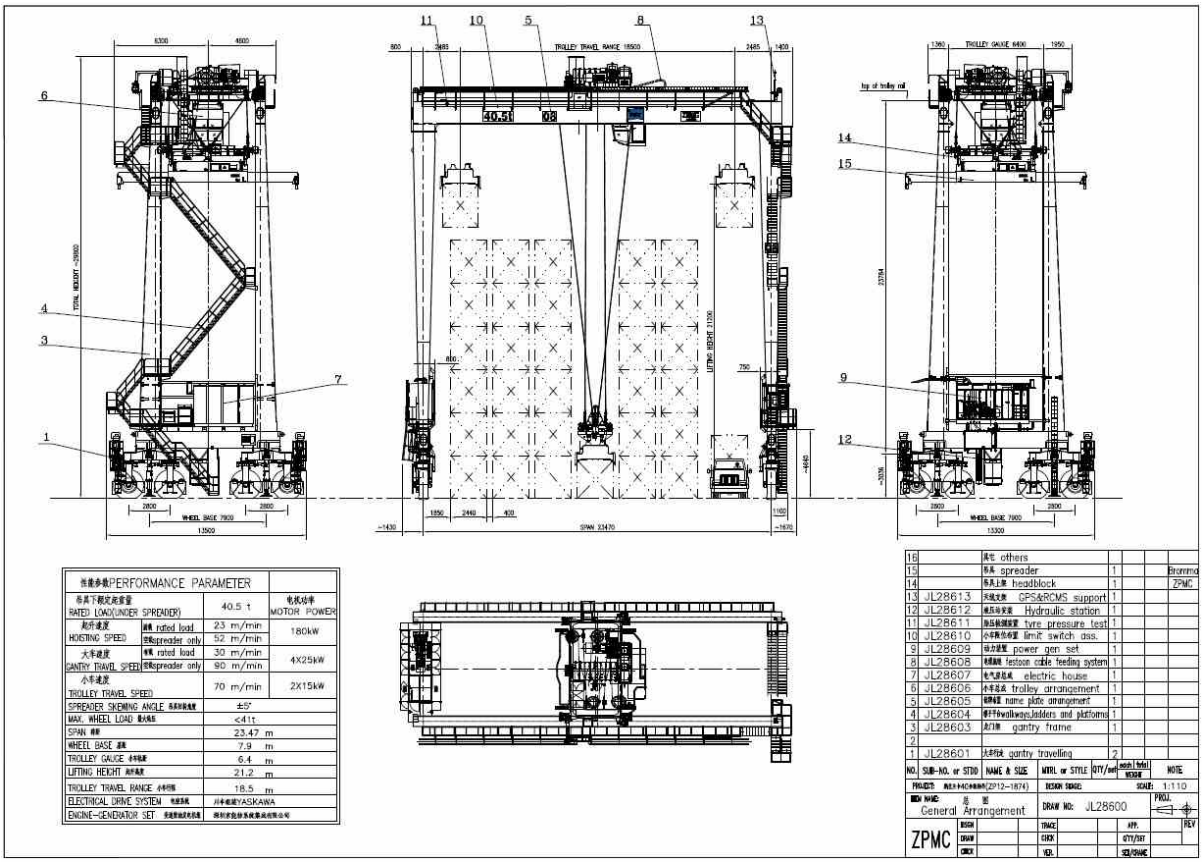


Fig 2-1 the general arrangement of RTG

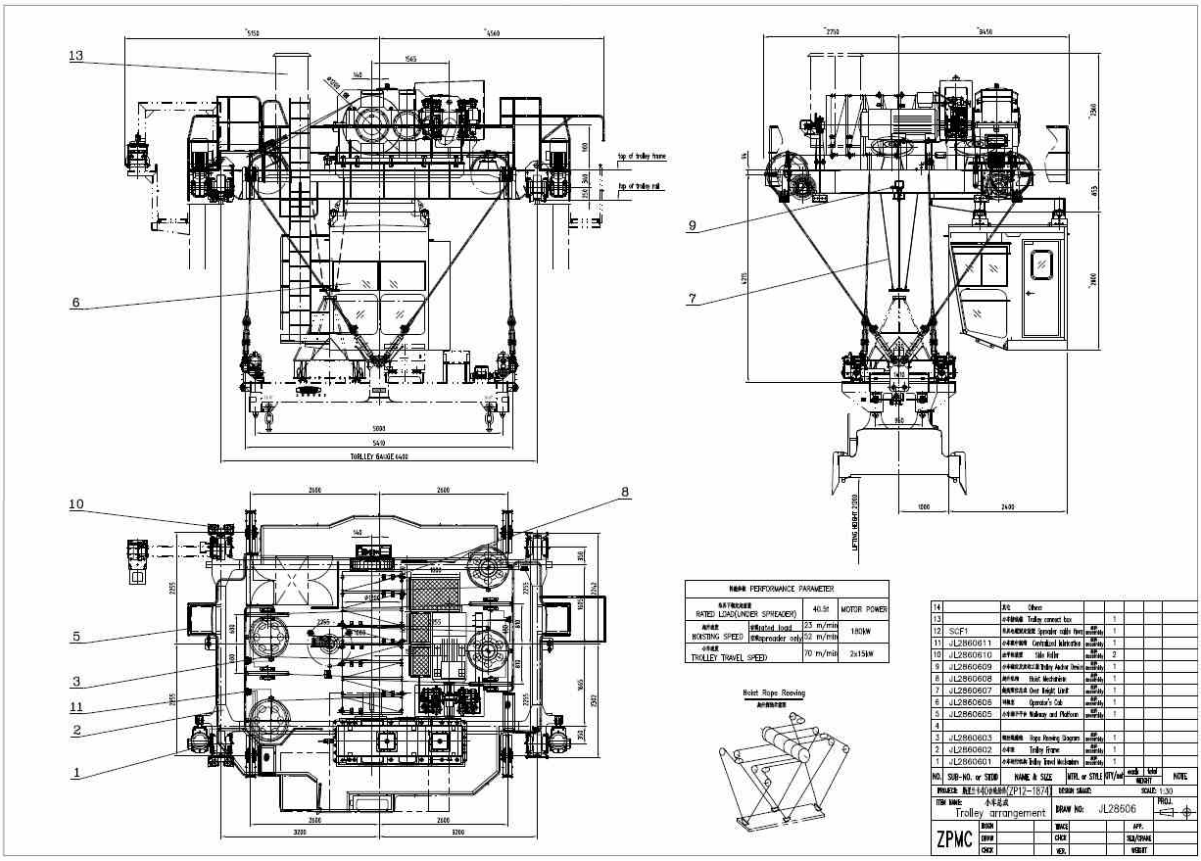
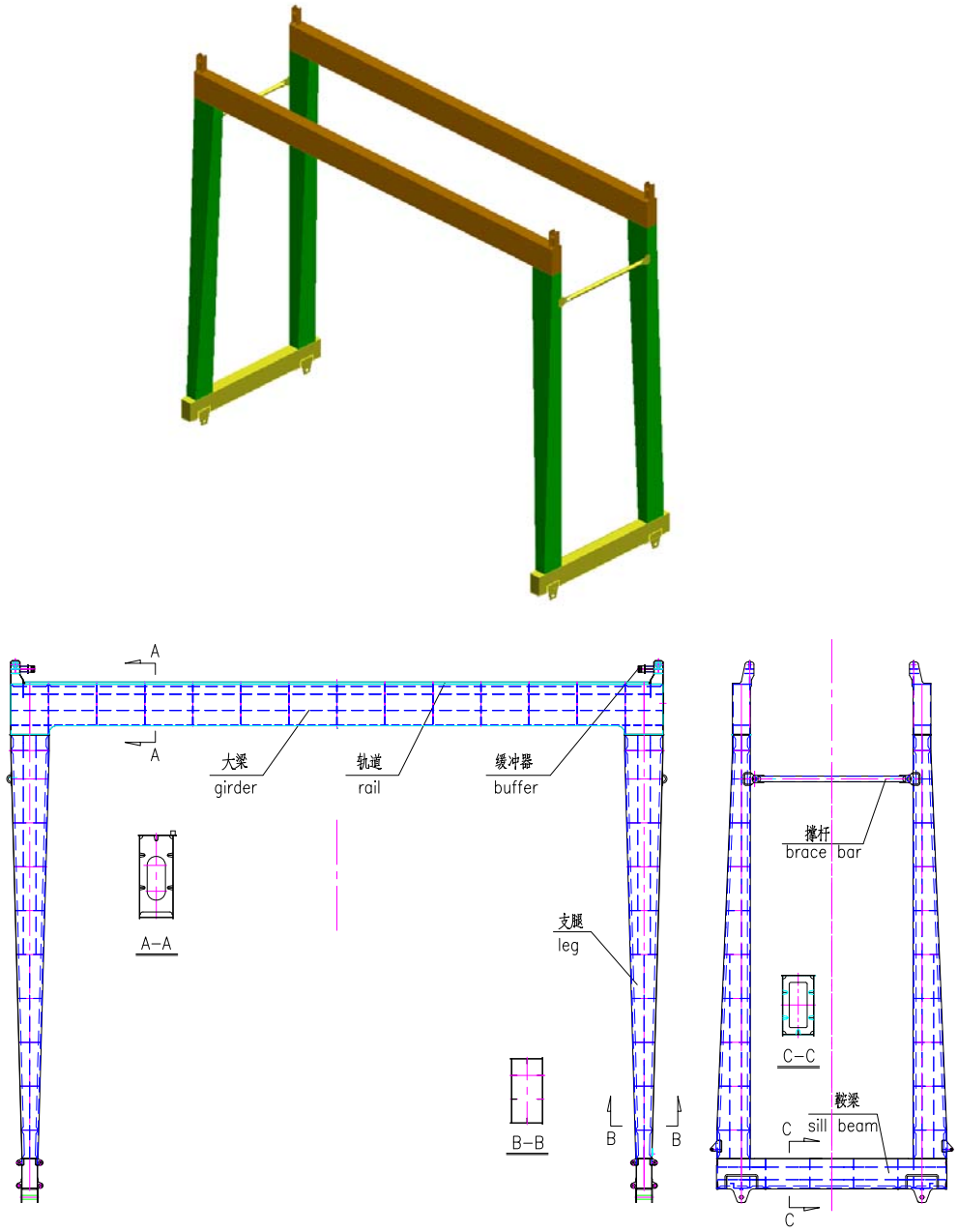


Fig 2-2 the trolley arrangement of RTG

2.4 Main components

2.4.1Gantry frame

Shown as in Figure 2-3. Gantry frame is the main structure of the RTG, it consists of two girders, four legs and two sill beams. Most of the structure member section is box section except some in tubular shape. The brace bar on the upper legs connects the structure as a whole frame. The trolley rails are arranged on the girders, we used I-shaped steel or square bar as the rails. The rails will be fixed on the girder by rail clips or welding. The buffer support at the end of girder can also be used as lifting ear for the RTG.



Pic 2-3 Gantry frame

2.4.2 Gantry traveling mechanism

Gantry traveling mechanism has two models, 8 wheels and 16 wheels. It consists of driving device, steering device, carriage, slewing bearing, equalizer beam and tubeless tyres. 8 wheels model divide into two wheels diagonal driving and four wheels four corners driving. 16 wheels model are all four corners driving, half of the wheels are drive wheels.

The gantry usually working along the yard in straight line, and also the gantry can turn 90° to transfer the stacking yard.

As an option, RTG can rotate around the vertical center line as the arbitrary angle 0-360 °(pivot function), to change orientation of operator cab.

2.4.2.1 Eight Wheels Gantry traveling mechanism

8 wheels traveling mechanism include 8 tubeless tyres. Each of the two tyres together with the frame, slewing bearings, hydraulic or electric steering device, equalizer beam form four sets of bogies (driving and driven), below the four legs.

The equalizer beam connects the sill beam by equalizer pin. Two sets of bogies on the diagonal side of gantry frame are the drive bogie. Outside of the tyre is drive wheel, equipped with independent driving mechanism. It consists of motor, reducer, sprockets and double strand bushroller chain.

Tyre size 21.00-35(40 Ply), rim model is 35155022/35155023. Tyres are equipped with protective railing. Four outside of the corner frame equipped with anti-collision device.

This project has four sets of drive wheels.



Figure 2-4 8 wheels gantry

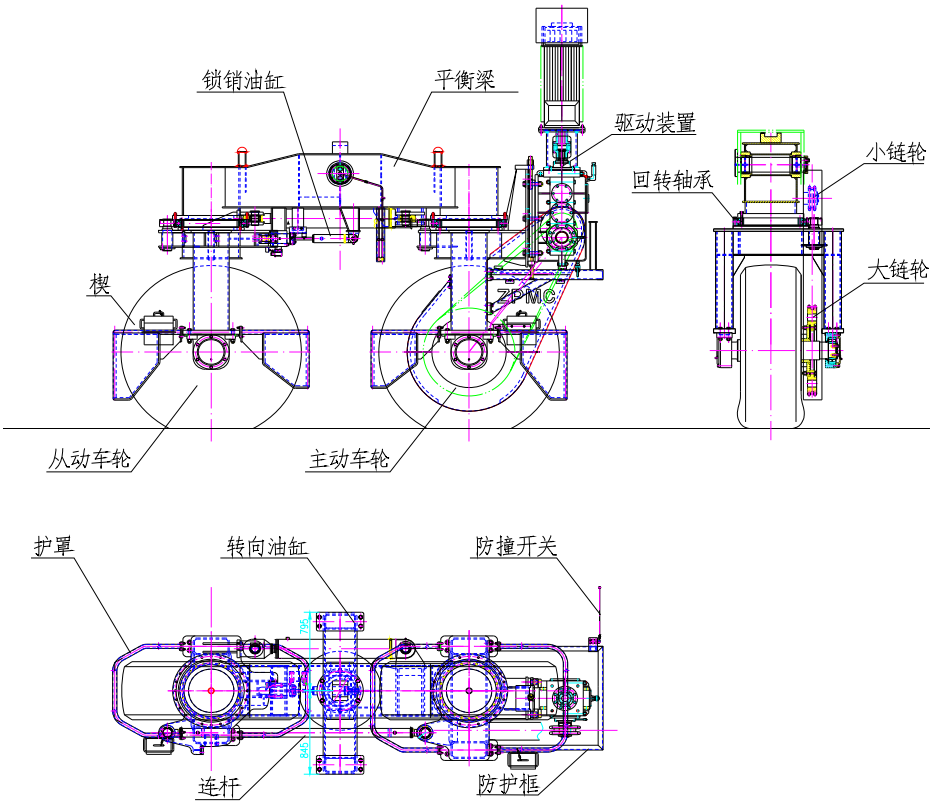


Figure 2-5 8 wheels drive gantry

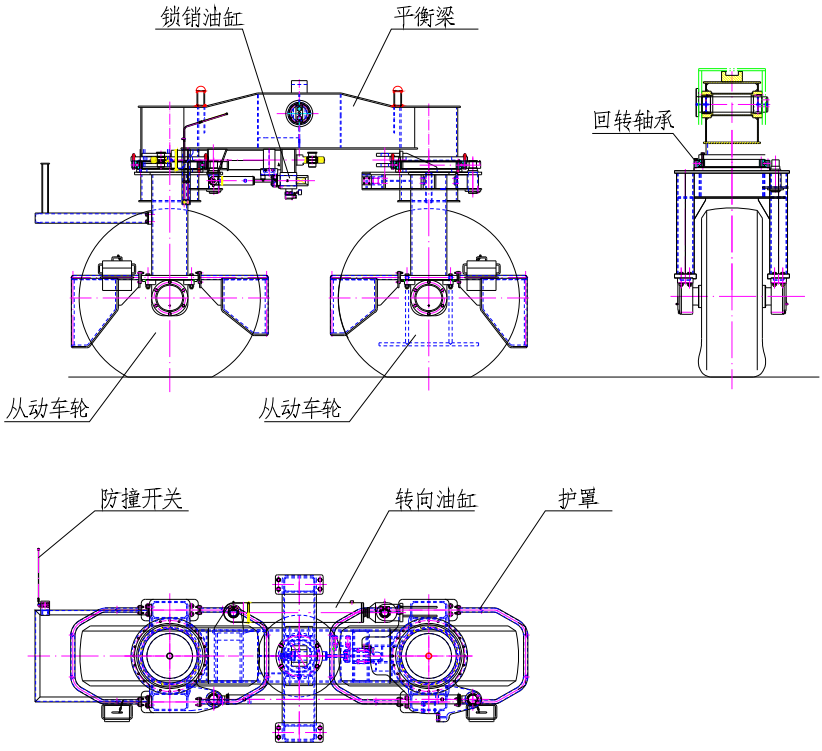


Figure 2-6 8 wheels idle gatry

2.4.3 Trolley frame

Trolley frame is one of the main structure of the RTG. It directly bears the lifting load and transfer the load to the gantry frame through the wheels. The trolley frame is composed of two I-shaped girders and two end beams. The frame will be mounted with wheels, driving device, hoisting device, operator cab, and other auxiliary equipments to be a complete trolley.

2.4.4 Hoist mechanism

2.4.4.1 Hoist driving mechanism

The hoist mechanism locates on the trollery. It consists of motor ①, high speed coupling② and reducer④. The drum⑥ use a lower speed coupling ⑤to connect with the reducer. One side of the reducer has two high speed brakes③. The high and lower speed couplings transfer the driving force from the motor to the hoist drum, to lifting or lowing the spreader.

The hoist mechanism is shown as Fig 2-10. High speed and low speed shafts of the reducer are on the same level. The upper and lower housings of the reducer are split from the center line of the gears and subdivision surface is sealed with sealant. At the bottom of the low speed shaft frame is provided with an oil discharge valve and oil level indicator, a scale is arranged on indicator for easier observation of lubrication oil. The handhole cover in the high-speed shaft frame is convenient for cleaning the inner cavity of reducer. The cover is equipped with air breather, and observation hole. The main hoist drum is made of steel plate and of welded construction. The pitch diameter of drum is above 30 times of the nominal diameter of wire rope. The drum has four sets of grooves, One side of the drum is connected to the reducer by a drum coupling, another side of the drum is sitting on the trolley frame through the bearing seat.

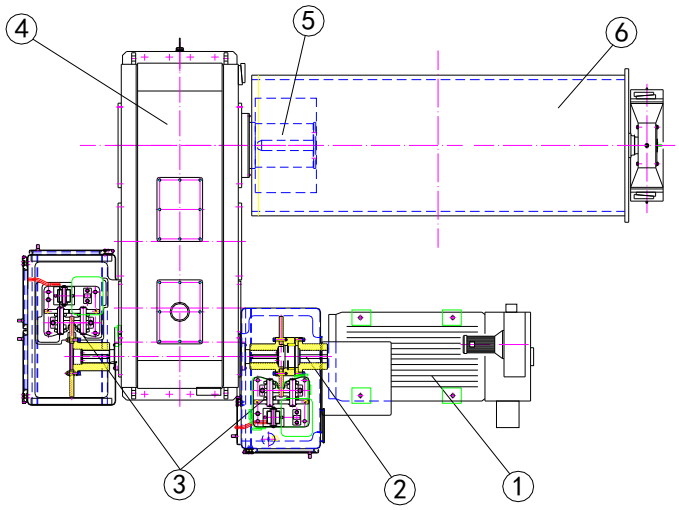


Figure 2-10 Hoist driving mechanism

① hoist motor ② hoist reducer ③high speed coupling
④ hoist drum ⑤ drum coupling ⑥high speed brake

Motor’s forced ventilation is accomplished by a cooling fan with filter. Cooling fan at the driving side

forced the filtered cooling air into the motor. Each motor is equipped with an encoder. 🕮 More detail instruction please refer to the motor supplier's manual.

Torque of the hoist brake 3680Nm!

At the end of the drum, two switches are connectted to the drum shaft, one is overspeed protection switch, another one is cam limit switch. The overspeed protection switch will be activatedwhen the motor speed reach 115% of normal speed. The switch will cut off the power of motor, dead stop all brakes. The cam limit switch controls the position of spreader. It's the backup switch of the motor encoder.

2.4.4.2 Wire rope reeving system

Main hoist device use the wire rope to lifting and lowering the spreader.



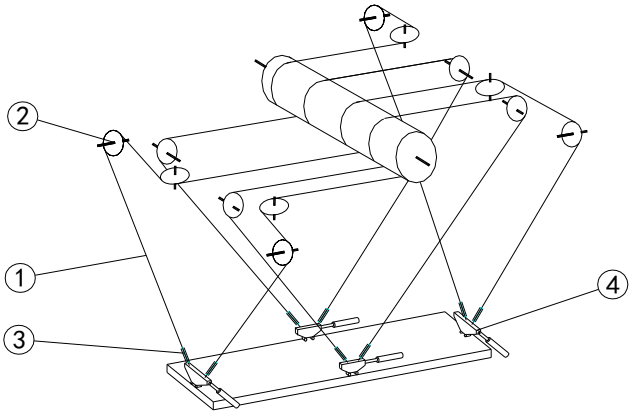




Figure 2-12 8 ropes reeving system

8 ropes reeving system

As shown in Fig 2-14, the auxiliary antisway wire ropes are directly replaced by hoist wire ropes. The force of hoist rope is far bigger than auxiliary antisway wire rope force. 8 ropes reeving system use 12 sheaves to enlarge the dip angle of the wire ropes. One end of the ropes are clamped on the drum, another end of the ropes go round the sheaves on the trolley, and then two ropes a group connected into midpoint of headblock's four edges though adjusting turnbuckle, and form four equal isosceles triangles. Because of the eight wire ropes are diagonal connect with the headblock, their horizontal force prevent the spreader sway in both gantry and trolley direction. So the 8 ropes reeving system don't need additional anti-sway device.



① wire rope
② sheave
③ adjusting screw
④ push rod

Figure 2-14 8 ropes reeving

2.4.4.3 Mirco-motion of the spreader

The spreader can skew, trim and horizontal side shift through different sides combined movements of hydraulic or electric push rod. Operate the reset switch, can recover the normal position. Parameter of the fuction shown as Table 2-2.

Table 2-2 parameter of mirco-motion

Reering type	**Fuctioin**	Driving device	**Parameter**	Remarks
8 ropes reeving device	Spreader skew +/-5	Hydraulic or electric push rod	±5°	Standard confiuaration
	Trim(left & right)		±2.5°	This function is not available for 8 rope system
	Gantry direction side shift	Hydraulic or electric push rod	±200mm	Optional
	Trolley direction side shift	Hydraulic or electric push rod	±200mm	Optional

(1) Skew and horizontal sideshift fuction of the 8 ropes spreader

The hydro-cylinder mounted between headblock and spreader or the electric push rod install on the spreader can achieve the skew and horizontal side shift fuction of the 8 ropes reeving system. The headblock consist of main frame, cable basket, junction box and electric push rod. Through the cylinder push the spreader to achieve skew and horizontal shift fuction. Shown as in Figure 2-18.

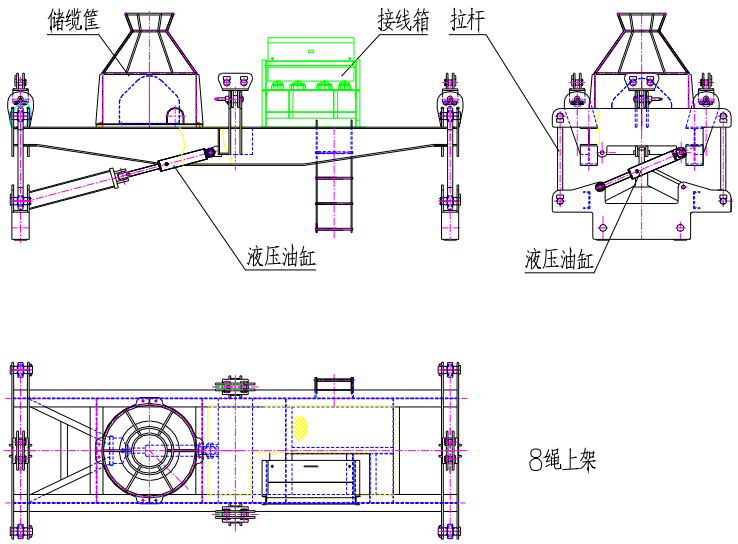


Figure 2-18 8 ropes spreader skew and horizontal side shift device

a. When the cylinder on both ends of the frame into a reverse action simultaneously, the spreader can skew ±5°.

b. When the cylinder on both ends of the frame does the same action simultaneously, the spreader can shift along trolley direction.

c. When the cylinders in the middle of frame action simultaneously, the spreader can shift along gantry direction.

The driving device of skew and horizontal side shift normally performed by the three sets of cylinders or four sets of electric push rods. Cylinder without mid-frame shown as Figure 2-19. Cylinder with mid-frame shown as Figure 2-20. 🕮 More details please refer to the supplier's manual.



Figure 2-19 cylinder without mid-frame



Figure 2-20 cylinder with mid-frame

When the mirco-motion at max angle, the cylinder or the push rod can not be further movement, the switch on the control console can return it to the initial position.

2.4.4.4 Load sensor

The crane is equipped with four load sensors. These load sensors connect to the four wire ropes of the main hoist reeving device, they measure the tension of each wire rope separately, and allow the PLC and CMS calculate and record the load and load eccentricity. The model of load sensors is shown in table 2-3. Basic element of weighing system comprises four sensors and an amplifier. Each load sensor will detect tension of one rope. Four tension signals come into analog signal by amplifier. After conversion the analog output signal to the digital meter in operator cabin and display the load in tons.

The crane monitor the load sensor though the PLC. Under normal operation conditions, the system will identify the size of lifting rigging connected to the spreader. Programmable logic controller calculates the allowable eccentrically load according to the size of lifting rigging. If over load, it'll stop the hoisting and over load alarm sound out from PA. At this time, the crane could only lower the goods in limit speed.

Specification of the load sensor shown as Table 2-3.

Table 2-3 Specification of the load sensor

Pin shaft type sensor	4 ropes	9t	Load 40Lt
		11t	Load 50Lt
		14t	Load 65Lt
	8 ropes	16t	Load 40Lt
		20t	Load 50-65Lt
Pressure sensor	4 ropes	4.5t	Load 40Lt
		5.5t	Load 50Lt
		7t	Load 65Lt
	8 ropes	4.5t	Load 40Lt
		5.5t	Load 50Lt
		7t	Load 65Lt

2.4.4.5 Hoist limit switch

At the end of the drum, two switches connect on the drum shaft, one of them is overspeed protection switch, another one is cam limit switch(as shown in Figure 2-22). The overspeed protection switch will be activated, when the speed reach 115% of normal speed. The switch will close the power of motor, dead stop all brakes. The cam limit switch controls the position of spreader. It's the backup switch of the motor encoder.



Fig 2-22 Hoist overspeed protection switch and can limit switch

Normal hoist stop position controled by the pulse encoder installed in the hoist motor. Hoist motion's zero position set at the hoisting terminal place. The zero

position will be setted when the headblock hoist at highest point and hitting the hammer limit switch under trolley.

1) Hoist over height stop
Hammer limit switch is located under the over height position. This switch is connected with emergency stop circuit, when this limit switch activated, it'll be emergency stop. The "limit bypass" selection switch located in the electrical room can release the limit state. The hammer limit switch shown as in Fig 2-23.

2) Hoist normal stop
Programmable software limit switch. When it activated, the switch will start ramp stop.

3) Hoist smart slow down
Programmable software limit switch (fuction). The position of this smart slow down is depending on the setting of hoist slow down. More detail definition about "smart slow down" please refer to the electrical manual.

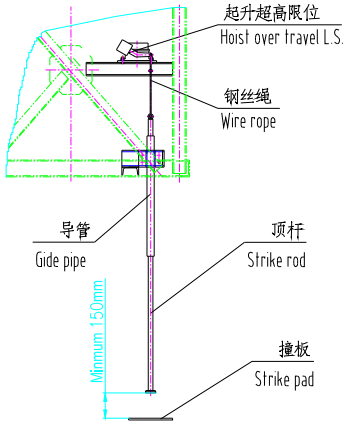


Fig 2-23 Hammer limit switch

4) Lowering smart slow down
Programmable software limit switch (fuction). The position of this smart slow down is depending on the setting of lowering slow down.

5) Lowering normal end stop
Programmable software limit switch (fuction). When it activated, the switch will start ramp stop.

6) Hoist stop check
Cam limit switch. If the encoder hasn't stop the hoisting speed, then activate this switch will stop the hoist speed.

7) Hoist slow down check
Cam limit switch. If the encoder hasn't decelerate the hoisting speed, then activate this switch will decelerate the hoist speed.

8) Lowering slow down check
Cam limit switch. If the encoder hasn't decelerate the lowering speed, then activate this switch will decelerate the lowering speed.

9) Lowering terminal inspect
Cam limit switch. It's the mechanical fuction backup of the over lowering protection.

2.4.5 Trolley travel sysytem

The trolley travel system consists of driving device and wheel supporting device, is mainly responsible for the horizontal transport of the container. Four non-flange wheels support the trolley frame, and driving device make the trolley moving along the rail. Because of the trolley wheels are non-flange, we have two sets of side guide rollers in both ends of one side trolley frame, and four groove shaped blocks under end beam of trolley, to avoid the trolley derailment.

2.4.5.1 Trolley driving device

Trolley driving device composed of motor, coupling, reducer and transmission device. According to the driving principle, it can be divided into wheel direct driving and rack and pinion driving. Wheel direct driving is using the friction between the wheel and rail to driving the trolley moving. Rack and pinion driving is using the mesh between the pinion and rack to driving the trolley. When the trolley is driving by wheel direct driving, the friction between the driving wheel and rail shall be big enough to avoid trolley skid.

A) Wheel direct driving device composed of motor, coupling, brake, reducer and universal drive shaft connect with two trolley wheels. There are two types, one is single wheel direct driving and another is dual wheels direct driving. Single wheel direct driving refers to Figure.2-24, dual wheels direct driving refers to Figure.2-25.

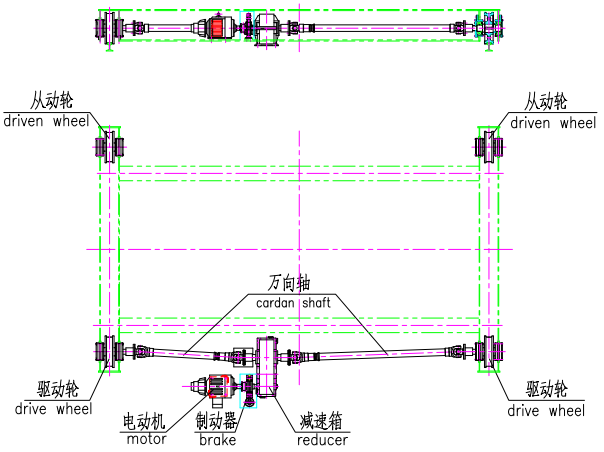


Figure 2-24 Signle wheel direct driving device

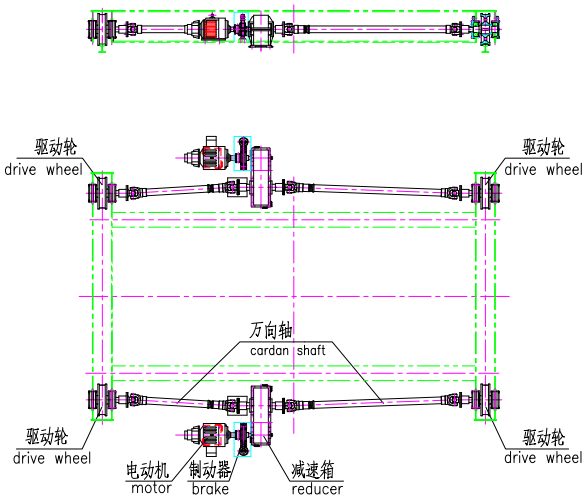


Figure 2-25 Dual wheels direct driving device

B) Rack and pinion driving device composed of motor, coupling, brake, reducer, universal drive shaft, driving pinion and rack. The rack and pinion will drive the trolley directly. Refer to Figure 2-26.

C) “Three in one” driving device belong to wheel direct driving device. It Combine the motor, brake, and reducer into one unit and direct install on the wheel shaft to driving the wheel. It has the advantages of tight configuration, light weight and easy maintenance etc. More attention shall be give to the synchronous driving of trolley wheels on different rails. Refer to the Figure 2-27.

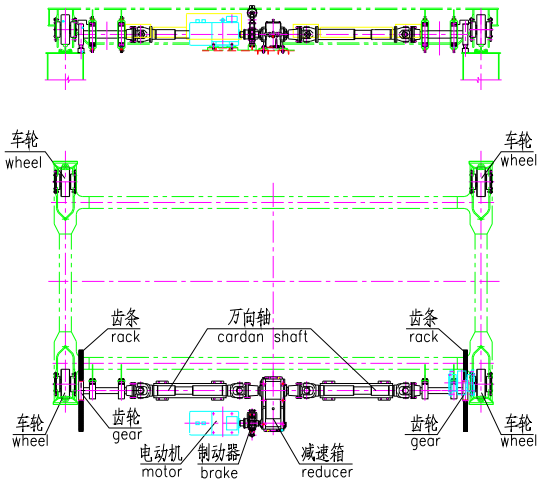


Figure 2-26 Rack and pinion driving device

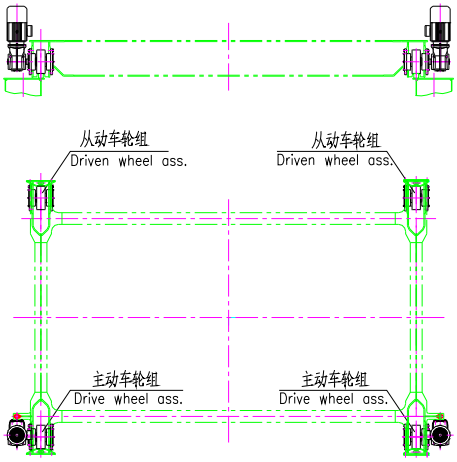


Figure 2-27 Three in one driving device

The trolley brake which released by electro-hydraulic thruster and braked by spring can stopthe trolley. More details about the brake, please refer to the Brake's manual.

Torque of the trolley brake setting as 130Nm!

2.4.5.2 Trolley wheel assembly

Trolley wheel divided into driving wheel and driven wheel:

A) Driving wheel composed of wheel, driving shaft, bearing, excentric sleeve and end cover. Refer as Figure 2-28.

B) Driven wheel composed of wheel, driven shaft, bearing, excentric sleeve and end cover. Refer as Figure 2-29.

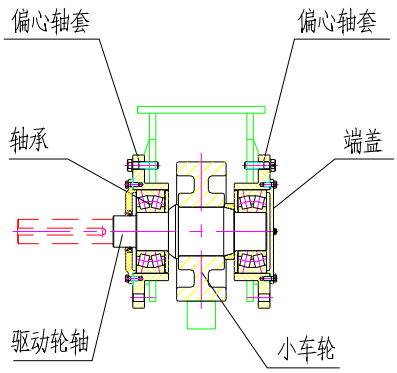


Figure 2-28 Driving wheel group

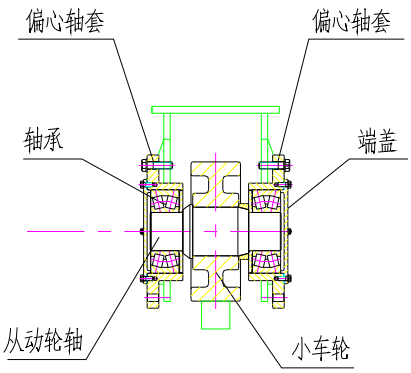


Figure 2-29 Driven wheel group

2.4.5.3 Trolley limit switch

In order to make the trolley travelling safely in the effective working aera, the limit switchs are setted in both sides of the trolley rail. The slow down limit switch can slow the trolley speed, terminal limit switch can stop the trolley and over travel limit switch can finally stop the trolley. Trolley limit switch arrangement refer to Figure 2-30.

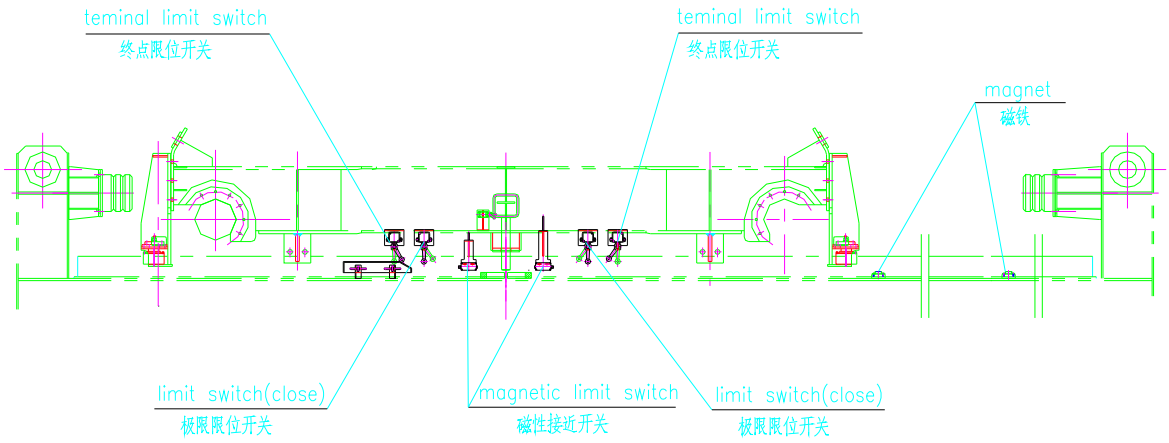


Figure 2-30 Trolley limit switch arrangement

1) Forward over travel stop

Mechanical limit switch, setting at the end of girder, and connect with emergency stop circuit. When this limit switch is activated, it'll be emergency stop. The "limit bypass" selection switch located in the electrical room can relieve the limit state.

2) Backward over travel stop

Mechanical limit switch, setting at the end of girder, and connect with emergency stop circuit. When this limit switch is activated, it'll be emergency stop. The "limit bypass" selection switch located in the electrical room can relieve the limit state.

3) Forward normal stop

Mechanical limit switch. When this limit switch is actived the trolley will stop, it's normal stop, no need bypass.

4) Backward normal stop

Mechanical limit switch. When this limit switch is activated the trolley will stop, it's normal stop, no need bypass.

5) Forward slow down limit switch

Proximate limit switch. Suppose the trolley pass the slow down limit switch as 100% speed then the trolley slow down, until 10% speed travelling

6) Backward slow down limit switch

Proximate limit switch. Suppose the trolley pass the slow down limit switch as 100% speed then the trolley slow down, until 10% speed travelling.

7) Parking position

No dedicated limit switch. At parking position, normally use the painting marker on the girder to guide the operator parking the trolley.

8) Entrence door limit switch of the operator's cabin platform。

In the platform of the operator's cabin, a proximate limit switch is setted at the entrence door. When the switch is triggered, the operation of the trolley will be banned.

2.4.6 Operator's cabin

In order to provide the best view for the driver, operator's cabin is positioned under the trolley frame, and connect the trolley with shock absorber. Left side of the operator's cabin install the energy chain or festoon system. The trolley will drive the cabin and cable together. The driver in the cabin can observe spreader's loading conditions. Arrangement of the operator's cabin shown as in figure 2-35.

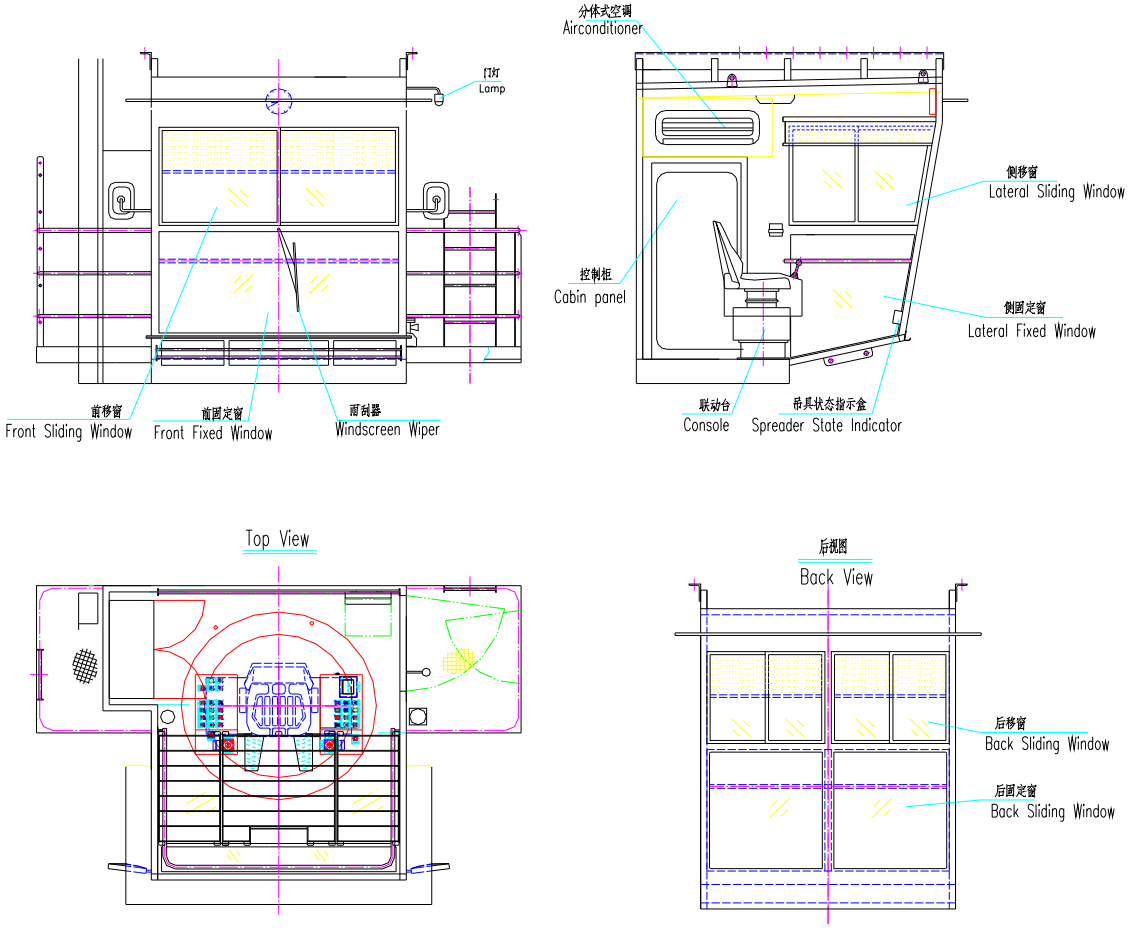


Figure 2-35 Arrangement of the operator's cabin

The cabin is designed according to the ergonomics, it provides a confortable and convenient working condition for the operator. So the operator can fully concentrate on heavy work, and not confused by unfomfortable external environment. The cabin equipped with adjustable seat, and it could be adjusted to suit the driver's weight, height and distance to the front window. So that drivers can comfortably observe the position of container, operate all control switch.

All the main functions of RTG can be achieved in the operator's cabin. The arrangement of the operator's cabin shown as in Figure 2-35. Front, lower and side of the cabin have wide window, the window is safe laminated glass. The cabin has good light, vision and spacious room. The driver seat is in the front of the middle. The seat is ergonomic design, can be adjusted according to needs, and the backrest can change the angle. The control console is set both side of the seat. In order to ensure the driver comfort and spiritual work, the cabin is equipped with air conditioning, heating, telephone and other facilities. The windows can be cleaned within the cabin and the floor window is equipped with safety fence. In front of the driver seat provide with a safety rod, it's the armrest when emergency stop.

All of the main handle, indicating lamp, switch, and some other control buttons are conveniently arranged in the left and right sides console. The driver can operate the auxiliary screen, spreader lockpin indication screen, computer and monitor without leaving the seat. The control console shown as in Figure 2-36.



Figure 2-36 Control console

The driver can easily communicate with other personnel on the RTG through the internal communication system in operator's cabin.

2.4.7 Main power equipment

🕮 Details about the main power equipment, please refer to the Supplier's manual.

2.4.8 E-Room

As shown in Figure 2-42, E-room is mounted on the sill beam. E-room provides a clean working environment with air conditioning for the drive control cabinets and other electrical equipments. The E-room installed of conventional mechanical locks and electronic card lock system. Only the authorized person can enter the E-room. Even the RTG is power off, the people can enter the room through key of mechanical lock. Consider the escape safety requirements, no door can be locked from the inside.



Figure 2-42 E-room

E-room adopts a double-layer structure, the interlayer is filled with insulation material. The normal temperature of the room control between 20℃（68℉）and 30℃（86℉）through two air conditionings.

The temperature switch install in the room setting at 30℃（86℉）. If the temperature over this value, it'll be recorded and warning the operator through a flash signal. This fault will cause the hosit and trolley operated in slow speed.

2.4.9 Energy chain system

The trolley use the cable energh chain or festoon system to provide the power.

Energy chain system as shown in Figure 2-43. The cable chain running smoothly in the stainless steel guide, with less influenced by the wind.



Figure 2-43 Energy chain system

This project use IGUS's energy chain, type is 5050RHD.30.300.0. The chain connects the trolley with L bracket. The bracket should avoid interfere with girder's platform. As shown in Figure 2-44.

Because of both trolley and energy chain has the installation and operation error, so we use floating connection between them. Shown as in Figure 2-45.



Figure 2-44 Connection bracket of the energy chain

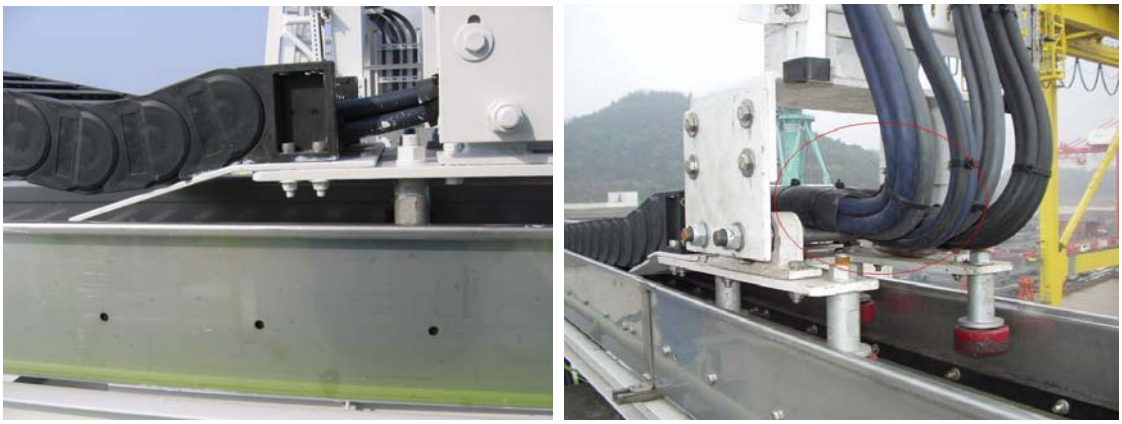


Figure 2-45 Connector of the energy chain

More details about the maintenance of the energy chain, please refer to the suppler's manual.

The festoon system shown as in Fugure 2-46.

More attention shall be paid to protect the cable from wind if festoon system is used.



Figure 2-46 Festoon system

2.4.10 Auxiliary facilities

2.4.10.1 Side guide rollers

In order to ensure the trolley moving along the rail safely, two ends of one trolley beam install with a pair of guide rollers.

Each set of side guide roller consists of a bracket, two eccentric shaft, two rollers and bearings etc. The gap between the guide roller and side of rail can be adjusted by eccentric shaft. The side guide rollers as shown in figure 2-48.



Figure 2-48 Side guide rollers

2.4.10.2 Trolley anchor device

In order to prevent the trolley from moving by high winds when the RTG not working in a long time, two anchor devices have been set in both side of the trolley frame. When the trolley pariking in the anchor position, and the wind speed exceed the warning, the archor pin need to plug. When the pin plug, the limit switch will forbid the crane to be operated. If the trolley is not anchored and storm wind come it'll be very dangerous. Even if it has the brake, the trolley could move along the rail. Once the trolley moving, it'll be enormous kinetic energy and may cause the accident

When the wind speed above 25m/s, the crane will out of service. The RTG must be in anchoring state.

2.4.10.3 Buffer

In order to fully guarantee the trolley safety, buffers are installed at the end of the girder to prevent the trolley from running out of the girder. The buffer shown as in Figureure 2-50. The bufferinstalled on the rigid support.



Figureure 2-50 Buffer

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Chapter 3 Crane maintenance instructions

Summary：

The service life of crane and auxiliary components greatly depends on the purchaser's **proper use, safe operation and timely maintenance**.

This Chapter gives a system expatiation on how to maintain correctly and timely. Crane maintenance mainly includes the following five parts: **cleanliness, fastening, adjustment, lubrication and inspection**. In order to fully achieve its potential service life, timely maintenance is very important for the crane and auxiliary components.

3.1. General

Crane Maintenance is composed of **Inspection, Routine Maintenance** and **Repair.** This Chapter basically expands upon the operating maintenance and routine inspection of the crane.

The inspection and routine maintenance is used to check whether the crane is in a safe state and to place the crane into a condition where a high confidence of reliable operation can be developed. The main work of the inspection and routine maintenance includes cleanliness, lubrication, fastening and connections, spot-check and periodic inspection.

It is basically a visual inspection and some functional/detective tests. If, in doing so, an adequate assessment is not possible, a disassembly of crane parts has to be undertaken to get access to the hidden parts. Chapter 4 “The repair Instructions” will illustrate crane repair procedures in detail on components which need disassembly and disengagement.

The cautions and instructions about inspection and maintenance are outlined, aiming at giving a general idea on how to carry out an inspection and a routine maintenance.

Inspection of cleanliness
Inspection of fasteners and connections
Inspection of concentricity (alignment) periodically
Inspection of lubrication
Inspection and preventive actions

In addition to this maintenance manual, all subcontractors have manuals for their own components. To maintain such components, please refer to their manuals. The list of these manuals is in Table 3-1 as follows.

Table 3-1： Manual from Subcontractors（Mechanical Part）

Reducer	ZPMC(G)/SEW(H/T)
Genetator	CAT
Motor	Zhongyuan
Brakes	ZPMC
Drum coupling	ZPMC
Motor coupling	ZPMC
Headblock	ZPMC
Spreader	BROMMA
Power Chain system	IGUS
RTG hybrid power system	ZPMC
Wire rope	TOKYO
Tyre	SIMEX
Energy chain	IGUS

3.2. Cleanliness

The dust, oil dirt, water, impurity interfusing into the lubricant, rust and etc. may erode, speedup abrasion and depress even destroy partial original performance like reduce the braking torque, bring the electrical system short circuit or other malfunction, infect the leak-proofness of the hydraulic system and even destroy the hydraulic parts and bring severe result like system malfunction. Cleanliness is not only to keep a friendly environment but also to keep the machine super performance and prolong its lifetime. So in the routine maintenance the impassive contamination to the parts and equipments should be timely removed and keep the equipment leak tightness, without dust and oil dirt. The following are the routine methods and typical examples to keep the crane clean.

3.2.1 Major methods of cleanliness

3.2.1.1. Dry cleaning

"Dry cleaning" used here means adopting physical methods to remove the contamination from the surface without water or solvent. Shown in Figure 3-1, the main methods including:

1) The cleaner can directly remove the dust and small dirt from the body surface. To create a clean environment and improve the service life and stability of equipment, the place where mechanical and electric equipments concentrate like machinery house, electric house and cab should be cleaned periodly.

2) Air compressor and air blower can directly remove the powder, iron chipping and other waste falling off from body surface easily. Its application including: cleanness of the faying face of the precision components such as bearing and deadeye, and wiping off dust from the surface of complicated shape part, electric element and other elements .

3) Cleaning by machining：

   a. Use wire brush or electric polisher with copper brush. This method can remove the flotation rust or metal surface treatment before repairing paint.

   b. Cleanness by finish machining such as turning or grinding. This method only is adopted in renovating the defective surface such as bulge or sunken in the brake disc, brake lining and so on.

3.2.1.2. Using cleanser

The cleanser is needed when the oil dirt on the surface is difficult to remove. In choosing the cleanser, advert to the problem that the solvent or cleanser based on halogenated hydrocarbon like trichloroethylene and dichloromethane will bring chemical reaction (injecting little water into the trichloroethylene, hydrochloric acid will generate) with the aluminum part, austenitic steel and galvanized part, and the part will be oxygenized. On extreme condition, such reaction will bring explosion. So the solvent or cleanser based on halogenated hydrocarbon shouldn't be used when cleaning the metal parts of the crane. In addition, the acid and alkali liquor shouldn't be used to clean the mental parts of the crane either, or it will destroy the parts directly. To cope with the common dirt, such cleanser as general gasoline bought in the market and industrial benzene can be used. As for the dogged dirt, use the cleanser such as dimethyl benzene.



Industry Cleaner



Air Compressor and Air Blower





Wire Brush、Copper Brush and Corner Grinder

Figure 3-1：Routine dry cleaning method

⚠	**Attention: Necessary protective equipments such as the goggles, gauntlet and costume should be worn to prohibit hurting to the body by powder, scrap iron and even the cold wind.**
⚠	**Attention: Some cleansers such as methylbenzene, dimethyle benzene, acetone, and trichloroethylene are poisonous. The related instruction book should be read carefully before using it.**

3.2.2 Main cleaning work

The following are some typical examples of cleanliness：

3.3.1.1 CLEANLINESS OF COMMON PARTS

1) Every rotating pin shaft of brake should be kept clean and free in rotation. No paint contamination and rust will be permitted (shown as Figure 3-2).



Figure 3-2：Oil dirt and rust on the brake disc must be removed

2) The cooling fan filter of the motor should be kept clean and properly ventilated. As to the DC motor, the commutator should be kept clean.

3) The internal of gear reducers should be kept clean. The bottom of the housing must be cleaned when oil is replaced.

4) The trolley running rail surface as well as both sides of rail should be clean and no foreign matter is permitted。

5) The gantry runway surface should be clean, and no foreign matter is permitted.

6) The gantry parts which easier to be contaminative should be cleaned periodically.

7) All glass in the operator's cab, E room should be cleaned frequently to ensure a good view for the operators.

8) Indication lights should be cleaned if there are powder dust and other contamination to ensure that the operators and maintenance personnel can identify colorful lights clearly.

9) All rear mirrors of the operator's cabin should be kept clean.

10) Remove excess grease and oil resulting from lubrication of the crane components, both as part of the lubricating activity and following a short period of operation.

11) The parts should be cleaned before reassembling.

3.3.1.2 CLEANLINESS OF THE ELECTRIC ELEMENTS AND SYSTEM

1) The E room should be kept clean. Doors and windows should be closed, and powder dust inside control panels and other places should be cleaned by vacuum cleaner. Pressure fan should be operated at all times.

2) Using dust collector to clean the place easy to sorb dust like the controlling panels and computer screens in the electric and CMS house (as figure Figure 3-3).



Figure 3-3：PLC controlling panel

3.3.1.3 Cleanliness of the hydraulic system

Clean oil is the life of the hydraulic system. If impurity trespass the small damping holes or the gap in the precision component, the component will be pulled damaged and the oil roads will be jammed, which will bring danger to the safety running of the hydraulic system. Generally the impurity inbreak methods include: unclean hydraulic oil, unclean oil filling tools, carelessness in filling oil and maintenance, impurity slipping from hydraulic elements and air or water entered into the hydraulic system.

A. The way to prohibit the impurity from inbreak

1) Before fill oil, the hydraulic oil should be filtered, the oil filling tools should be clean and the filter placed at the entry of the oil tank should not to be removed in order to speed up filling. To prohibit the solid or fibre impurity from inbreak, the oiler should keep the glove and clothing clean.

2) In maintenance when the oil filling cover, the filter cover, inspecting holes or the oil pipe is taken down, keep away from the flying dust. The cover of the oil tank can not be opened until the dirt around the tank cover has been removed and the impurity at the connection place has also been removed after loosing the bolts (to refrain the water from infiltration, the water can not be used to clean the impurity).

3) Choose the cleaning materials without fibre impurity dropping off and hammer with rubber beating surface if they have to be used. Hydraulic elements and hoses should be cleaned and blow dry before assembling. The filter also should be cleaned when oil changes. The type of cleaning oil should be the same as that of the system, the oil temperature should be kept between 45～80℃.not lower than 25℃. In order to completely remove the impurity in the system, keep cleaning oil flow in large capacity. Hydraulic system should be cleaned over three times and after each cleaning the cleaning oil used should be sent out completely before it cool down. Before fill new oil, the filter should be cleaned and new filter element should be adopted.

B. Keep the air out of the hydraulic system

The entrained air in the hydraulics will cause elements cavitation erosion, make the compressibility increased and the operation instability, reduce the work efficiency and make the actuator “crawl”. In addition, the air will also make the oil oxidation and speeding up its metamorphism. To keep the air out of the hydraulics the following list should be paid attention to:

1) Random check should be conducted after maintenance and changing oil, and then begin normal work.

2) The mouth of the inlet pipe in the pump can not come out of the oil lever and keep the inlet pipe being well sealed.

3) To keep well obturation for the driving axis of the lubropump, oil seal certified commended by the supplier should be adopted and ensure the function of airproof and oil proofing.

C. Prevent water penetration into hydraulics

If the oil contains excessive water, that will make the hydraulic elements corroded, the oil emulsified and metamorphosed, the tenacity of lubricating film reduced and abrasion speedup. If the water in the oil is exceeding standard, the hydraulics needs to be clean thoroughly and the oil should to be replaced.

D. Inspection of the filter element

Filter is the cleaning equipment used in the hydraulic system. Inspect the squalidity degree of the filter and the characters of the impurity on the filter element to provide evidence for further fault diagnosis.

3.3 Fastening

3.3.1 Grade of fasteners

Although all bolts used on the crane are based on Chinese standard, the Chinese standard for bolts is same as the ISO standard. The bolts are also equal to the American metric bolts in specification and have parallelism relation in grade. So it is very easy to find suitable bolts for replacement. TABLE 3-2 is another comparison table for bolt grade in different standards.

The strength grade is usually marked in the head and end of the bolt (shown in Figure 3-3). Bolts used in the crane are usually greater than or equal to grade 8.8 for normal carbon steel and alloy steel bolts, so less grade bolts can not to be used in the maintenances.

It must to be noted that the grade classification as mentioned above is suited for the bolts with the material carbon or alloy steel, and it is different from the stainless-steel bolts. Detail information please refers to other related data. Generally stainless-steel bolts with grade A2-70 or ASTM316, nuts and washers are used when the bolts are less than M14 in the crane, so the same grade bolts shall be used in maintenances.

Table 3-2： SAE Grade and ISO Grade

ISO898-1	SAE J429	ISO898-2	SAE J995
Bolt	Bolt	Nut	Nut
Grade 4.8	Grade 1	4	
Grade 5.8	Grade 2	5	Grade 2
Grade 6.8		6	
Grade 8.8	Grade 5	8	Grade 5
Grade 9.8		9	
Grade 10.9	Grade 8	10	Grade 8

Fasteners used for metal structures should be ISO Grade 8.8 and 10.9 (ASTM A325 or ASTM A490) high-strength bolts. The bolt head and nut are usually bigger than the common ones and the matched bolt, nut and two washers are supplied as a complete set (shown in Figure 3-3, a set of bolt connection with grade 10.9). There are firm requirements on the bolt application, and only some simple knowledge is offered in this chapter, for more detail information please refer to the related criterion. As for the bolts ASTM A325 and ASTM A490，the《Specification for structural joints using ASTM A325 or A490 bolts 》written by AISC can be referred to. Pretightening torque is required for many bolts and it is realized by using a calibrated torque wrench. As listed in Figure 3-44 are some typical torque wrenches.

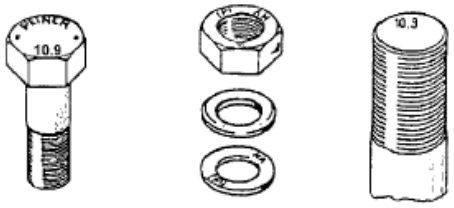


Figure 3-3：10.9 Grade fastener for structure



Hand torque wrench　　Hydraulic torque wrench　　Electric torque wrench

Figure 3-4：Typical torque wrench

3.3.2 Inspection of bolts

Due to alternating loading in working the bolts may be loosened. In order to reduce stress amplitude, the bolts shall be pretightened to set value.

Periodic inspection is aimed to insure the bolts loaded uniformly. Replace the bolts having yield and been lengthened or the deficiency-worried onebefore fatigue broken.

Specification should be carried out during the downtime of the crane. There are two methods about the fastener specification： visual check and using wrest wrench，and visual check is mostly used

3.3.2.1. Visual check

Using the following methods to visually checking, refer to the Figure 3-55:

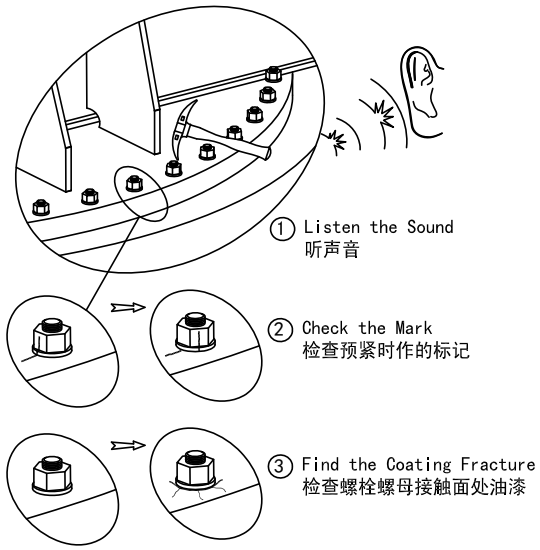


Figure 3-5：Methods of Finding a Loosened Bolt

1) The sound of a pointed hammer hitting the head of a bolt can indicate the looseness/tightness of the bolt.

2) The position between the nut and the connecting part should be visually checked (during the fabrication in the ZPMC's facility, the positions between some critical nuts and connecting parts will be marked out with a marking pen). If the marking positions are deviated, it indicates that the bolts are loose.

3) The paint film of nuts and connecting parts can also be visually inspected. If the paint film has fracture, the bolts should be inspected.

3.3.2.2. Use the torque spanner

A. Common fasteners（excluding the Fasteners used for metal structures）

1) Before inspection, the crane should be off working and make the bolts idly without working stress.

2) If needed, remove all the dirt and paint on the fasteners. Spray a thin layer of oil-lubricant rust remover (CRC or other equate product) on the nut and washer. The lubricant rust remover is functioned after 10~20 minutes.

3) Mark the original location by a marking pen on the nut and on the joint surface of the structure and the washer (shown at Figure 3-6).

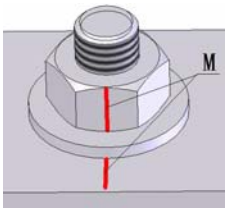


Figure 3-6：Marked bolt

4) An initial tension of 75% of the final tension shall be developed.

5) Check the nut: if the nut can be turned, make it out by a marking pen or painting and refasten or replace it later on.

6) If needed, the bolt joint should be replaced.

7) Repeat the process 3)-6) to inspect the other bolt joints.

8) The rotating nut should be refastened after the loosing the bolt and according to the methods mentioned in the section 3.3.3.3-A.

9) The non-rotating nut should be refastened according to 10) ~11) step by step.

10) Turn the nut about 30°by a torque spanner to loosen the nut.

11) Refasten the nut to its pretightening torque requirement.

12) Remove all the dirt and use the proper solvent to clean bolts, nuts, washer and places around the joint surface of the structure, and then repair the damaged paints according to paint repairing process.

B. Fasteners structures（**take the flanged connection as an example**）

1) Before inspection, make the crane under stopped state and make the bolts idly without working stress (without loading).

2) If needed, remove all the dirt and paint on the fasteners. Spray a thin layer of oil-lubricant rust remover (CRC or other equate product) on the nut and washer. The lubricant rust remover is functioned after 10~20 minutes.

3) Use a marking pen to mark out the original position on the nut and the surface between the connection part and the washer (shown in the Figure 3-11).

~~4)~~ An initial tension of 75% of the final tension shall be developed by a marked torque wrench.

5) Check the nut: if the nut can be turned, make it out by a marking pen or painting and refasten or replace it later on.

6) Repeat the process 3)-6) to inspect the other bolt joints.

7) Account all the nuts that marked rotation and should be replaced.

8) If the account of the bolts marked to be replaced exceeds 20%, use the same grade bolts to replace them.

9) If the account of the bolts marked to be replaced under 20%, replace all the marked bolts and the bolts on both sides.

10) The non-rotating nuts should be refastened according to 11) ~13) step by step.

11) Turn the nut about 30°by a torque spanner to loosen the nut.

12) Refasten the nut to its pretightening torque requirement.

13) Use proper solvent to clean the bolt, nut, washer and the joint surface of the structure. Then repair the paint according to the "paint repair process".

3.3.2.3. Inspection period for the fasteners

Table 3-3：Inspection Period for the Fasteners

Inspection Type	Inspection Period （before the first maintenance）*	Inspection Period （after the first maintenance）*
Visually check	Per 3 months （except special positions）	Per 6 months （except special positions）
Use torque spanner	10% spot-checked (the sixth month or 50000 work cycle)	the 5th, 15th、25th、30th year
Replace fasteners	If needed	If needed

* The first maintenance here gives priority to the first expiration of the 6th month or 50000 in work cycle.

After the crane is put into operation for six months or 50000 work cycle(which comes earlier), the tightness of bolts should be checked according to Table 3-3： Because the bolts are easy to be loosened in the primary stage, so an all-around inspection should be carried out. High-strength bolts should be spot-checked by 10% for each group. If any one bolt is found loose, all bolts within this group should be thoroughly checked. In addition, the bolts bearing vibrating or varying load and the effective anti-loosening methods have not been adopted, the inspection period should be shortened to avoid accidents. Electrical wiring and connecting bolts of small electrical parts, such as bar wiring connection in control panels, motor wiring, junction box wiring, various kinds of limit switches, encoders and cam limit switches (inside and outside), usually have smaller area and size. Impact and vibration during the operation of the crane may cause the bolts to loosen. Once the bolts are loose, the electrical control will fail. Therefore, the inspection of fastening condition for this kind of bolts must be carried out, and the wiring condition should be inspected periodically.

3.3.3 Process for bolt's replacement and tightening

3.3.3.1. Replacement standard

The bolts under the following condition should be replaced：

1) The bolts required to be replaced mentioned in the above section.

2) The bolts with such defects like impairment, distortion, slide tooth, lacking tooth, corrosion, big change of the screw roughness and so on.

3) If the bolts following the standard GB1228、GB1229、GB1230 ASTM325 and ASTM490 are hot dipped galvanizing, they are not allowed to be used repeatedly, and should be replaced once loosened after being final tightened. According to the standard AISC only the bolts by black finished can be used again with engineer's permission. In addition, when the adjacent bolts are loosened by tightening bolts, they are not needed to be replaced.

3.3.3.2. Tightening torque for the bolts

As listed in the table 3-4 is the tightening torque for the bolts, studs and screws. if no special requirement, the data in the table can be used directly. Pretightening torque Mo = K*Fo*d (K is the torque factor, Fo is the pretightening force and d is the bolt diameter). The friction factor of the nut and washer, the nut and bolt changes will cause the torque factor change correspondingly, so even when the required pretightening force is unchanged, but the torque factor is changing. Generally under dry connection, the torque factor K=0.145, if a little lubricant or MoS_2 is filled, K=0.12. For the important connection, the torque factor is got by spot-check of the each bolt or that of the same connection and measuring through experiments. Before finishing tightening the fact torque factor of all the high-strength bolts or other similar connection should be measured.

Table 3-4：Tightening Torque of Stainless Steel Bolts with Small Size (Grade A2-70 级)

Size	M5	M6	M8	M10	M12	M14
Tightening torque (Nm)	3.2	5.5	13	26	46	74

Table 3-5：Tightening Torque of Major Bolts

Note: * = without lubrication ** = with a little lubricant or MoS2 lubrication

Size	Tightening torque (Nm)				Tightening force (N)	
	8.8s*	Grade 8.8s**	10.9s *	10.9s**	8.8s	10.9s
M 12	66	54	92	76	37766	53109
M 14	105	87	147	122	51520	72450
M 16	163	135	229	190	70336	98910
M 18	225	186	316	261	86016	120960
M 20	318	263	448	370	109760	154350
M 22	433	358	609	504	135744	190890
M 24	550	455	774	640	158144	222390
M 27	805	666	1132	937	205632	289170
M 30	1093	905	1537	1272	251328	353430
M 33	1488	1231	2092	1731	310912	437220
M 36	1911	1581	2687	2224	366016	514710
M 39	2473	2046	3477	2878	437248	614880
M 42	2851	2360	4009	3318	468168	658362
M 45	3818	3160	5369	4443	585109	822809
M 48	4593	3801	6459	5346	659973	928087
M 52	5937	4913	8349	6909	787401	1107282



The torque listed in the table is only suit for the connection surface without special requirements. The bolts in such connection are only bearing pretightening force, and the working force is vertical to the bolts and transferred through the friction from the connection surface. For those bolts bearing not only the pretightening force but also the working force, for example the bolts used in the bearing pedestal and the motor base, the pretightening force should be recalculated (the tightening torque is often less than the one listed in the table), please refer to the related drawing.

3.3.3.3. Bolt fastening and replacing process

A. Fastening process of the general blots (excluding the high strength bolts for structure) during the first installation or replacement.

1) Ensure the washer be planished and clean without oil dirt and burr。

2) Clean the connecting surface of the structure and washer and ensure the surface be smooth.

3) Clean the bolts and nuts.

4) Set well the washer and ensure the inclined plane touch the bolt head or the nut.

5) An initial tension of 75% of the final tension shall be developed. To ensure the flange bolts load equally, fasten the blots in the order from the rigid joint to free end or form the center of the bolt group to around. If the connection steel plate warps and the connecting bolts are fastened from the two ends to the middle, the middle of splice plate may plumps up and can not keep closely which will make friction forces lose partly. For detailed fastening process refer to Figure 3-. Make marks on the large hexagon head and high strength bolts to identify their first fastening to avoid missing.

6) Finally fasten the blots to 100% required torque. The fastening order is similar to the first one. Generally use two kinds of color to distinguish the first and final fastening.

B. Replacement of the high strength bolts for structure（flange connection）

1) To ensure the same torque factor, the bolts replaced one time should be the same batch products.

2) Ensure the bolts be enough in quantity, have the correct strength degree and tightening torque. Revise the torque wrench repetitiously before using.

3) Ensure the washer be planished and clean without oil dirt and burr。

4) Clean the joint surface of the structure and washer and ensure the surface be smooth.

5) Clean the bolts and nuts.

6) Use a new bolt to replace the old one which marked to be replaced and an initial tension of 75% of the final tension shall be developed immediately. Pay attention that there should be only one bolt to be lossened and disconnected at anytime. Ensure the blots to be replaced and fastened one by one.

7) The old bolts being taken down should be destroyed purposely to prevent from using again.

8) After all the bolts being replaced, the final tension shall be taken to all the new bolts in the order shown in Figure 3-.

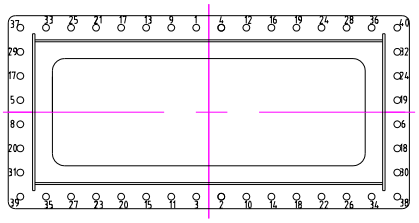


Figure 3-7：Fastening order of the flange bolts

3.3.4 Anti-loosing methods for bolts

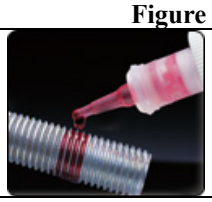
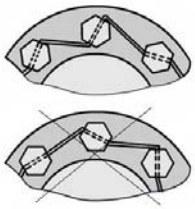
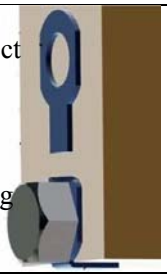
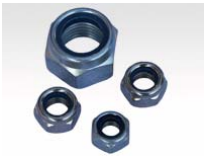
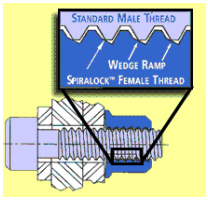
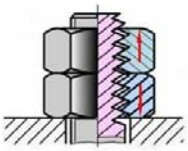
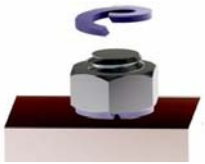
It is very important to adopt some methods to prevent bolts from loosing. If the bolts are loosened, the structure strength will be reduced and the downfallen bolts will bring danger to persons. Table 3-6： lists the main anti-loosening methods for the bolts used on the container crane.

3.3.5 Pin connection

Pins shall be well greased prior to each installation. After the installation, this process is to be repeated at all accessible spots of the pin.

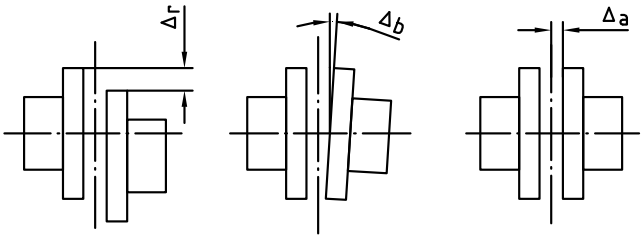
- If snag rings are used to secure pins, the snap rings must only be mounted once.
- New snap rings shall be used after a dismounting of the pins.

Table 3-6：Anti-loosening methods of the bolts used on the container crane

METHODS	Instruction /application	Figure for example
using anti-loosing glue	The anti-loosing glue is often used in the fields with limited place, for example used on the fixing bolts of the anti-abrasion block for the wire rope.	
Lock washer and Fastening with wire rope	Both of these methods are comparatively reliable and have been widely used. Such methods are commended to be used in the fields without space restriction for example on the running or moving parts like braking disk, sheave and so on to prevent it from falling or flying off when the bolts are loosened.	 Correct Wrong 
Nylon anti-loosing nut SPL nut Double nuts	Nylon anti-loosing nut is the most widely used and it is reliable to anti-loose when matching with bolts. SPL nut has a strict requirement on the retightening torque and the nuts can be reused. Double nuts are not used commonly on the crane for inconvenience.	 
Spring washer or other anti-loosing washer	Spring washer is not very reliable and usually generates yellow water when used outdoors. Usually they are not used in ZPMC'S crane. But in some outsourcing or unimportant parts the stainless steel washers are still used.	 

3.4 Inspection of alignment

For rotary components such as couplings, inspection of alignment should be completed periodically. The alignment of couplings has great influence on the mechanism running and life. Therefore, the coupling alignment should be checked periodically. In the first year that the crane is brought into service, the inspection cycle should be shorter, e.g., once per three months. After the mechanism get stable in the service, the inspection cycle can be lengthened suitably. This inspection mainly contrapose the motor coupling and drum coupling of main hoist, and trolley travel drive and the motor coupling of gantry travel drive. The standard of alignment is determined by the request of motor's alignment and the style of coupling. **Figure 3-3** shows the main check items of coupling's alignment. Table 3-7 shows the standard's detail of coupling's alignment.



Radial misalignment Δr

Gap(angular displacement) Δb

Axial displacement Δa

Figure 3-3 The check items of coupling's alignment

Table 3-7： The standard's detail of coupling's alignment

Part	Style of coupling	Δr （mm）	Δb	Δa （mm）
Main hoist high-speed coupling	flexible coupling	≤0.08	0.18 mm	＝Δr

Trolley high-speed coupling	flexible coupling	≤0.08	0.18 mm	= Δ r
Gantry high-speed coupling	flexible coupling	≤0.08	0.10 mm	= Δ r
Main hoist drum coupling	Gear coupling	---	10’	---

3.4.1 Redress methods for alignment of motor and gear reducer

3.4.1.1 Use the laser alignment measuring instrument controlled by microcomputer.

Please see the following diagram and procedures.

1) Coupling needn’t be taken apart in the inspection. The laser generatrix is fixed on the shaft on one side of the coupling and the receiver is fixed on the other side. (See Figure 3-40). Both the generatrix and the receiver should round together.

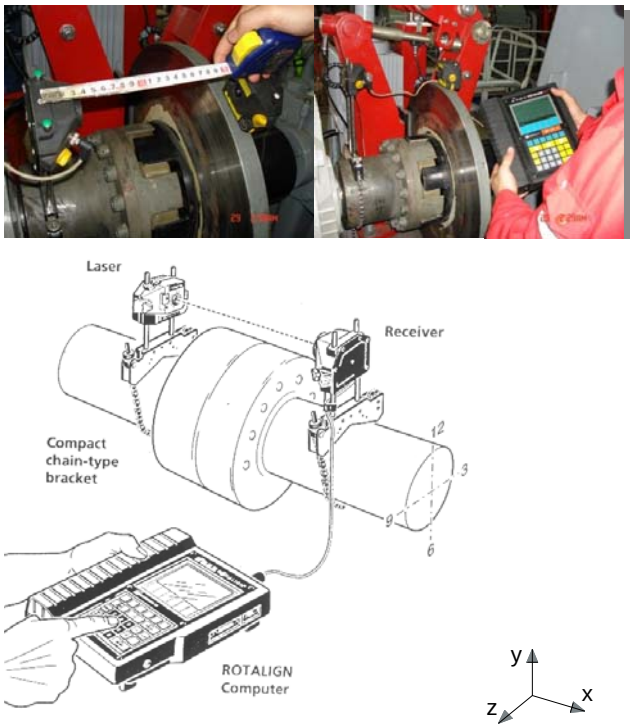


Figure 3-40 Alignment Check by Laser Measuring Instrument

2) Input correlative dimensions into the microcomputer, such as the dimensions of the motor pedestal, and the distance between the coupling center and the motor shaft ends (See Figure 3-43).

3) Before rotating, the initial value should be reset to zero.

4) Turning both the motor shaft and the reducer shaft only up to 60°-75°

5) Both the angular deviation of δ_z and the radial deviation of δ_y will show on the panel. The accuracy of the readings will be up to 1μm

6) The microcomputer will automatically calculate the value of displacement of the motor and the difference of the shim thickness which are advantageous for an operator to re-align the motor. The microcomputer has memory and printing function.

7) Add or reduce the shims to decrease the windage in vertical direction, adjust the fitting bolts or blocks for motor pedestal to decrease the windage in horizontal direction. See Figure 3-5.



Figure 3-5

8) Maybe the step 2~7 has to be repeated until the windages meet the standard.

3.4.2 Inspect the alignment with a dial indicator

Figure 3-6 shows the alignment measure method with a dial indicator.

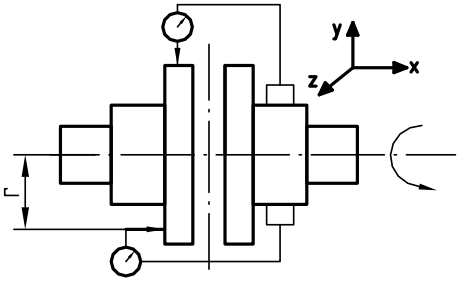


Figure 3-6 The measure method with a dial indicator

1) The indicator base is set on the shaft face of inner half-coupling and the finger contacts the other one.



Figure 3-74

2) Initially, set the reading to zero. Suppose that the starting point is 0°, then the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read after the coupling rounds 270° (please note the positive and negative signs). The circumferential jumping value in the vertical direction is $\Delta r_y = \delta_{180°}/2$.

3) The circumferential jumping value along the horizontal central line is $\Delta r_z = (\delta_{90°} - \delta_{270°})/2$. The maximum circumferential jumping value is $\Delta r = \sqrt{\Delta r_y^{\ 2} + \Delta r_z^{\ 2}}$.(See Figure 3-44).

4) Similar as shown in the Figure 3-, make the finger contact the end face of the other half-coupling or surface of the brake dick, and adjust the value to zero. After turning 270°, the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read. And $\Delta a_y = \delta_{180°}$, $\Delta a_z = \delta_{90°} - \delta_{270°}$, then $\Delta b_y = arctg\Delta a_y / r$, $\Delta b_z = arctg\Delta a_z / r$.



Figure 3-14

5) Estimate the thickness of shims under the motor pedestal and the translate in horizontal direction by Δr_x and Δr_y. And estimate circumrotation in direction Y (the windage of thickness between the front and back shims) as well as the circumrotation in direction X (the windage of thickness between the left and right shims) by Δa and Δb. Refer to 3.4.1.1 step 7) and adjust the estate of motor to meet the standard.

3.4.3 Adjustment methods of alignment for drum coupling

1) The indicator base is placed on one side of the coupling and the finger contacts the outer ring of the other side of the coupling. Refer to Figure 3-8

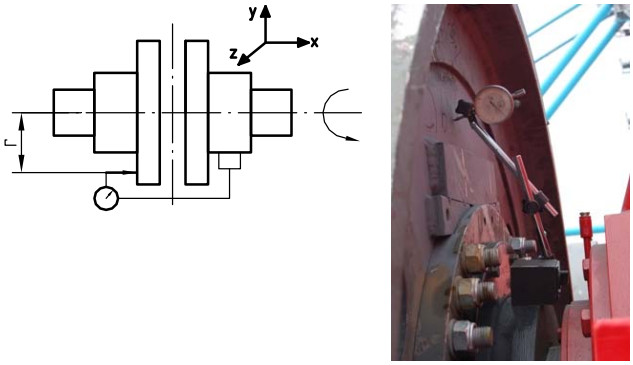


Figure 3-8：Inspect alignment of low-speed coupling with dial indicator

2) Initially, set the reading to zero. So the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read after turning 270° (please note the positive and negative signs).

3) Survey the radius r of measure-point.

4) Similar as shown in the Figure 3-8, make the finger contact the end surface of the other coupling or surface of the brake dick, and adjust the value to zero. After turning 270°, the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read .And $\Delta a_y = \delta_{180°}$, $\Delta a_z = \delta_{90°} - \delta_{270°}$, then $\Delta b_y = arctg\Delta a_y / r$, $\Delta b_z = arctg\Delta a_z / r$.

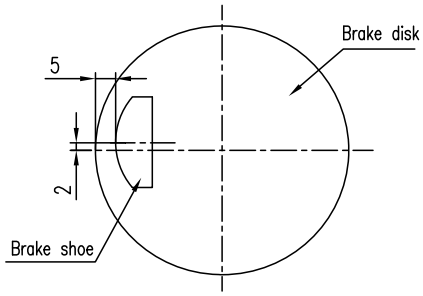
5) Adjust the thickness of the shims under the drum bearing pedestal and the bearing pedestal's position to meet the standard.

6) The drums are supported by the low-speed shaft of reducer and in the center of it, so there are only angular deviation and axial deviation, no radial deviation.

3.4.4 Alignment of brake-disc

1) When the brake is open, the clearances between the two brake linings and disc should be the same. $\delta_1 = \delta_2 = 1 \sim 1.5$ mm. Refer to Figure 3-19

2) δ_1 and δ_2 should not be less than the minimum value of 1mm set for the brake. Refer to Figure 3-9.



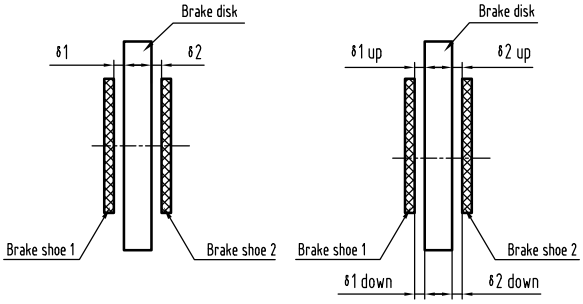




Figure 3-99： Clearance excursion of brake

3) The up and down clearance of brake disk should be the same, which means $\delta_{1\,up}=\delta_{1\,down}$, $\delta_{2\,up}=\delta_{2\,down}$. Refer to Figure 3-9.

4) Inspect the height of the brake to align its central line with the brake disc center. The deviation should not be more than 2mm. The horizontal deviation of the brake lining from the edge of disc should not be more than 5mm. Refer to Figure 3-9.

5) Inspect the position of the brake base in the motor axial direction. The deviation of the central line of disc thickness should not be more than 0.5mm. Refer to Figure 3-99

6) The contact area between the linings and the disc should not be less than 70% when brake is closed.

3.5 Lubrication

3.5.1 Summarization

Lubrication is a very important maintenance task. It will determine the normal service and endurance life of the components. Proper lubrication practice will help to avoid premature wear and extend the endurance life of the components. Lack of lubrication will destroy the components. For example, if an anti-friction bearing is short of lubrication, it will be damaged in a very short time. It is very important to monitor the temperature and sound level of the components to determine whether the components get enough lubrication. The lubrication of wire rope will be described in Chapter 4.

Attention: All used oil and grease must be collected and disposed properly!

3.5.2 Lubricant

For well effect of lubrication, the type of lubricants must be selected properly. Consistency should be maintained on lubricants. Do not mix up different lubricants together. If the lubricant supplier, type or grade is changed, clean the bearing, reducer, pin, and other devices thoroughly to remove the previous lubricant. If different lubricants are mixed together, please contact the lubricants supplier.

The lubricants and grease used in the crane are listed in table 3-8~3-10. And the comparison of those products is also listed in the tables.

Table 0-8 GearBox Oil

ISO (40 ºC, mm2/s)		Name and code name		°C pour point	°C Flash point	Place	Code name
ISO VG 220	220	GB5093-95	L-CKC	-8	200	gantry reducer	A1
		Shell	Omala	-18	238		
		Shell	Omala LPP*	-27	210		
		Shell	Omala HD	-48	245		
		Mobil	Mobilgear 630	-24	240		
		Mobil	Mobilgear XMP	-24	272		
		Mobil	Mobilgear SHC XMP	-45	244		
ISO VG 320	320	GB5093-95	L-CKC	-8	200	Hoist reducer, trolley reducer	A2
		Shell	Omala	-15	238		
		Shell	Omala LPP*	-27	210		
		Shell	Omala HD	-45	242		
		Mobil	Mobilgear 632	-18	265		
		Mobil	Mobilgear XMP	-18	268		
		Mobil	Mobilgear SHC XMP	-38	242		

* ZPMC special purpose lower temperature gearbox oil

Table 0-9 Hydraulic lubricants

ISO (40 ºC, mm2/s)		Name and code name		°C pour point	°C Flash point	Place	Code name
ISO VG 32	32	Shell	Tellus	-30	209		B1
		Shell	Tellus T	-42	242		
		Shell	Tellus S	-30	207		
		Mobil	DTE 20	-27	220		
		Mobil	DET 10M	-45	210		
		Mobil	DTE Excel	-33	222		
ISO VG 46	46	Shell	Tellus	-30	218	Emergency brake pumping station	B2
		Shell	Tellus T	-39	246		
		Shell	Tellus S	-30	218		
		Mobil	DTE 20	-27	232		
		Mobil	DET 10M	-42	216		
		Mobil	DTE Excel	-33	226		

Table 0-10 Grease

NLGI class	Name and code name		°C pour point	Work temperature °C	Place	Code name
-	Lithium based grease	GB7324-94 No.2	175	-20~+120	Heavy Duty Bearing Grease/ Equalizer hinge point/ Open Gear/ chain	C1
1			185	-20~+100		
1	Shell	Alvania EP1	190	-25~+120		
	Mobil	Mobilux EP 1				
-	Lithium based grease	GB7324-94 No.2	175	-20~+120	Wire rope	C2
-			90	-10~+60		
1	合成钙基润滑脂	SH/T0372-95 3 号	185	-20~+100		
-	Shell	Alvania EP1	190	-25~+120		
1	Shell	Malleus GL				
	Mobil	Mobilux EP 1				

NLGI class	Name and code name	°C pour point	Work temperature °C	Place	Code name
1 2	Shell Alvania RL1 Mobil Mobil Polyrex EM	180 288	-30~+120	High speed bearing/ Electric Motor Bearings Grease(ref. to the motor manual)	C3

3.5.3 Lubricant Replacement Standard

The deterioration of the lubricant lies on the work frequency and the ambience.

In inspection of the lubrication or in the use of lubrication, replace the lubricant whenever sizable quantities of any of the following conditions are found:

(1) Rubber-like deterioration

(2) Impurity

(3) Metal powder and badly discoloring

(4) Emulsification

(5) Separation of mineral oil and saponification radicel (deterioration due to oxidation).

In addition, sample specimen termly as follows to determine whether the lubricant should be replaced: take approximate 20 milliliters of lubricant from the machine and consult the lubricant manufacturer or deliver it to lab to check .The detail of standard is listed in TABLE 3-11, replace lubricant or append additives according to the test result if any standard is met. Refer to sampling program to Section 3.5.5.

Table 3-11：The standard of lubricant replacement

Test items	the standard of **replacement**	Test method
kinematics viscosity variety ($40^{o}C$)	Increase 15% or decrease 20%	ASTM D 445
acid number	decrease 35% or increase ≥0.4mgKOH/g	ASTM D 974
normal heptane thawless	≥0.1%	IP 316
sheet copper erodibility	Less than Grade 2a	ASTM D 130
Chroma variety	Chroma ≥2 than new lubricant	ASTM D 1500
humidity	≥0.1%	ASTM D 1744

3.5.4 Lubrication Procedure

(1) When the reducer oil is changed, drain the used oil while it is still warm. The tank floor must be cleaned out with oil to remove deposited silt, metallic particles and oil residue. Use the same type of oil to keep normal operation.

(2) When bearings are re-lubricated, pump the new grease into the oil pocket through grease fitting. Please clean out the old extruded grease after re-lubrication. During the first shift of operation after re-lubrication, the lubricating points should be cleaned.

(3) For lubricating motor, emergency brake, reducer and other outsourcing parts, please refer to their manuals.

(4) Wire ropes should be under good lubrication condition at all hours. Lubricating will prevent the wire rope from corrosion and reduce the friction in rope strands, between the wire ropes and the sheaves and/or drums. The type of lubricant and application frequency varies with the rope construction, the operating conditions and its functional application. .

(5) The quantity of oil/grease should be suitable.

(6) For gear reducers (splash lubrication), the oil level should be kept a little higher than the central level between the highest and lowest levels because the oil level will become lower when the gear reducer is working.

(7) For bearings and other relatively sliding parts, it is natural that the old grease flows out of the seal from one side when the new grease is pumped in from other side.

(8) Oil and grease complementary cycle. It has relationship with the ambience, sealing condition, working grade and operating frequency of each part.

(9) For reducers not used frequently, bottom valves should be opened termly to inspect the moisture content of the oil.

(10) At the early stage of the crane is put into operation, three samples should be collected from the replaced old oil to analyze the percentage of metal powder in it. Compare and analyze the lubricant to find out the working status of machine.

3.5.5 Lubrication Analyze and Monitoring

An effective oil conditioning and monitoring program can help extend the equipment life, reduce scheduled and unscheduled downtime, reduce maintenance costs and extend lubricant change intervals.

The condition of oil is very important for proper operation of a device. A Sample for each component in the table below should be taken at least once a year. An analysis of the sample should be carried out as a minimum:

1) Kinematical Viscosity

2) Particle Count

3) Moisture content inspected by Karl Fisher

4) Metal grains inspected by Spectroscopic

3.5.5.1 Basic principles of sampling of lubricant

1) Sample from the same point every time to the best of

one's abilities.

2) Sample from the machine used latently and sample when the temperature of lubricant is near that at working time.

3) Sample from the lower reaches of working group. Refer to Figure 3-20. Don't sample from the points listed below:

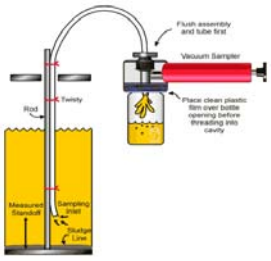


Figure 3-10: Step 1

Where there is only a little lubricant or lubricant is not ambulatory.
Where remnant and water conglomerating (for example, in the relief valve)
The end of blind-tube
The lower reaches of filter
The drainage valve on the bottom of the tank

4) Don't sample while the lubricant has just been changed.

5) Keep clean of the sampling tool (dry, no remnant).

3.5.5.2 Sampling tool in common use

Table 3-12: Sampling tool

	
sampling bottle	sampling canal
	
sampling pump	sample label / sampling point label

3.5.5.3 The steps of sampling

1) Take off the cover of a clean, dry sampling bottle and screw the bottle into the pump.

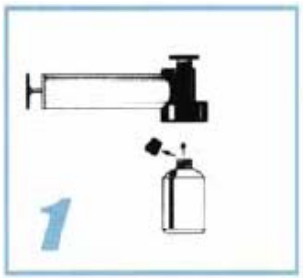


Figure 3-112: Step 1

2) Shear the ends of a new section of pipe to 45 degree.



Figure 3-22: Step 2

3) Tuck the pipe in the pump until the pipe into the bottle through the top of the pump, and then tweak the screw to fix the pipe.

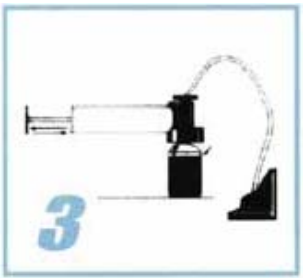


Figure 3-12: Step 3

4) Tuck the end of the pipe into the sampling point, keep the bottle upward. Absorb the lubricant to the bottle with the pump, and remain 25mm air clearance.



Figure 3-13: Step 4

5) Turn off the bottle and tweak the cover to avoid pollution.

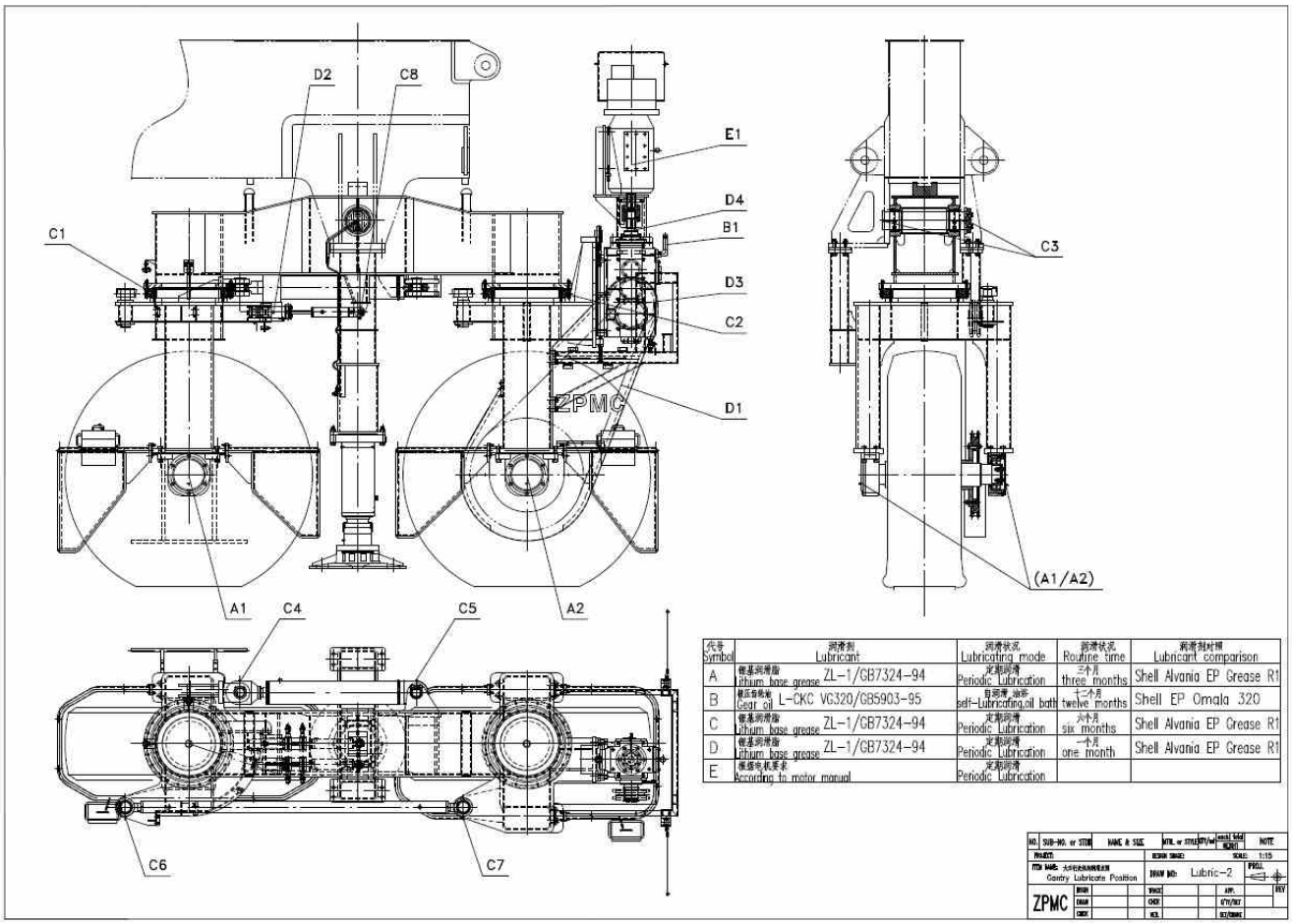
6) Fill in the sampling label clearly and stick it to the sampling bottle. The collection of lubricant is completed.



Figure 3-23: Lable

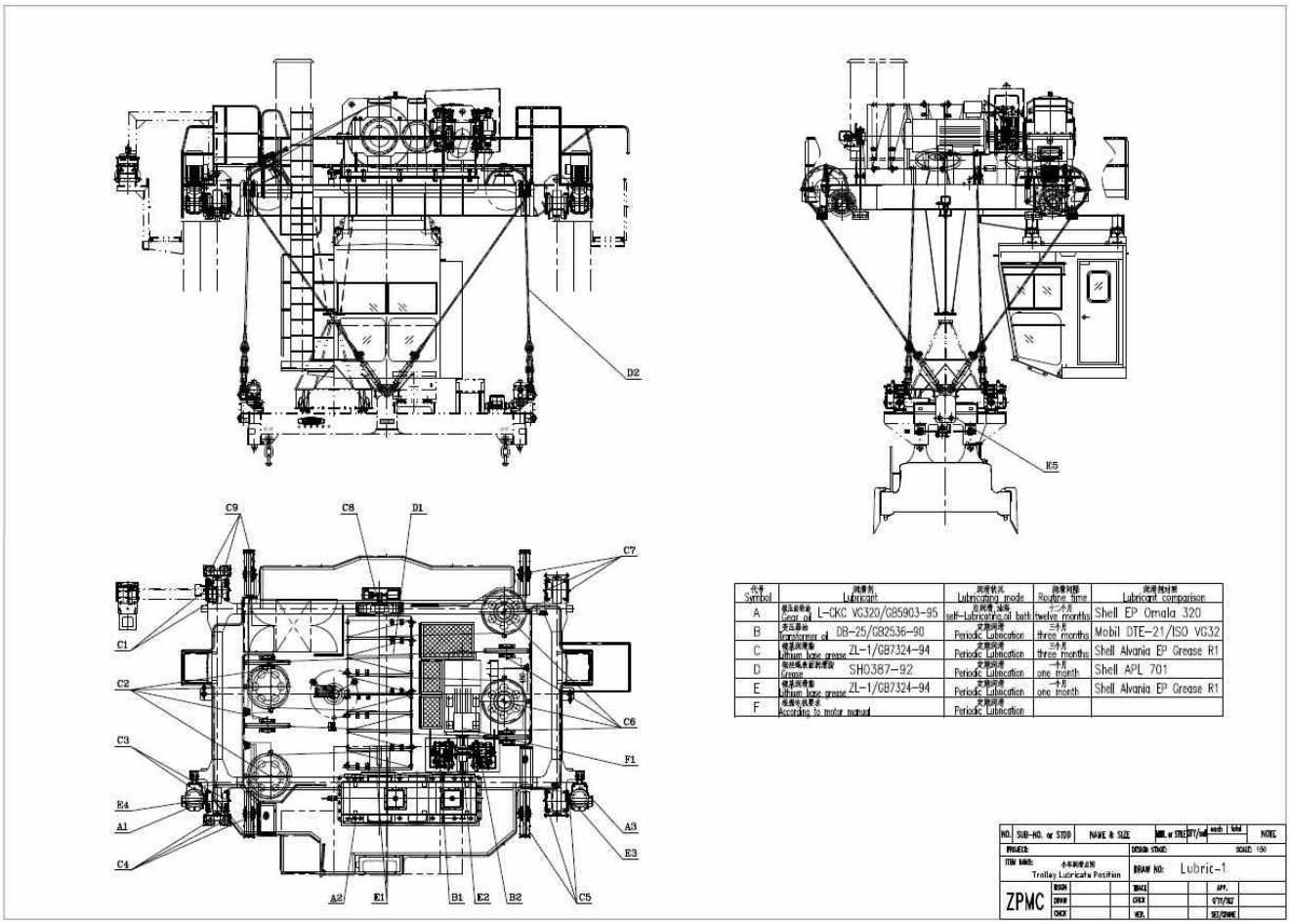
3.5.6 Lubricating points layout

Figure 3-24 and 3-25gives the assembly to be lubricated, the amount of lubricating points and the style of bearing at lubricating point. There are relationship between the figures and Table 3-13.



Symbol	Lubricant	Lubricating mode	Routine time	Lubricant comparison
A	Lithium base grease ZL-1/GB7324-94	Periodic Lubrication	three months	Shell Alvania EP Grease R1
B	Gear oil L-CKC VG320/GB5903-95	self-Lubricating,oil bath	twelve months	Shell EP Omala 320
C	Lithium base grease ZL-1/GB7324-94	Periodic Lubrication	six months	Shell Alvania EP Grease R1
D	Lithium base grease ZL-1/GB7324-94	Periodic Lubrication	one month	Shell Alvania EP Grease R1
E	According to motor manual	Periodic Lubrication		

Figure 3-14 Gantry lubricating points layout



Symbol	Lubricant	Lubricating mode	Routine time	Lubricant comparison
A	Gear oil L-CKC VG320/GB5903-95	Self-Lubricating oil bath	twelve months	Shell EP Omala 320
B	Transformer oil DB-25/GB2536-90	Periodic Lubrication	three months	Mobil DTE-21/ISO VG32
C	Lithium base grease ZL-1/GB7324-94	Periodic Lubrication	three months	Shell Alvania EP Grease R1
D	Grease SH0387-92	Periodic Lubrication	one month	Shell APL 701
E	Lithium base grease ZL-1/GB7324-94	Periodic Lubrication	one month	Shell Alvania EP Grease R1
F	According to motor manual	Periodic Lubrication		

Figure 3-15 Trolley lubricating points layout

3.5.7 Lubricating frequency

The lubricants and lubricating frequency are listed in Table 3-13. Lubricating frequency has relationship with the weather such as temperature and rainfall, and the service condition of the crane at the terminal. The lubricating frequency listed in the table can be used in common condition. The maintenance people should observe whether the lubricating is enough during the cranes service, to protract or shorten the lubricating periods properly. Refer to Figure 3-14: Gantry Lubricating points layout and Figure 3-15: Trolley Lubricating points layout while using the table.

The lubricating frequency listed is based on ZPMC's experience and common condition. Please keep an intact note or log. Based on the maintenance note and well lubricating condition, the lubricating frequency can be redefined or changed

The interval of lubrication is at most 3 years commonly. After this period, the lubrication should be changed even if the lubrication is not metamorphosed. Refer to the lubrication suppliers' recommendation.

Sample the hydraulic oil and gear lubrication while the recommended replacing period listed in table 3-13 is reached. If the sample shows that the lubrication needn't be changed, the lubrication can be used while monitoring should be reinforced. What's more, gear lubrication should be sampled every 2 years even if the replacing period is not reached.

The lubrication of forestay is very important for keeping well working status. If there is rust on the friction surface between the pin and stay bar, appended bending moment will be engendered in the stay bar while it is turning, and flaw can be engendered and expanded. So please remove ancient grease and add new grease every month.

Table 3-13: Lubricating frequency

symbol

D	Daily	3M	Every three month or～1000 hours
W	Weekly	6M	Every six month or～2000 hours
M	Monthly ～or 300 hours	A	Annually or～4000 hours
H	According to working time (hour) or else		
Abbreviation of lubrication （Refer toTable2-8～Table2-10）			
A1	ISO VG 320	A2	Mobil SHC 632
B1	ISO VG 32	B2	ISO VG 46
C1	EP 2	C2	Chesterton
C3	Mobile(refer to supplier)		
Abbreviation Of Lubrication Points Types：			
Ⅰ	Hydraulic	Ⅱ	Gear
Ⅲ	Common anti-friction Bearing	Ⅳ	Motor Bearing
Ⅴ	Sliding Bearing	Ⅵ	Chain drive
Ⅶ	Gear Coupling or Spline	Ⅷ	Wire Rope
Ⅸ	Others such as spring, seal…		
🕮	Refer to drawing / correlative independent part maintenance manual		

Lubric-ation point P	**period**							**lubricant (code name)**	**Lubrication point style / remark**	**Amount/ RTG**	🕮
	D	W	M	3M	6M	A	H				
1				●				C1	Gantry wheel shaft support	16	
2							●	A1	Gantry reducer	4	Refer to reducer maintenance manual
3					●			C1	Equalizer pin/slewing bearing/steering system	32	
4			●					C1	Chain/ reducer bearing/pin	20	
5							●	C3	Gantry motor	4	Refer to YASKAWA motor maintenance manual
6							●	A1	Main hoist reducer / trolley reducer	3	Refer to reducer maintenance manual
7							●	B1	Brake	2	Refer to brake maintenance manual
8				●				C1	Sheave/trolley wheel/drum bearing support/side roller/floating frame	29	
9			●					C3	wire	8	Refer to wire maintenance manual
10			●					C1	Reducer bearing/drum coupling	5	Refer to reducer maintenance manual
11							●	C3	Main hoist motor	1	Refer to YASKAWA maintenance manual

Attention: To oil seal lubrication, add grease according to the direction of nameplate near the oil fitting, to avoid redundant grease come into the tank and mix with the gear lubrication.

3.6 Prevention and inspection

In order to ensure reliable crane operation and prolong the crane's life, each component on the crane must be operated correctly, and a proper maintenance schedule must be followed. To keep up the operating status and operational safety of the crane, please accomplish the maintenance of the crane at regular intervals listed in the rear of the chapter. The crane routine maintenance and inspection can be started with the following aspects.

3.6.1 Summarize

The inspection is used to judge whether the crane is in a safe and reliable state. It is basically a visual inspection and functional test. If an adequate assessment is not possible by this, further tests have to be taken, e.g. nondestructive testing. If necessary, disassemble the crane parts to observe the hid parts. When performing inspection and maintenance on the crane, the appropriate preventive actions should be taken according to TABLE 3-13.

The abnormal temperature, sound, or libration can be the omen of failure, abrasion or mangle. The following sections list the detail.

Table 3-143： Preventive action (mechanical)

Inspection Items	Contents of Inspection	Preventive Action
Gear Gear coupling	Grease on gear Alignment Gear engagement and wear	Apply grease if poorly greased. Realign if out of tolerance Adjust the attachment if engagement is not proper. Replace excessive wear ones
Sliding surface of bearing, sheaves, wheel, pins, lever pins, etc.	Lubricating condition	Apply if necessary so that respective lubrication points are lubricated as required.
	Wear condition	Replace if wear is excessive
Bolts, nuts, keys, pins, shaft coupling bolts	Looseness, Attachment condition of split pin retainer	Tighten sufficiently. Replace if quality is deteriorated
Gear box	Lubrication Temperature sound	Add oil if the amount of oil is low. Replace if quality is deteriorated. Check abnormal sound
Wire ropes	Lubricating condition, broken wires, abrasion	Apply grease as necessary if grease is insufficient or degraded.
Wire rope end termination	Condition of wire end	Tighten loose bolts with specified torque. See Section 3.4.7
High strength bolts, and welding for steel structure	Looseness Crack, damage	Replace if loose. Repair if crack or damage is observed.
Brake lining	Wear, fitness to brake disc	Replace if wear is excessive.
Brake disc	Wear of sliding surface	Repair if sliding surface is worn or rough.
Rail and rail clip	Bolt looseness, Rail abrasion, Rail distortion	Tighten loose bolts with specified torque, Replace excessive wear ones, Realign
Twist lock for headblock and spreader	Abrasion, crack	Replace excessive abrasion ones or ones with crack
Sheave, drum, friction-block	Abrasion	Replace if wear is excessive
Other consumable parts(such as seal, bulb）	Abrasion, damage	Replace if wear is excessive, Repair or replace damaged
Steel structure	Crack, paint, etc.	Refer to Steel structure maintenance manual
Safety device	Any abnormity	Repair while there is abnormity.

The crane and its additional equipment, primarily the safety devices, have to be checked for proper condition and function depending on work use. It requires special supervision of: − emergency-stop button and control; − drive & brake; − electrical equipment; − limit switch and overload protection.

3.6.2 Inspection of temperature rise

Temperature rise is a phenomenon. Maintenance personnel who well understand the temperature rise condition can judge the working condition of the crane. If there are no otherwise specifications, the temperature rise of the bearing should not exceed 60 ℃ (140℉) and the absolute temperature of the oil should be less than 85 ℃ (185℉). Infrared thermoscope is in common use. See Figure 3-16. The following temperature raise locations on the crane should be carefully inspected:



Figure 3-16 Infrared thermoscope

(1) High-speed rotation supporting locations, such as the motor bearings and the high-speed shaft bearing of reducers mainly used in main hoisting and trolley.

(2) Heavy-load and slow speed supporting locations, such as reducer low-speed shaft bearing, drum bearing pedestal, wheel bearing and sheaves bearing.

(3) Hydraulic actuator temperature rise, such as the hydraulic thruster of the brake (especially hydraulic thrusters of main hoist and trolley travel mechanisms), the relief valve and the throttle valve.

(4) Hydraulic oil reservoirs, such as the multifunction pump, emergency brake, gantry wheel turning and spreader hydraulic power unit.

Normally, the inspection of temperature rise condition can be achieved by hand touch and judging if the temperature is abnormal. When there is abnormal temperature rise, it is often accompanied with abnormal noise. If there is abnormal temperature, specialized thermometer should be used to measure and to aid in finding the cause of temperature rise. Only after the problem is solved, should the operation be resumed.

Generally, the causes of temperature rising are:

- ☞ Improper lubrication
- ☞ Improper gear meshing
- ☞ Bearing clearance is too loose or too tight (or bearing damage)
- ☞ The alignment coupling can't meet standard.
- ☞ Brake not fully released or not centered (one side released but the other side still touching the disc)
- ☞ Hydraulic thruster oil internal leakage
- ☞ Abnormal operation of the relief valve

3.6.3 Inspection of abnormal noise

The main cause of mechanical noise is vibration, impact noise, friction noise and structural transmitting. Maintenance personnel who understand the noise change can judge the operating condition of the crane through daily noise observation. Generally, the abnormal noise may be caused by improper gear meshing, bearing damage, loose bearing clearances, shaft bending, loose bolts, interference or motor fan damage, lack of lubrication, etc.

Noise can be identified/isolated by using an industrial stethoscope.

Noise measurement directly with ear may be harmful to ears, so only do this when necessary and for a short period.

Generally, the abnormal noise and temperature rise happen simultaneously, such as when the bearing clearance is too big or the brakes are not fully released, and so on. Therefore timely noise inspections can avoid many equipment failures.

3.6.4 Inspection of abnormal vibration

Vibration is usually related to noise and element fatigue. Close attention should be paid to the abnormal vibration during crane inspection, such as for the motor base, gear reducer base and drum support rigidity, base rigidity, bolt pre-tightening, misalignment of motor and gear reducer, dynamic balance of coupling and static balance of drum.

The inspection sequence is generally started with the check of alignment (radial and axial vibrations). If the vibration still exists after adjustment, the coupling should be removed and the motor checked separately.

After causes of the motor and loose bolts are eliminated and dynamic balance of coupling are checked, the coupling can be connected and the high-speed shaft bearing of the gear reducer and should be driven and checked. Generally, the problems caused by improper gear meshing and lack of adequate rigidity

of the installation base can be found during test running.

If the vibration happens after the crane is put into operation for a certain period, loss of alignment, loose bolts, excessive bearing clearance or improper gear meshing usually cause the vibration problem.

3.6.5 Inspection of crack

Crack inspection sometimes can prevent failure or disaster of a crane at its early phase. The inspection of crack consists of inspection of the driving parts, the connecting parts and the structural parts. The cracks are normally caused by fatigue and happen on locations of fatigue source, stress concentration, and repeated stress effect.

(1) Driving parts: such as driving shaft, wheel, gear, sprocket, sheave, coupling, braking wheel (disc), bearing and so on.

(2) Connecting parts: such as fixed shaft, wheel shaft, sheave shaft, structural connecting pin shaft, headblock connecting pin, spreader twistlock and so on.

(3) Structural parts: welding seam and base material.

The method of structural inspection can be visual inspection, which can be judged through the surface paint film. If the base material or welding seam is cracked, the paint film will be also cracked. Telescope can be used to inspect pre-determined regularly inspecting locations. If there is any suspicion after visual inspection, MT or UT inspection can be applied. 🕮Please refer to the Steel Structure Maintenance Program for details. Driving or connecting parts can also be checked during maintenance when they are knocked down.

3.6.6 Inspection of rail & rail clips

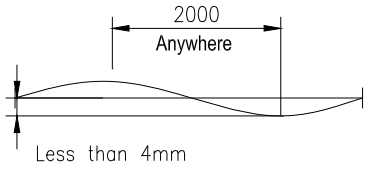
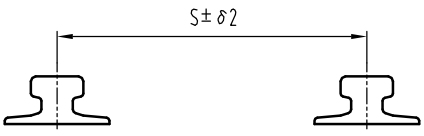
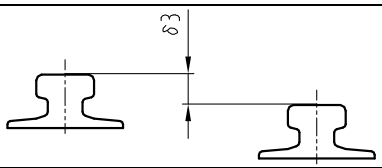
3.6.6.1. Rail installation tolerances

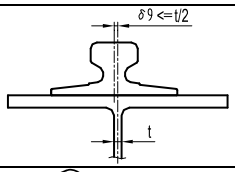
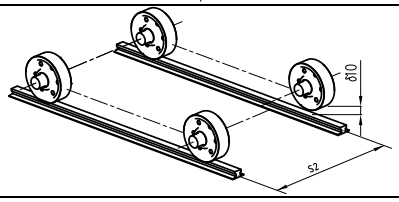
When designing a complete container handling system, the problems of the crane rail track never appear to be insignificant, and that is important to crane's using.

Failures of rails, to a great extent, can be attributed to improper lying of rail during foundation work or the crane or trolley may not be tracking accurately. If the crane is used with curved or uneven leveled runway rails, its structural members can be twisted causing high stresses. Accordingly, the bolts in joints may be loosened and the bases of the various machines on the steel structure may be distorted. Thus, crane performance may be impaired.

In order to prevent such troubles, at the time of annual inspection, measure the level, straightness, span, etc. of the trolley travel rail and gantry span and keep their records. Also, carefully inspect the trolley travel rail for cracks. If measured dimensions are not within the allowable tolerance in Table 3-15, immediately repair or replace the trolley travel rail.

Table 3-15: Permissible Tolerances for Trolley Travel Rails

Inspection items	Measuring Method		Maintenance tolerance
Lateral Straightness		Measured by a transit or piano wire	$\delta 4 = \pm 4$mm
Span (S)		Measured by a steel tape measure	$\delta 2 = 10$mm
Difference in height		Measure by a transit or a water level	$\delta 3 = \pm 10.3$ mm 0.15% of span
Slope or grade	Less than1/200		

Inspection items	Measuring Method		Maintenance tolerance
	in the area without camber when there is not trolley. in the area with camber when there is trolley with full load. *Refer to Girder structural drawing for camber information.		
Center distance between rail and web		Measured by a steel ruler	δ9≤=t/2=±6mm
Levelness		Measure by a water level	0.1% of span σ_{10}≤6.9mm

3.6.6.2. Common failures of rail support, rail clip and rail installations

A. Rail support system:

1) Rail clip devices:

a. Loosening of the bolts

b. Breakage of bolts, studs, or weld fillets

c. Elongation of holes

d. Rotation of clips

B. Rail:

1) Excessive wear or flattening of the rail head

2) Wear of edges of rail head

3) Rail breakage

4) Deformation at joints caused by wheels running over gaps

3.6.6.2.1Causes of dilapidation

Figure 3-17 shows the acting force of the trolley. The crane operation causes repeated wheel loading of the rail resulting in fatiguing cycles. Fatigue failure occurs due to:

① High stress in the rail support,

② Rail movements causing fatigue in the rail attachment components.

Figure 3-18 and Figure 3-19 shows the main abrasion form.

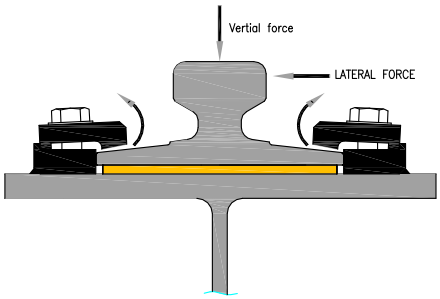


Figure 3-17：Rail under stress

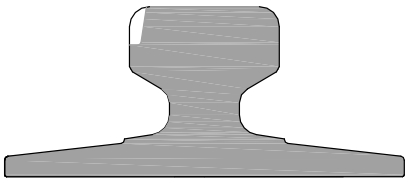


Figure 3-18：Side wear of rail

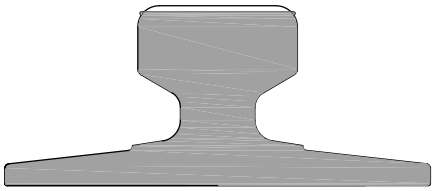


Figure 3-19：Surface wear of rail

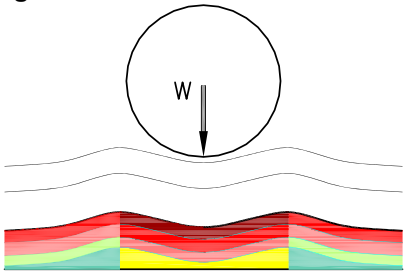


Figure 3-20：“bow wave” effect

High stresses caused by:
- Wheel flanges riding on the rail
- Compression stress due to concentrated wheel load
- Bending stress due to eccentric loading
- Lateral stress due to "crabbing", trolley motion and swinging live load.

Rail movements caused by:
Longitudinal due to:
- Bending of the girder under load
- Thermal expansion
- Crane gantry or trolley accelerating and braking

Lateral due to:
- Crane or trolley "crabbing"
- Swinging of live load
- Gantry or trolley accelerating and braking forces

Vertical due to:
- Rail "bow wave" effect under wheel load
- Height difference of rail at interface of two adjacent girders
- Twisted rails

Rotation due to
- Twisted rails
- Load eccentricity

- Steering effects of crane on rail head

3.6.6.3. Trolley rail clip installation and inspection

1) Inspect the bolts of rail clip to estimate whether they are loose. Screw down bolts if needed, and insure the exact position of stud. The process of installation about ZPMC trolley rail clip is listed hereinafter for reference.The installation of Gantrex clip can refer to Gantrex manual

2) Welding of base. Weld the base all around with low hydrogen electrodes and 4mm fillet welding. Spot welding first, and then weld the base. Especially pay attention to clamp the base when welding to avoid the moving in upright direction and keep the contact of base. Refer to **Figure 3-21**.

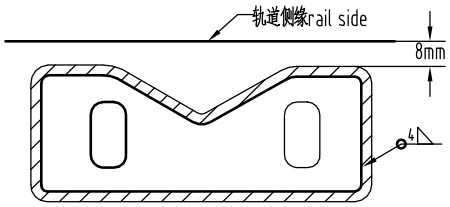


Figure 3-21：Welding of base

3) Installation of bolts and clip. Set the bolts into the holes of clip base, circumvolve it 90° and keep it lying in the wimble holes reliably. After confirming the bolts are oriented reliably (pay attention to the orientation line in the section of bolts), install clip and screw down nut manually. (It's important to screw down manually to ensure the correct location of the head of bolts in its head room.) Refer to Figure 3-22.

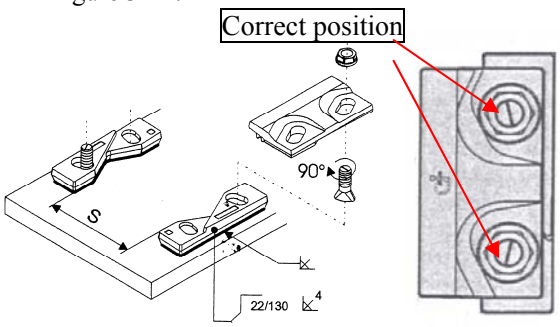


Figure 3-22：Installation of bolts and clip

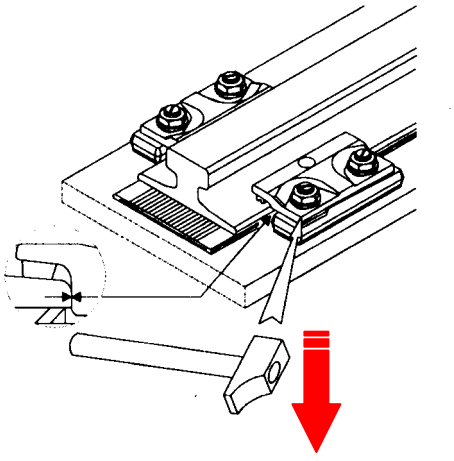
4) Cleaning up of base bolt hole. Clean up the rubbish (with air pump) and slag (with welding hammer) in the base bolt holes before set up the clips. Refer to Figure 3-34.



Figure 3-23 Cleaning up

5) Tighten the clip. Knock the clip slightly with hand hammer to keep the close contact of clip and rail side. Then, complete the tightening of the nuts to **150Nm**. In case of further painting protection, the use of solvent is prohibited. Refer to .Figure 3-24

If, for any reason, a nut must be loosened after mounting, you must verify the correct position of the bolt shaft before re-tightening the nut.



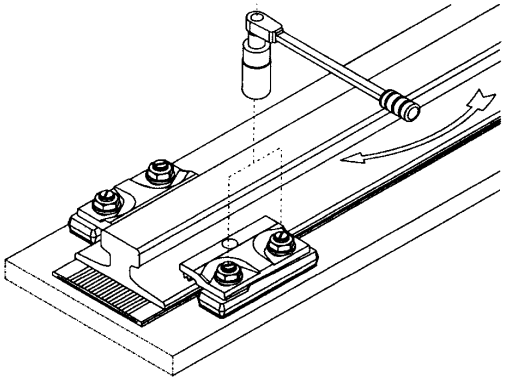


Figure 3-24： Tighten the clip

3.6.7 Inspection of gantry travel tires

The gantry travel is indispensable for operation by this machine. Therefore, it is important to maintain the travel wheels in normal state all times. The following passages describe the inspection procedure of the wheel.

The gantry tires is easy influenced by foreign hard things, so inspection of tires need be careful. Include tires and steel ring. **Ref to Figure3-35 and 3-36**

The following items should be checked during the daily, monthly, and annual periodical inspections

1) Air pressure. The normal air pressure is 10kg/m2. the air pressure should between 9.6kg/m2 10.4kg/m2。

2) Check the surface of tires,and need to clean foreign thing,if find damage,and the depth of damage is more than 10mm,should more attention to check air leakage.if find air leakage,change the tires.

3) Gantry Acceleration will strike the tire, especially turning tires without jack up device will wear the tires,if find air leakage,change the tires.

4) Check the cycle value. if find air leakage,change the value.

5) No break on the surface. If more than 3 places disfigurement with dimeter over 40mm, or 1 places disfigurement with dimeter over 100mm, the tyre shall be changed.

6) Surface of tire may weared with long time use.when the decorative pattern reduce two thirds,change the tire.

7) Chaping. The Surface of tire may chaped.when the depth of chaping is more than 1mm,change the tire.

8) Check the tire contact with the steel ring, if find air leakage, maybe the tire have quality problem or steel ring is damaged.check the seal ring,if find rust, deformation or damage,change the steel ring.



Figure 0-35 gantry tires



图 0-36 tire

3.6.8 Inspection of trolley travel wheel

The trolley travel are indispensable for operation by this machine. Therefore, it is important to maintain the travel wheels and rail in normal state all times. The following passages describe the inspection procedure of the wheel.

The following items should be checked during the daily, monthly, and annual periodical inspections.

(1) Wear in treads and flanges

(2) No flanges riding on rail

(3) During traveling, both sides moving smoothly in parallel

(4) Abnormal noise or vibration

(5) Wear in wheels, bearings, etc., developing evenly for both legs.

(6) Cracks or repair welding cracks in wheels is not allowable.

(7) Disfigurement with area over 3cm2 and more than three points on the same surface, the wheel shall be replaced.

(8) The rigidity of RTG is weaker than STS, make the wheel without flange in order to reduce side force to avoid rail gnawing, the wear of two side wheel will different because of offset track.when the diameter difference of two sides of the wheel over 3mm,change the wheels.

After inspection, if the wear of trolley travel wheel reaches the values indicated in Table 2-16, replace the wheel.

The items listed above should be checked periodically in peacetime, monthly and yearly.

Table 2-16： Wear Limitation of Trolley Wheel

	Original dimension	Wear Limitation D	difference of the 2 driving wheel diameter D_1'- D_1'（0.5%D）
	500mm	485mm	2.5mm
	630mm	610mm	3.15mm

3.6.9 Inspection of rope sheaves, drums, idler and slap block

3.6.9.1. Sheave

If the following symptoms are found, check the wire rope for wear and replace the sheave at the same time of replacing the wire rope.

(1) The flange is damaged or deformed

(2) The wear in the flange is 10% or more of the diameter of the wire rope used.

(3) A trace of the wire rope lay can be observed at the bottom of the sheave grooves

(4) The wear in the sheave groove is 15% or more of the diameter of the wire rope used.Refer to Figure 3-25.

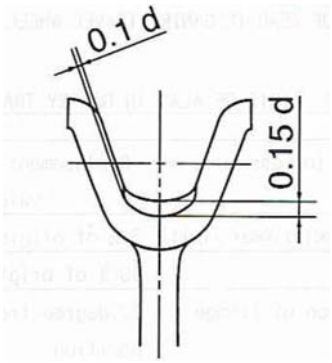


Figure 3-25 limits of wear in sheave groove

Refer to Figure 3-25 for limits of wear in sheave groove. The symbol "d" indicates the diameter of the wire rope used.

3.6.9.2. Drum

Periodically check the drum groove for corrugation and excessive wear.

The wear limit in the drum groove is 20% of the diameter of the wire rope used. Refer Figure 3-26.

There should be no crack in drum groove, wire rope fixing point and drum axis.

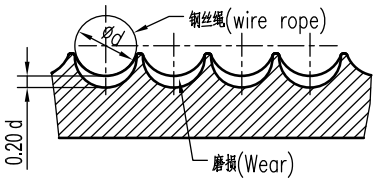


Figure 3-26 Limits of wear in drum groove

3.6.9.3. Inspection of rope support roller and Slap Block

The inspection of the rope support roller is described as follows:

(1) Verify that roller rotates freely.

(2) Determine the anti-friction material is peeling, aged or worn out. The wear resistant roller ring shall be replaced when the amount of wear reaches 50% of its original thickness. Make sure that the worn roller ring has been replaced before the wire rope penetrates the thickness of the roller to expose the base metal of the roller shaft. If this happens, the wire rope will rub against the metal to cause the abrasion failure of both the wire rope and the roller shaft.

3.6.10 Inspection of the spreader

As a direct performer of loading container, the load handling device's reliability is very important. Please refer to the spreader maintenance manual to get the information about the detail of the spreader's maintenance standard. Following routine inspection to the load handling device are suggested in handing over or taking over:

1) The reliability of twistlock connected with headblock and spreader.

2) Check the structure in the headlock and spreader to eliminate damage, incomplete fusion or weld crack in the parts easily to be stroked. Check lubricating condition on the slide surface of flex arm and slide rail.

3) Lubricating and movement condition of key parts such as twist lock pin and peak-pin, reliability of twist lock pin locknut.

4) The tightness of all the limit-board of spreader, the integrity of limite switch, the correctness of signal.

5) The reliability of cable used in spreader, and whether there is sudden damage.

6) The abrasion and damage condition of the head of twist lock of spreader.

7) The damage condition of guide-flipper of spreader.

8) Any leakage of hydraulic system and oil lever of oil tank.

3.6.10.1 Inspection of twist lock pin

The twist lock pins become weaker against crack due to wear and hardening caused by extended use. Therefore, perform the following inspection once a year.

Perform the MT test or tint check. Repair or replace the parts on condition

listed hereinafter.

1) Crack, crevasse or hair-crack on the face of the head of twist lock pin, shaft and the connect root of the lock head;

2) Damage or wear of the head of twist lock pin;

3) Deform or crack on the screw segment of the tail of the pin.

Always Inspect A and B area with special care!

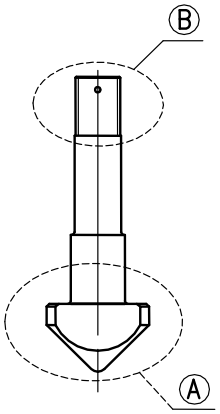


Figure 3-27 Annual check of Twist lock pin

It's not permitted to repair the disfigurement in the hook and twist lock pin by welding. Replace the failure part if there is un-repairable disfigurement.

3.6.11 Inspection of hydraulic thruster brake

Brake should be inspected periodically with items listed hereinafter:

1) Whether the brake lining and brake disk is clean and not covered with oil dirt.

2) Whether the arm bars of the brake are flexible.
3) Inspect whether the belleville spring is broken or hydraulic oil is leaked to the belleville spring through inspection window of the brake cylinder. Repair or replace the failure part if the break or leak happened.
4) Inspect whether the push rod of the limit switch is loose and the action is right.

Attention! Hydraulic thruster brake cylinder has especial structure inside and connecting proprietary technology. So don't disassembly it yourself and avoid belleville spring collapsing and hurting the people. Replace the fault cylinder as one piece.

5) Inspect the abrasion condition of the brake lining. If the attrited thickness of the brake lining is over 40% of the original thickness of the friction material, a new brake lining should be set.

3.6.11 Inspection of brake disc

DANGER! Any grease on the brake disc will cause the substantive drop of the brake torque.

Inspection of the brake disc is very important to the safety of the crane operation.

Periodically check if the brake disc is contaminated. Any oil or grease on the disc must be cleaned with detergent or solvent immediately.

Also check the brake disc surface roughness. If the disc is damaged, scoring or cracked, if anything is found as shown in Table 3-17 , the brake disc should be replaced.

Table 3-17: Standards for Brake Disk Changing

	Main hoist	Trolley drive	Main hoist emergency brake
Brake disc thickness	30mm	30mm	40mm
Wear limitation	27mm	27mm	36mm
Convex or concave on the disc surface	2mm	2mm	3mm
Brake lining wear limit	Attrited thickness should not more than 40% of base thickness. Interface between brake lining and brake disk should not less than 75%. The brake lining with adust face or aging should be replaced.		

3.6.12 Inspection of shaft and axle sleeve

Some parts of the crane, such as equalizer pin of gantry is equipped with shaft + axle sleeve (copper sleeve commonly, sometimes copper sleeve with plumbago lubrication). Check if the pin and bushing are well lubricated and rotates relatively freely. The relative rotation always happens between the bushing and the pin. The bushing should stand still in the housing, so no wear should occur on the outer diameter of the bushing. The abrasion form of the shaft and axle sleeve and the measure should be taken are listed hereinafter.

1. Sort according to the abrasion form which is shown in Figure 3-28.

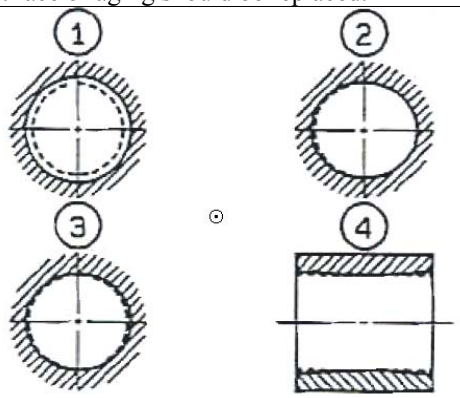


Figure 3-28 Abrasion form of shaft hole

1) symmetrical wear
2) wear induced by unilateralism load
3) wear induced by bidirectional load
4) wear only on the both sides

2. Measure while the hole is worn.

1) replace the worn axle sleeve

2) deal with the worn shaft holes conform to the steps listed hereinafter
a. bore the hole after repair welding
b. enlarge the diameter of pin, axle sleeve, and shaft hole
c. replace axle sleeve

If the wear of the bushing reaches the amount shown in Table 3-18, replace it.

Table 3-18： The Wear Limits of Bushings

D (mm)	Wear limita-tion of dia (mm)
10~24	0.8
25~40	1.2
41~60	1.6
61~100	2.0
101~160	2.5
161~250	3.0

3.6.13 Inspection of oil seal

Please check the oil seal if leakage is founded according as the description listed hereinafter.
1) whether there has been appropriate seal set in the leakage point
2) Whether the brim seal is damaged by keyway, spring, screw, rough surface or the brim of shaft
3) Whether the seal is set in reverse direction.
4) Whether the deformation of seal is caused by setting with unsuitable tool.
5) Whether the venthole is jammed.

3.6.14 Bearing unit inspection

In the crane's working, judge the state of the bearing housing during operation, then take the following measures if abnormality is observed.

(1) If vibration is found:

If the vibration is severe, first check the retaining bolts for looseness and verify that there is no looseness between the bearing unit and the installation base. Tighten the bolts as necessary. If the vibration remains after the above mentioned procedure, check if the causes of other rolling element vibration such as bent shaft exist. If no abnormal part is found, replace the bearing unit since the excessive clearance due to wear inside the bearing is suspected.

(2) If abnormal temperature rise is found.

If the temperature of the housing rises over 20°C (36°F) after subtracting the ambient temperature, carry out the inspections below.

1) Check if any causes of the temperature rise exist, such as no relief clearance due to the one-side free bearing, overloading, etc.

2) If no abnormal condition is found, replace the bearing since the severe condition such as an initial state of bearing damage etc. is suspected. Especially the temperature rise with abnormal noise is considered to be a sign of developing damage. Replace the bearing immediately in such a case.

(3) If abnormal noise is found.

If abnormal noise is heard from the bearing unit, carry out the inspections below.

1) Measure the bearing block temperature. If the temperature is less than 20°C (36°F) after subtracting the ambient temperature, check the retaining bolts for looseness and verify that there is no looseness between the bearing unit and the installation base. Tighten the bolts if necessary.

2) If the abnormal noise remains after procedure 1, the following causes are suspected. Immediately replace the bearing judging from the abnormal noise degree and the machine importance.

- ◆ Slight wear appears on the track surface and/or rolling element of the bearing.
- ◆ Inclination toward noise.
- ◆ Foreign materials inside the bearing

3) Measure the bearing block temperature. If the temperature is over 20℃ (36℉) after subtracting the ambient temperature, replace the bearing unit since the developing peeling of the track surface or rolling element is suspected.

3.6.15 Inspection procedure of key and keyway

The key and keyway suffer various kinds of damage according to the used condition. If damage is found on inspection, immediately replace the key. The damage types and their cause are as follows:

(1) Shearing of key

The key may be sheared if the abnormal torque that is larger than key strength is applied in such a case. The keyway often may be deformed as well, so re-cut the keyway and install the key with hardness of superior

grade.

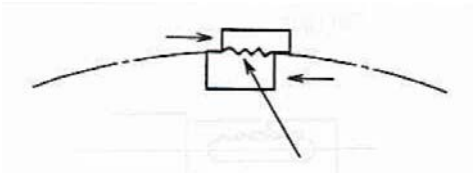


Figure 3-44 Shearing of Key

(2) Crack of keyway

The stress concentrates on the edges of the shaft or the keyway in the fillet engaged with the shaft. Therefore, cracks may be caused if the part suffers abnormal torque larger than their strength.

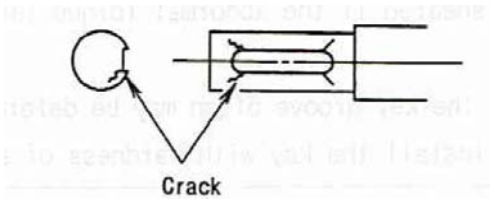


Figure 3-29 Crack of Key Groove

If the condition remains, the crack may be enlarged, resulting in keyway damage of shaft failure.

(3) Wear of keyway side

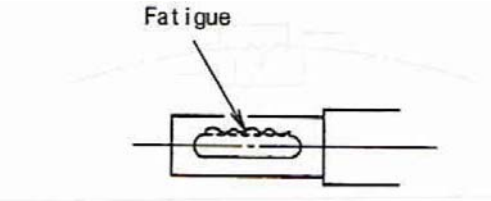


Figure 2-30 Wear of Key Groove Side

The fatigue and deformation may be caused if abnormal torque is applied to the keyway sides and abnormally high surface pressure is struck on the keyway sides repeatedly.

3.6.16 Inspection of consumable parts

Consumable-parts are those parts which may be depleted or worn out by use of the crane and **they do not belong to guaranty products**. During crane maintenance and inspection, pay special attention to the consumable parts listed as follows:

1) Inspection of wire rope for abrasion, wire breaks and other damage. ☞ Refer to the relative provisions in Chapter 5.

2) Check if there is any oil leakage from gear seal rings.

3) Inspection of braking shoe abrasion.

4) Inspection of carbon brush of conducting ring (such as DC motor and spreader cable reel slip rings).

5) Inspection of kryptol for trolley rail and wheel lubrication.

6) Inspection of all lamp bulbs.

7) Make sure the spare fuses are available.

8) Inspection of filters.

9) Inspection of vulnerable parts, such as the spreader flippers.

3.6.17 Inspection of steel structural members

	**The structural integrity of the crane requires that no attachments or fixtures of any kind be welded to any part of the crane without prior written approval of the responsible Structural Engineer.**
	**Please refer to Steel Structure Maintenance Program for more inspection information.**

In order to maintain the crane in good condition, continuing inspection is very important. When damages occur to steel structural members, this crane may be unusable and require detailed inspection. Not only machine damage but also accidents including human injury may arise.

Deformation of the steel structure members may be caused by improper usage of the crane, development of cracks, and loose bolts. Therefore, always follow the operating procedures described in the container crane operation instruction manual and conduct the regular inspections as described in this manual.

Inspection and diagnosis methods for metal structure：

1) Inspect and judge the metal structure by human sense and experience:

- Check to find out whether the hinge points and the interconnecting pieces run smoothly and flexibly, whether have abnormal sound and blocking phenomenon, whether have a good lubrication and a severe abrasion, whether the center axle have movement and so on.

- Check the bolts connection to eliminate loosening.
- Check the main connection pieces and welding to find out painting crimple, spalling or welding cracking.
- Check the components to find out global or local deformation and bending, folding twisting and damage phenomenon.

2) Period inspection through non-destructive detection should be carried out for the welding of FCM pieces (for detailed information refers to STRUCTURE AND MAINTENANCE MANUAL FOR RAIL MOUNTED GANTRY CRANES).

3) Use electrical logging stress method to test the strength and rigidity of the metal structure. Through careful and quantitative inspection to make a scientific judgment on the working condition of the metal structure.

The following things should be paid much attention to:

1) Check deformation, corrosion, crack, loose bolts, etc. at the time of regular inspection such as daily, monthly, and annual inspection.

2) The deformation of steel structure can degrade the alignment of the machinery installed on the structures, increasing operating noise, causing improper engagement of the gears, etc.

3) If some deformation of steel structures, development of cracks, loose bolts, etc. is observed, immediately report to the service office, and repair it under the supervision of the structural engineers.

4) In order to prevent corrosion of the steel structure, clean clogged drain holes. Repaint areas if there is any indication of rusting. Corrosion will reduce the thickness of the structural members, increase members stress, and may create cracks. In order to inspect cracks in the structural members, thoroughly remove the rust and paints from the area, then conduct the indicated testing.

5) The lubrication of the stay bar pins is essential to keeping the stay bars sound. If the rusting of the pin occurs on the stay bar friction surface, the additional bending moment will be induced on the bar and may cause the initiation and propagation of cracks. Therefore, the lubrication must be made periodically to remove old grease and to refill the new grease.

3.6.18 Inspection of all safety protections

Safety protections should be checked before any work.

1) Regularly inspect if the braking torque setting-up of all brakes. Make a practical plan to periodically verify the braking torque. All slow-down limit switches and over-travel limit switches. Ensure that the positions of all encoders, functions of all interlock protections are correct.

2) Regularly inspect the hoist over-travel limit switches and hook of service crane.

3) Visually inspect if all buffers mounting condition and wheel stoppers are reliable.

4) Replace any missing or illegible warning sign.

5) Refer to Chapter No.1 to find out the working condition of crane safety equipments like tie down device and so on.

6) Check the function of the operator's cab, and check the headblock to find out crack and loosing parts. Check the handrails to find out the loosed or deformation bolts.



Never tamper with any safety protection devices. It's about your own safety!

3.7 Routine Inspection Items

Confirming performance and ambient safety before operation.

Observe the Crane's condition and stop operation after any abnormal condition is found.

After operation has been completed, the Crane shall be stopped at the correct position and it should be confirmed that there is no abnormal condition.

The following maintenance and inspection records are attached to these Instructions. The results are necessary for the Crane repairing and modification and shall be carefully recorded on table 3-20

Table 0-20 Before starting operation

No.	Inspection Items	Inspection method
1	Obstruction within traveling range and the load traveling path	Visual
2	Oil leaking on spreader	Visual
3	Obstruction on trolley rail track	Visual
4	Wire rope jumping off the drum or sheave or contact of the rope with any obstruction	Visual
5	Lack of lubrication in wire rope, strand breakage, loosening of clamp bolts	Visual
6	Inspection of brake pads	Visual
7	Remedied condition of the defects found on previous day	Activation
8	Inspection of brake	Starting
9	Various instrument panels and indicator lamps	Starting
10	Activation of various operation joysticks and button switches	Starting
11	Activation of major limit switches (main hoist, trolley Spreader)	Starting

Table 0-21 During operation

No.	Inspection Items	Inspection method
1	Abnormal sound, heating, vibration	Visual
2	Abnormal current, voltage	Visual
3	Activation of the spreader	Visual

Table 0-22 After operation

No.	Inspection Items	Inspection method
1.	Inspection of spreader	Visual
1.1	Deformation and fracture of base frame	Visual
1.2	Hydraulic oil level	Visual
1.3	Deformation and fracture of twistlock pins and flippers	Visual
1.4	Damage and torsion on spreader cable	Visual
1.5	Loosening of securing bolts	Visual
2.	If the various joysticks return to off position	Visual
3.	Any oil leakage at various oil pumps and hoses	Visual
4.	If various bearings, gears and other lubricated parts are normal	Activation
5.	Any abnormal heating	Visual
6.	Any fracture or damage on various components	Visual
7.	If the power switch is disconnected	Starting

8.	If power pack is switched off	Visual
9.	If various components are clean	Visual
10.	If the operator's cab door is locked	Visual
11.	Operation log and record of container handling for shift transferring	

Table 0-23 The objective of this inspection is to find out defects between monthly inspections.

Inspection time	Inspection items	Inspection method
After operation	1. any loosening of bolts and damage of bearings	Visual
	2. any loosening of various pins, key boards and end stopper boards	Visual
	3. any torsion, damage, cracks of festoon cable and any loosening , corrosion of terminals	Visual
	4. activation of electric-magnetic contacts, knife switch, etc.	Activation

Table 0-24 Items Inspected every month, every annual

The objective of regular inspection is to inspect the major parts of the Crane in detail and to propose disassembly and repair when necessary and to find out trouble that are not found during routine and weekly inspection and propose repairing. Regular inspection is divided into monthly and annual inspection.

Category		Monthly Inspection	Annual Inspection
		Inspection Items	Inspection Items
Steel Structure	Steel Structure	1. Loosening of securing bolts	1. Corrosion of steel
		2. Fracture of welds	2. Peeling off of coating
		3. Deformation, fracture of steel	3. Wear of shafts, pins
		4. Grease for shafts, pins	
		5. Loosening of bolts on stairs, handrails, platforms	
Steel Structure	Trolley rail	1. Rail end wear and sinking	1. Rail span
			2. Elevation difference between the two rails
			3. Bending in vertical direction
			4. Bending in horizontal direction
			5. Slope
			6.Misalignment of rail joints
			7.Damageand corrosion of end bumpers
	Bearings	1. Abnormal noise, abnormal heating	1. Any cutting scrap in the oil (grease)
		2. Damage or fracture of the body	2. Clearance between the pin and bush
		3. Loosening of securing bolts	3. Wear of the bearing
	Shaft		1. Wear, fracture
			2. Loosening of the key

			3. Deformation of the key way
	Key		1. Deformation
	Gear coupling	1. Abnormal noise, heating	1. Alignment
		2. Oil amount (grease)	2. Supply of oil (grease)
		3. Oil leakage	3. Teeth backlash
		4. Loosening of securing bolts	4. Loosening of the key
Mechani-cal part	Gear coupling		5. Deformation of the key way
			6. Damage of fracture of teeth
	Wheel	1. Wear, deformation of the flange and tread surface	
		2. Deformation of the key, bolts loosening	
	Drum	1. Loosening of wire rope fixing	1.Wear on the rope groove
		2. Damage to the wire rope guide	
		3. Damage to drum securing bolts	
		4. Fracture of welds	
	Sheave	1. Revolving condition	
		2. Fracture	
		3. Rope groove wear	
		4. Damage to wire rope anti-slipping device	
	Wire rope	1. Broken wires in strand	
		2. Wear and external damage	
		3. Bending, deformation	
		4. Strand loosening	
		5. Greased condition	
		6. Rust and corrosion	
		7. Fixing condition of wire rope end	
Mechani-cal part	Spreader	1. Deformation an fracture of the base frame	1. Wear of the pins, shafts, etc.
		2. Damage, deformation and wear of twistlock pins, flippers	2. Peeling-off of coating
		3. Loosening of securing bolts	3. Corrosion of steel
		4. Oil supply	
		5. Center to center of locks	
		6. Activation	
	Grease Supply	1. Contamination of grease and aging, discolor of grease	
		2. Grease supply at lubrication points	
		3. Damage and oil (grease) leakage in piping	
Mechani-cal part	Hydrau-lic Part	1. Oil level and oil contamination	
		2. Filter cleanliness	
		3. Oil pressure	
		4. Abnormal noise, abnormal heating of oil pump, oil motor	
		5. Adjustment and action of valves	
		6. Action of oil cylinders	
		7. Damage to oil piping	
		8. Oil leakage anywhere	
		9. Loosening of securing bolts	

3.8 Causes of Common Troubles and Troubleshooting

3.8.1 Regular Inspection Item and method

Causes of general troubles in the Crane parts and components and their remedies are shown in the following table 3-28:

Table 0-28 Regular parts and componets troubles and and troubleshooting

Inspected components		Inspection Items		Inspection Method and apparatus	Inspection criteria	Inspection period
Item	description	Item	description			
1	Reducer	1	Lubrication oil amount	Visual, with oil scale	Indicated range on the scale	per month
		2	Sealing	Visual	Fixed joint: no leakage. Relatively moving joint: no drip	per month
		3	Running condition	Testing, hear and feel	no abnormal sound, abnormal heating	per month
		4	Bolt connection	Knock and test	no loosening	per month
		5	Oil quality	Chemical analysis	To specified by standard	3 months
		6	Gear meshing	Visual test	Tooth face smooth and clean normal meshing	3 months
		7	Wear of gear wheel	Measure	Wear is less than 10% tooth original thickness	Every year
2	Drum	1	Wear of the drum wall	Measure	Wear is less than 15% original thickness	Every half a year
		2	Wire rope clamp	Knocking	no damages	per month
		3	Rope grooves	Visual	no damages	per month
		4	Fatigue cracks	Visual	no damages	per month
		5	Bearing lubrication	Visual	Proper lubrication	per month
3	Wire rope	1	lubrication	Visual	Grease is not dried, distributed uniformly	per month
		2	wear	measure	Diameter reduced by <7%	per month
		3	Wire broken, strand broken	Visual	Broken wire <10% no strand is broken	per month
		4	Torsion or corrosion	Visual	no torsion, no corrosion	per month
4	Sheave	1	Cracks	Visual	no cracks	per month
		2	Sheave rotation	Hearing test	Maneuverable no abnormal noise	per month
		3	Uneven wear of the groove	Measure	Less than 3 mm	3 months
		4	Wear of groove thickness	Measure	20% original groove thickness	3 months

		5	Wear at groove bottom diameter	Measure	<50% wire rope diameter	3 months
		6	Damage	Visual	no damage	per month
		7	bearing	Hearing, Visual	no abnormal noise proper lubrication	per month
5	brake	1	Brake base frame	Visual	no cracks	per month
		2	Wear of thickness of brake pads	Measure	<50% original thickness	per month
		3	Spring	Visual Measure	no yield deformation	3 months
		4	Wear of shaft or bore	Measure	<5% original diameter	per month
		5	Hydraulic thruster	Visual	no oil drip, normal activation	per month
		6	Clearance when released	Visual Measure	> 0.6 mm < 1.5 mm	per month
5	brake	7	Bolts, nuts	Knocking test hammer test	no loosening	per month
		8	Operation condition	Testing and observe	Normal operation, reliable, no abnormal heating, smell, and action	per month
6	Brake disc	1	Brake disc body	Visual	no cracks	per month
		2	Wear in disc thickness	Measure	< 4% original thickness	3 months
		3	Unevenness of disc surface	measure	< 1.5 mm	3 months
		4	Friction surface	Visual	no cracks	per month
7	coupling	1	Coupling body	Visual	no cracks	per month
		2	pins	Visual	no damage, normal wear	per month
8	Wire rope	1	Wire rope end, screw lever	Visual	no abnormal condition, no cracks	per month
		2	Bolt connection	Visual, Knocking test	Normal, no loosening	per month
		3	Wire rope Support rollers	Visual	Normal operation, no deformation at deport, no cracks	per month
9	lubrication points	1	Grease amount	Visual	To meet requirement	per month
		2	Quality of grease (oil)	Visual	To meet regulation	per month
		3	Nipples and tubing	Visual	No damage, unblocked	3 months
10	Trolley wheel	1	Wheel body	Visual	no cracks	per month
		2	Wear of wheel flange thickness	Measure	< 50% original thickness	3 months
		3	Bending deformation of wheel flange in thickness	Measure	< 20% original thickness	3 months

		4	Wear of tread in thickness	Measure	< 15% original thickness	3 months
		5	Ellipticity	Measure	< 1 mm	3 months
11	Rubber tyres	1	Inflation pressure	Measure	10kg/m^2	per week
		2	Surface of the tyres	Visual	no racks	per week
		3	Wheel bearing	Visual, testing	Proper lubrication, no abnormal	per week
		4	Connection bolts	Visual, knocking test	Normal condition, no loosening	per week
12	Diesel engine	1	Air filter	Visual	clean	per week
		2	Lubrication	Check	Foreign matter and water content meets standard	per month
		3	Water tank cover	Visual	no missing, no damage	per week
		4	Diesel fuel tank cover	Visual	no missing, no damage	per week
		5	Fan belt	Test	Tension meets requirement	per month
		6	Sealing	Visual	no leakage	per week
		7	Abnormal noise	Hearing	no abnormal noise	per week
13	Alternator set	1	Resistance of insulation	Measure	>1 MΩ in cold state >0.5MΩ in hot state	3 months
		2	Wire connection	Visual, Knocking test	No damage, no loosening	per week
		3	Filtered cover of the ventilator	Visual	No dust & foreign matter	per week
		4	transmission bolt	Testing	No defects, tension meets specified	per month
		5	Running condition	Test run	No abnormal noise no abnormal heat	per week
14	Trolley horizontal wheel	1	Horizontal wheel body	Visual	No defects such as cracks	per month
		2	Clearance between the guiding aril	Measure	3~4 mm	per month
		3	Bolt connection	Visual Knocking test	No abnormal condition, no loosening	per month
		4	Lubrication	lubricate	Proper lubrication	per month
		5	Wear	Measure	Meet technical requirement	3 months
15	Universal transmission	1	Connection	Visual. knocking	No loosening correct	per week
		2	Lubrication	Lubrication	Proper lubrication	per week
		3	Parts of shaft	Visual	No cracks	per week
16	bumper	1	Bumper body	Visual	No damage & cracks	per month

		2	Support	Visual, Knocking	No cracks, firm connection	per month
		3	Spring	Visual	No abnormal condition	per month
17	Headblock twistlock	1	Cracks, damage	Visual, NDT	No cracks, no damage	per year
		2	Wear	Inspection	As technically required	3 months
18	Spreader telescopic	1	Load bearing	Visual Measure	No damage no wear	per month
		2	Rail	Visual Measure	No defects, wear within permitted range	per month
19	Cable carrier	1	Cable clamp	Visual Measure	Perfect no loosening	per month
		2	Roller	Visual	Proper lubrication, no damage	per month
		3	Support bracket	Visual	Perfect	per month
		4	Bolt connection	Visual, knocking	No loosening	per month
		5	cable	Visual	No damage	per month
20	Spreader twistlock	1	Plastic deformation	Visual	No plastic deformation	per month
		2	damage	Visual	No damage	per month
		3	Turning angle	Testing	0°, 90°	per month
		4	wear	Measure	As technically required	per year
		5	cracks	Visual	No cracks	per year
21	Steel structure	Box-section members	welds	Visual Telescope rulers	No cracks, ultrasonic and X-ray test if necessary depth of corrosion <10% weld height	3 months
			Plate surface	Visual, rulers	No cracks, local deformation <3% depth of corrosion <10%	
		Walk-way stair	Oil (grease) contami-nate damage	visual	Normal-walking No damage	per month
		railings	damage	Visual	No damage	per month

3.8.2 Causes of Common Troubles and Troubleshooting

Causes of general troubles in the Crane parts and components and their remedies are shown in the following table 3-29:

Table 0-29 Causes of general troubles and their remedies

Troubles	Reasons	Remedy
High temperature	Lack of grease or no grease	Add sufficient grease
	Contamination in the bearing(s)	Clean the bearing with kerosene and fill with grease
Noise in rolling bearing is increased	Poor installation misalignment of the bearing or over tightened and jammed	Check correctness of installation and adjust
	Damaged or worn bearing elements	Replace bearing
Periodic chattering noise of the gearing in the reducer, particularly from the idle gear wheel	Excessive error of the teeth pitch. Teeth lateral backlash exceeds standard value	Remedy, re-assemble
Sharp friction sound and clatter sound in the reducer	Clearance of gearing is too small , misalignment of the gear wheel, sharp and thin edge on the teeth top, teeth face uneven due to wear	Remedy, re-assemble or replace
Uneven meshing, continuous knocking Sound heard from any point on the reducer casing and vibration of the reducer	Defects on tooth lateral surface (laminated texture)	Replace

Troubles	Reasons	Remedy
Knocking sound in the worm gear reducer, alternate high and low and the period coincides with that of gearing rotation sound	Excessive axial backlash of the worn or the teeth of worm gear are seriously worn, pitch circle of the fear is eccentric in relation to the shaft, accumulated error of the circular pitch of the gear	Replace Repair or re-assemble
Heating of the reducer	Wrong oil level. More than 50° C temperature difference between housing and enviroment	Oil level shall be kept between the two pointer scales
High noise of shock in the gear reducer	Excessive wear of gear wheel	Replace gear wheel
	Foreign matter in the casing	Remove the foreign matter
	bearing is damaged	Replace bearing
	Foreign matter in the bearing	Clean the bearing
Bearing overheated in the reducer	Contamination in bearing	Disassemble and clean the bearing
	Too small clearance	Adjust
	Lack of grease (oil)	Add grease (oil)
Brakes can not hold the load (excessive braking distance of the trolley or gantry)	Oil (grease) on the brake disc or brake pads	Clean up the brake element jam and grease the pivots
		Clean brake disc and brake pads with kerosene
	Jam in the joints of lever system	Clean the brake, remove jam and add grease to the joints
	Excessive wear of the brake pads	Replace brake pads
	Main spring damaged or loosened results in insufficient tension	Replace spring or adjust the nut to make the spring tension sufficient
	Loosening of the lock nut of the lever, lever's play	Adjust the lever and tighten nut
Burnt spots on brake disc	Uneven clearance between brake disc	Adjust clearance to make it even,

and rapid wear of brake disc	and brake pads, friction appears when released and the disc is heated, auxiliary spring is damaged or bent	replace auxiliary spring
Brake is easily deviated from the adjusted position	Adjustment nut is not tightened or lock nut is not tightened, thread of the nut is damaged	Adjust brake tighten nut and lock nut, or replace defective nut

Troubles	Reasons	Remedy
Rapid wear or frequent breakage of wire rope sheave does not rotate	Contamination and lack of grease	Clean and lubricate the wire rope
	Lack of lubrication between shaft and bush	Disassemble and lubricate parts and add grease
Too long starting time of brake(s)	Too high resistant force at various pivot points of brake(s)	Clean up every pivot point and grease
	Too low oil pressure or oil piping blocked	Clean piping and fill oil with oil as necessary
	Internal oil leakage	Clean and check hydraulic thruster for noise and replace damaged parts
No pressure in hydraulic thruster, no braking	Orifice of pressure relief value is blocked	Clean pressure relief value
	The solenoid is reversibly connected and the thruster can not be pushed downward	Adjust wiring
	Pump fitting is not tightened	Tighten the fitting of the pump
	Filter is blocked	Clean the filter
Pressure in hydraulic thruster can not reach operating pressure value	The plane of the control lever of the adjustable device of pressure relief is not leveled; control lever has deviated from the center of nozzle	Find out the failure point and correct it
Too long braking time	Insufficient current in the solenoid of electro-hydraulic adjustment device	Increase current within permitted range
Release time by the hydraulic thruster is too long, residual pressure too high	Small distance between the control lever and the nozzle	Slightly tighten the upper end nut of the cross-spring
	Too small diameter of the nozzle bore	Enlarge the bore diameter
The brake can not be released	Too high tension of the main spring	Adjust the tension of the spring
	Thruster is damaged	Repair or replace hydraulic thruster

CHAPTER 4 REPLACEMENT OF WEARING PARTS

4.1 GENERAL
4.1.1 Attention
4.1.2 Tool list for replacement
4.2 REPLACEMENT OF BEARING BLOCK
4.2.1 Preparation
4.2.2 Replacement procedure
4.3 REPLACEMENT OF MOTOR COUPLING
4.3.1 Preparations
4.3.2 Replacement procedure
4.4 REPLACEMENT OF DRUM COUPLING
4.4.1 Preparations
4.4.2 Replacement procedure
4.5 REPLACEMENT OF MOTOR BRAKE
4.5.1 Preparations
4.5.2 Replacement procedure
4.5.2.1 Repalcement of whole brake
4.5.2.2 Replacement of thruster
4.5.2.3 Replacement of brake spring tube
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4.6 REPLACEMENT OF GANTRY WHEEL
4.6.1 Preparations
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4.7.2 Replacement procedure
4.7.3 Adjusting of trolley wheel
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4.9 REPLACEMENT AND ADJUSTING OF HORIZONTAL TROLLEY WHEEL
4.9.1 Preparations
4.9.2 Disassembly of horizontal trolley wheel
4.9.3 Adjusting of horizontal trolley wheel

Chapter 4 Replacement of wearing parts

Summary:

The RTG is worn, fatigued, distorted, eroded and degenerated inevitably while being used. Proper operation and timely maintenance can ensure each parts of crane to achieve expected lift. So when the parts of crane achieve normal life or wear limitation, how to repair the crane reasonably is the very important aspect of modern RTGs management.

This chapter will introduce the replacing methods of RTG's main components elaborately by illustrating pictures after expected lives of which are achieved.

4.1 General

4.1.1 Notice

This chapter includes main components of RTG such as: bearing block、motor coupling、drum coupling、motor brake、drum brake、trolley wheel、gantry wheel、main hoist sheave、wire rope and so on.

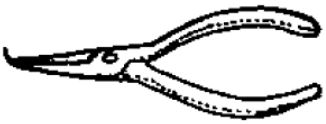
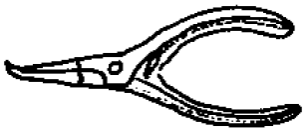
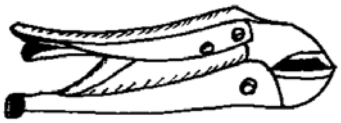
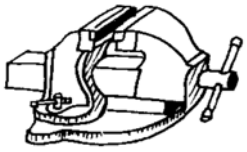
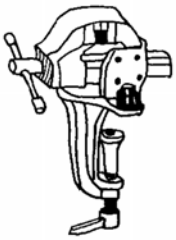
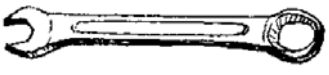
The personnel who carry out the replacement work must be acquainted with the structure of components & the operation step and have the ability of coping with the abrupt problem. Before working, it is essential that set down a processing procedure in detail according to this chapter, have a pre-work meeting, prepare safety precautions，point someone to take responsible for safe coordination, and strictly obey the safety rule request in chapter 1.

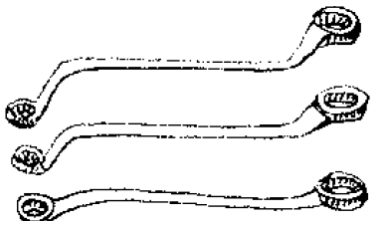
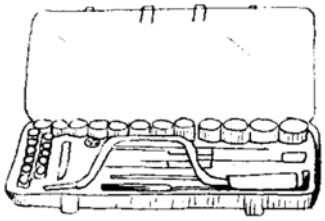
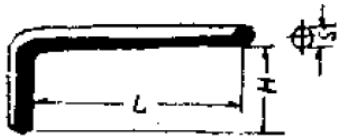
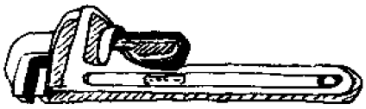
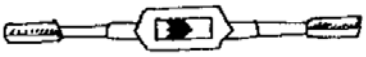
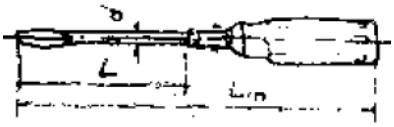
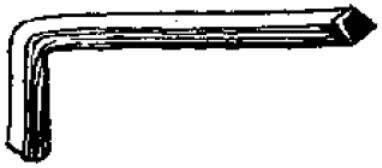
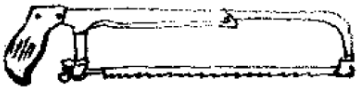
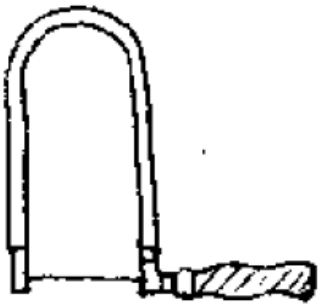
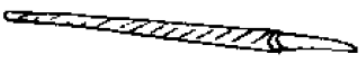
	**Danger! Be care of stumbling!**
	**Keep away from hanging weight**
	**Watch your hand!**

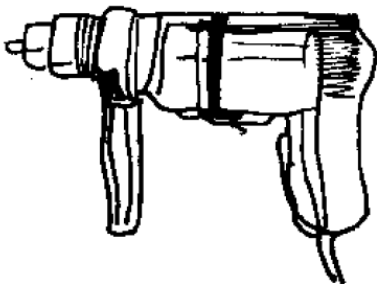
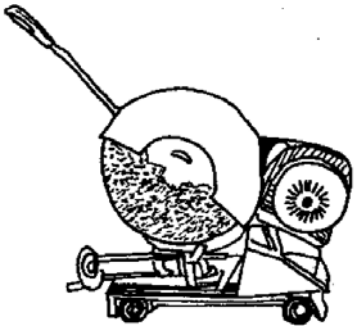
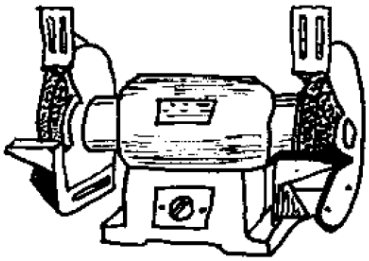
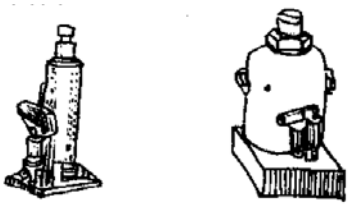
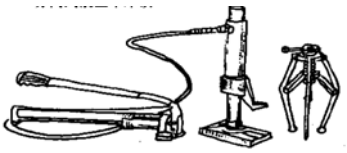
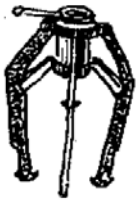
4.1.2 Tool list for replacement

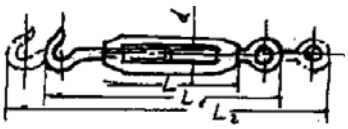
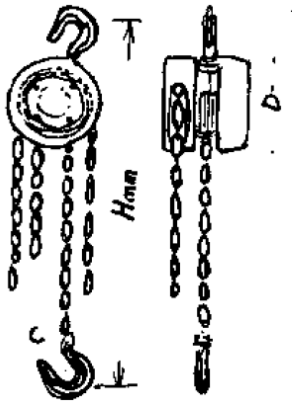
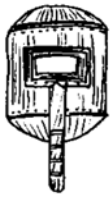
Table 4-1 List some tools in common for replacement, the pictures are only for reference.

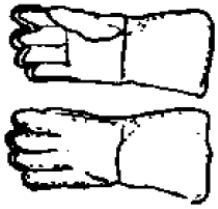
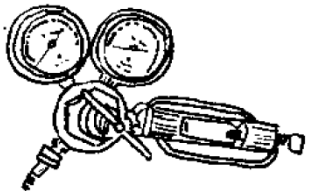
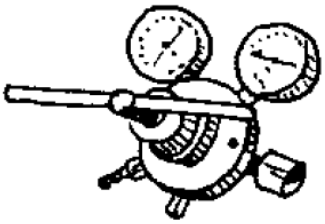
Table 4-1 Tool list for replacement

wire cutters		slip joint pliers	
needle point pliers		internal circlip pliers (bending tip)	
outer circlip pliers (bending tip)		vise grip pliers	
bench vice		table vice	
double spanner		combination spanner	

double offset ring spanner		manual socket spanner	
L-key		torque wrench	
ratchet wrench		pipe wrench	
round die stock		tap wrench	
slotted head screw driver		cruciform screw driver	
L-key with handle		hack saw	
little hack saw		scroll saw	
flat file		round file	

engineer's hammer		flat punch	
hand electric drill		section cutting machine	
straight electric grinder		angle electric grinder	
table grinder		hydraulic jack	
screw jack		separated hydraulic jacking machine	
separated hydraulic jacking machine accessory (puller)		extension tube	

open hook lifting sheave block		open chain lifting sheave block	
closed hook lifting sheave block		closed chain lifting sheave block	
thimble for wire rope		wire rope shackle	
wire rope knob		wire rope knob (CO type)	
wire rope clamp		hand chain block	
welding head		hand shield	

welding gloves		welding shoe cover	
injector torch		injector blowpipe	
acetylene regulator		oxygen bottle	
oxygen regulator		coverd arc electrode	acidic electrode alkali electorde

4.2 Replacement of bearing block

The bearing blocks are included in main hoist mechanism，disassembly drawing shown in

1. Assembly drawing shown in

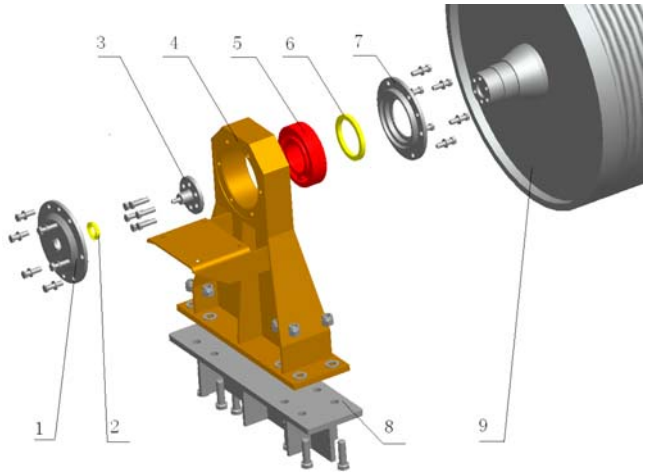
2.

4.2.1 Preparations before replacement

Before replacement, carefully research the drawings, understand the bearing block constitution. Personnel and tools needed on **错误！未找到引用源。**1.

Table 4-1 personnel and tools for bearing block replacement

Personnel/quantity		Tools/quantity	
lifting worker	1	lift crane or service crane	1 (6 tons at 25m height)
mechanic worker	2	jack	1 (5 t)
electric worker		chain block	2 (2 t)
	1	puller	1
		line	some
		other mechanic tools	some



1 outer end cover 2 seal 3 limit switch shaft 4 bearing block

5 bearing 6 seal 7end cover 8 base 9 drum

Figure 4-1 disassembly drawing of bearing block

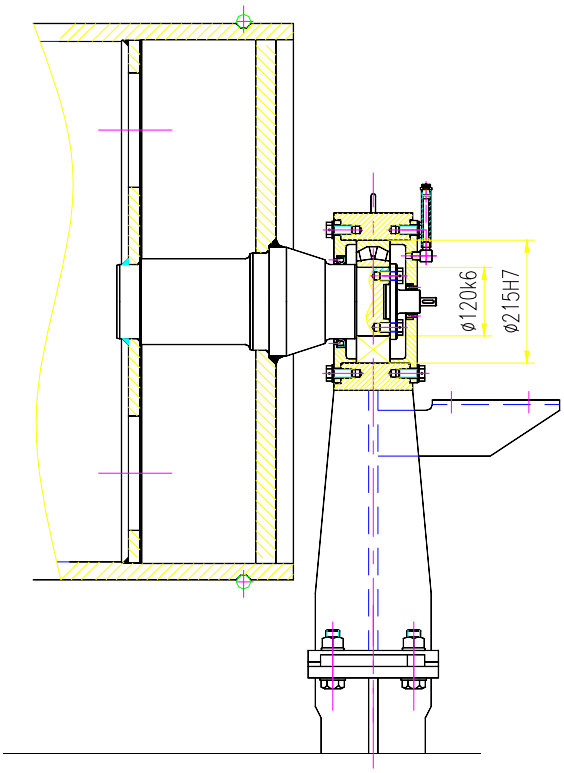


Figure 4-2 bearing block assembly drawing of main hoist

4.2.2 Replacement procedure

To avoid the force on drum caused by self weight of wire rope, use chain block to tie down the wire ropes on reliable item such as post of Machinery Room and so on.

Lower the spreader to the ground and slacken the Main Hoist ropes.

Lock out the crane's motion which affect safety by performing lock-out and tag-out procedure.

Disconnect the limit switch device at the drum end.

Remove the oil collecting plate at the other side.

To avoid the force on drum caused by self weight of wire rope, use chain block to tie down the wire ropes on reliable item on the trolley. Slacken enough rope, so that the drum can be moved uprightly.

Lift the side of bearing block with crane or RTG's service crane. The method as below also for consideration: put a wood or bracket under the bearing block side of drum then use a 5t jack to support drum. For safety, add woods as the drum is jacked up, and tuck wedge block at both sides of drum to avoid rolling. The jack point should be on the

support beam of trolley, or else thick plate should be put under jack.

Remove grease tube, outer end cover

Remove limit switch shaft.

Remove the bolts of inner end cover

Remove the base bolts

Lift the drum up slightly by lift or jack, make the bearing block unloaded. Support the drum with wood or skid reliably.

Remove bearing block. Use crane or service crane, or chain block by temporary staging to lift the bearing block upright a little and move it transversely slowly. If the fit is tight, use puller to get the bearing block disconnected. Puller could be fixed to the bolt holes of base, then put jack between drum shaft and puller to get the puller move. Finally take the bearing and cover out.

Inspect oil seals. Oil seals should be replaced if necessary.

Install a new bearing.

Re-assemble the bearing seat together in the reverse order mentioned above. Before reassembling, wash the drum shaft and bearing block with specified detergent and dry them with cool wind.

Realign the drum couplings. Refer to related criteria. Verify the drum-mounted rotary limit switch setting and alignment.

Remove lock-out and tag-out devices and Pre-Op the crane. Startup and monitor the crane to verify its operation before returning it to service.

4.3 Replacement of motor coupling

4.3.1 Preparations

The motor couplings are included in main hoist mechanism, trolley traveling mechanism,，gantry traveling mechanism , which are all elastic spider couplings，shown in 错误！未找到引用源。3. Before replacement, carefully research the drawings, understand the motor coupling constitution，the drawings about motor coupling on .错误！未找到引用源。2, personnel and tools needed on 错误！未找到引用源。3.

Table 4-2: coupling drawings for refer

Main hoist motor coupling	JL120060501
Trolley traveling motor coupling	JL1200601
gantry traveling motor coupling	JL12001A

Table 4-3: personnel and tools for motor coupling replacement

personnel/quantity		tools/quantity	
lifting worker	1	lift crane or service crane	1 (3 tons at 25m height)
welder	1	chain block	2 (2t)
mechanic worker	3	puller	1
electric worker	1	torque wrench	2
		other mechanic tools	some



Figure 4-2 actual picture of motor coupling

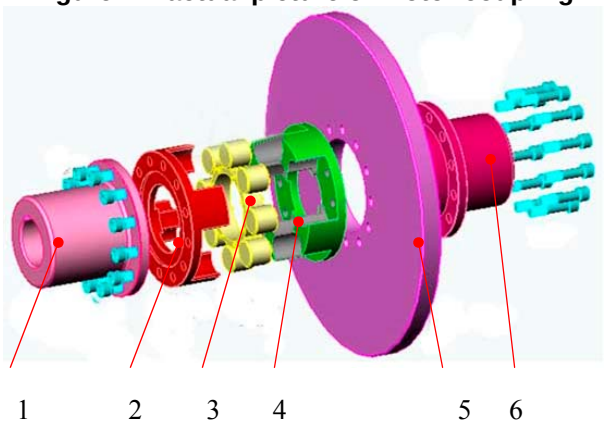


Figure 4-3 disassembly drawing of motor coupling

1 motor side half coupling 2 motor side jaw shape flange

3	spider elastic unit	4	reducer side jaw shape flange
5	brake disc	6	reducer side half coupling

4.3.2 Replacement procedure

Lower the spreader to the ground and slacken the Main Hoist ropes. If the trolley traveling motor coupling is replaced, the trolley need to stop at anchor position. If the gantry traveling motor coupling is replaced, the crane need to stop at the maintenance position, wheel is stopped by chocks.

Lock out the crane's motions which affect the safety by performing lock-out and tag-out procedure.

Remove the limit switch device at drum end.

To avoid the force on drum caused by self weight of wire rope, use chain block to tie down the wire ropes on reliable item on the trolley.

Remove the electrical connection from the motor brake, release the brake manually, then remove the brake by service crane.

Unfasten the bolts between motor side half coupling ,remove the bolts between motor side half coupling and jaw shape flange; remove the motor and half coupling by service crane.



Figure 4-4 step 6

Take out motor side jaw shape flange



Figure 4-5 step 7

Take out elastic spider unit

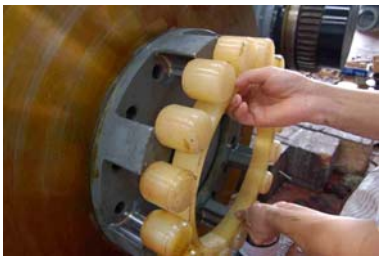


Figure 4-6 step 8

Remove the motor side half coupling by puller. Protect the motor shaft during the procedure



Figure 4-7 step 9

Remove the reducer side half coupling，Remove the jaw shape flange.



Figure 4-8 step 10

Remove the brake disc.

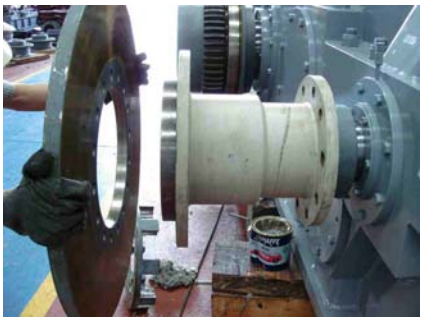


Figure4-9 step11

Remove the driven side half coupling by puller (jaw puller)，Protect the reducer shaft during the process.



Figure4-10 step 12

Next is the procedure of re-align the new motor coupling. Newly bought coupling should be anti-corrosion treated, sealed well.



Figure4-11Folded motor coupling

After unfold the cover, clean the anti-rust oil, heating the two half couplings by electric furnace, or by oil bath while no choice.

Use scraper to clean the anti-rust oil, to achieve required degree, each position should be cleaned thoroughly.



Figure4-12 step 14

When re-aligning, match the serial number between the half coupling and the brake disc.



Figure4-13 step 15

	Normally, the serial number in disc includes item number, mechanism number, and brake disc front or back (A or B). B and B aims at half coupling, A aims at half flange.

Thread the heating half coupling into the reducer shaft, for blocking proof the quicker the better, not more than ten minute.

Realign the motor coupling according to the related criteria, as for coupling allowable compensation rate, install separation, nuts torque, please refer to the coupling manual.

4.4 Replacement of drum coupling

4.4.1 Preparations

The drum coupling includes main hoist mechanism, which is curved tooth coupling, shown in 错误！未找到引用源。,4-15. Before replacement, carefully research the drawings, understand the drum coupling constitution，the drawings about drum coupling on 错误！未找到引用源。4. Personnel and tools on 错误！未找到引用源。.

coupling drawing for refer

Main hoist drum coupling	JL120060501

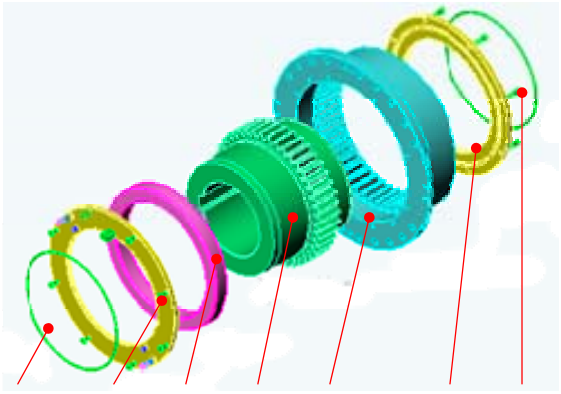
Table 4-5: personnel and tools for drum coupling replacement

personnel/quantity		tools/quantity	
lifting worker	2	lift crane or service crane	1 (3 tons at 25m height)
		chain block	2（2t）
painter	1	puller	1

			(hydraulic) jacking machine 200t)
mechanic worker	3	torque wrench	2
electric worker	1	other mechanic tools	some



Figure4-14 actual picture of drum coupling



1 2 3 4 5 6 1

Figure 4-153 disassembly drawing of drum coupling

1 seal 2 outer end cover 3 load ring

4 outer teeth hub 5 inner teeth flange 6 inner end cover

4.4.2 Replacement procedure

Lower the spreader to the ground and slacken the Main Hoist ropes.

Lock out the crane's motion which affect the safety by performing lock-out and tag-out procedure.

Remove the limit switch device at drum end.

To avoid the force on drum caused by self weight of wire rope, use chain block to tie down the wire ropes on reliable item on the trolley.

Lift the drum , then remove the bolts of bearing block base, take the drum with bearing block out from reducer shaft transversely slowly.

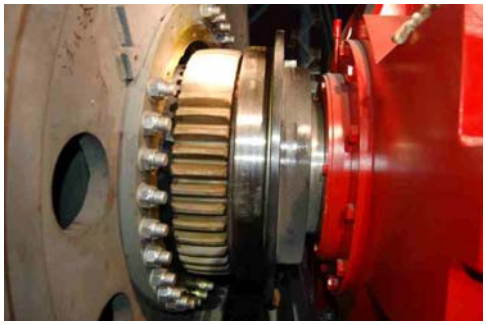


Figure4-164 step5

Disconnect the drum coupling, be careful about outer teeth hub and inner teeth flange if no abrasion.

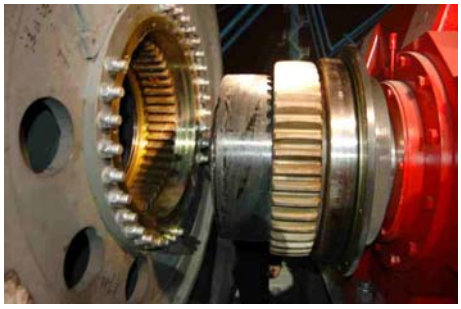


Figure4-17 step 6



Figure4-18 disassembled inner teeth flange



Figure4-19 disassembled outer teeth hub

Lay down the drum to a convenient place, unfasten the bolts.



Figure4-20 step 7

Hold the bolt hole of inner teeth flange with lifting crane or service crane, then move the flange out of drum horizontally slowly



Figure4-21 step8

Before disassembling the out teeth hub, at first estimate that if the hub is useful or not, If it is to be discarded as useless, cut it down from the key slot by gas cutting, then the coupling can be taken out easily.



Figure4-225 step9

During replacement of drum coupling, install large stud on the holes of outer teeth hub.



10 Figure4-23 step 10

Put proper washers to take advantage of the height of jack, make sure all the washers are solid, and avoid falling of washers.



Figure4-24 step 11

Put the jack on the washers reliably.



Figure4-25 step12

Put the flange to the most outside, get studs through the flange holes. Tighten the studs. During operation make sure the surface between flange and jack to contact fully and flatly. Flange is supported by bracket. Then start jacking up, the whole operation may need 1 hour.



Figure4-26 step13

Next is the procedure of realign the new drum coupling. Newly bought coupling should be anti-corrosive treated, sealed well.



Figure4-27 folded drum coupling

For the following cleaning and installation, first take down the inner teeth flange, outer teeth hub and abrasion indicator.



Figure4-28 step15

Clean the inner teeth flange and outer teeth hub thoroughly.



Figure4-29 step16

Heat the outer teeth hub in electric furnace, if no choice, oil bath also be accepted.



Figure4-30 step17

After preparations, start installation, first insert the abrasion indicator.



Figure4-31 step18

Insert outer teeth hub, for fear the hub cooling down, the hub should be installed ASAP, and to the right position, up to shaft shoulder of reducer.



Figure4-32 step19

After the hub is fixed, push the indicator to the corresponding position of coupling.



Figure4-33 step20

Reassemble the drum and outer teeth hub according to the method above, verify the serial numbers of both parts, which is important for accurate installation.

Realign the drum coupling and remark line at the abrasion indicator.

4.5 Replacement of motor brake

4.5.1 Preparations

The motor brake includes main hoist mechanism, which is electric hydraulic disc brakes. shown in **错误！未找到引用源。** some trolley traveling mechanism also adopt this type. Before replacement, carefully research the drawings, understand the motor brake constitution，the drawings about motor brake on **错误！未找到引用源。**, the needed personnel and tools on **错误！未找到引用源。**.

Table 4-6 brake drawings for refer

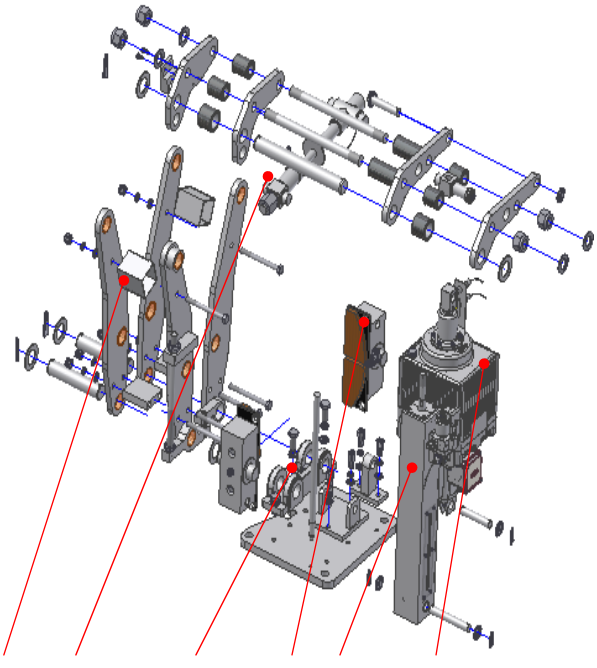
main hoist motor brake	JL1200605
trolley traveling motor brake	JL1200601

Table 4-7 personnel and tools for drum coupling replacement

personnel/quantity		tools/quantity	
lifting worker	1	hand chain block	1 (5t)
welder	1	other mechanic tools	some
mechanic worker	2	torque wrench	1
electric worker	1		



Figure4-33 actual picture of motor brake



1 2 3 4 5 6

Figure4-34 disassembly drawing of motor brake

1 brake arm 2 brake rod 3 balancing device

4 brake shoe 5 brake spring 6 thruster

4.5.2 Replacement procedure

Lower the spreader to the ground and slacken the Main Hoist ropes.

Avoid the crane's motions which affect safety by performing lock-out and tag-out procedure.

Only necessary，remove the whole brake., next we will introduce how to replace the whole brake and its accessories.

4.5.3 Replacement of whole brake

Remove the power wire from the thruster, remove the L.S, and release the brake by hand.

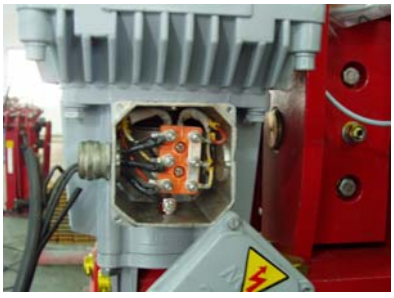


Figure4-35 step4

Remove the base bolts



Figure4-36 step5

Lift the brake away, mark the original position



Figure4-37 step6

According to the mark, fully install a new brake, relocation and adjusting is needed if necessary



Figure4-38 step7

Remove lock-out and tag-out devices and Pre-Op the crane. Startup and monitor the crane to verify its operation before returning it to service

4.5.3.1 REPLACEMENT OF THRUSTER

Remove the cotter pins of up & down pins of thruster.



Figure4-39 step1

Take out the pin of thruster head, lean the thruster down reliably.



Figure4-40 step2

Take out the pin of bracket.



Figure4-41 step3

Lift the thruster away. The thruster is very complicate, can't be fixed at spot, so if there is some problem of thruster, please get it to professional person.

Install new thruster.

4.5.3.2 REPLACEMENT OF BRAKE SPRING TUBE

Take out the pin of thruster head, lean the thruster down reliably as above expression.



Figure4-42 step1

Turn the torque adjusting nut anticlockwise to release the spring force.



Figure4-43 step2

Take out the elastic cylindrical pin on slotted nut, then remove the slotted nut of spring tube.



Figure4-44 step3

Take out the cotter pin from pin of spring base.



Figure4-45 step4

Take out the pin.



Figure4-46 step5

Lift the triangle plate.



Figure4-47 step6

Take out the spring tube.



Figure4-6 step7

4.5.3.3 REPLACEMENT OF SPRING

Install nut to brake rod.



Figure4-749 step1

Turn the brake rod clockwise, make sure the rod not press the sleeve at the bottom of tube.



Figure4-50 step2

Take out the sleeve at the bottom of tube with brass rod and hammer.



Figure4-8 step3

Turn the brake rod anticlockwise, until the spring force disappear, turn out the nut of brake rod, then take out brake rod and spring base, finally take out spring, then change a new one.



Figure4-52 step4

4.5.3.4 REPLACEMENT OF AUTOMATIC WEAR COMPENSATION DEVICE

Remove the cover.



Figure4-53 step1

Remove the cotter pin, loose the set screw of pin.



Figure4-54 step2

Take out the compensation stud above the cross shaft.



Figure4-55 step3

Loose the nuts of stud.



Figure4-56 step4

Pull out the stud.



Figure4-57 step5

Pull out pin.



Figure4-58 step6

Take out auto wear compensation device and triangle plate from the gap of brake arm.



Figure4-959 step7

Remove triangle plate, take down auto compensation device, then install a new compensation device.



Figure4-60 step8

4.5.3.5 REPLACEMENT OF BRAKE LININGS

Unlock the hand release lever. Turn clearance adjusting nut to let the brake open enough.



Figure4-1061 step1

Loose lining bolts.



Figure4-62 step2

Move the lining to the disc direction to get off the flat key, take out from upright direction.



Figure4-63 step3

4.5.3.6 REPLACEMENT OF HAND RELEASE LEVER

Loose lever bolts.



Figure4-64 step1

Take out hand release lever



Figure4-65 step2

Loose roller bolts.



Figure4-66 step3

Remove the roller.



Figure4-67 step4

4.5.3.7 REPLACEMENT OF BALANCING STUD

Loose the nut at bottom end of balancing stud.



Figure4-68 step1

Turn the upper adjusting nut to the bottom of thread.



Figure4-69 step2

Loose the balancing stud, till it exit from screw hole, remove the top nut.



Figure4-70 step3

Take out stud, and change a new one.



Figure4-71 step4

4.6 Replacement of gantry wheel

4.6.1 Preparations

Gantry wheels are included in gantry traveling, the gantry traveling has 8 wheels type and 16 wheels type. 8 wheels type has 2 wheels each corner, 16 wheels type has4 wheels each corner. The drawings about wheels see **错误！未找到引用源。**, Tools and personnel needed shown in **错误！未找到引用源。**.

Table 4-8 wheel drawings for refer

gantry traveling wheel	JL12001A, JL12001B

Table 4-9 personnel and tools for wheel replacement

personnel/quantity		tools/quantity	
mechanic worker	3	hydraulic jack	2（50t）
		professional tyre disassemble tools	1set
		torque wrench	1
		other mechanic tools	some
		fork lift	1

Preparations before replacement:

Jack up the crane with no suspended load in calm weather with spreader on ground.

Lock out the crane's motion which affect safety by performing lock-out and tag-out procedure.

Enclose the safety area with guardrails, keeping away independent person.

Move the trolley to the following position, to max travel opposite to the side of wheel need replacement:

- If the E-room and power house are at different sides, when jack up power house side wheels, trolley stop at max travel of E-room side; when jack up E-room side wheels, trolley stop at max travel of power house side.
- When power house and E-room are at same sides, when jack up power house side wheels, trolley stop at max travel of the other side, when jack up the other side wheels, trolley stop at max travel of power house side.

4.6.2 Idle wheel disassembly of 8 wheels gantry traveling

Stop trolley to right position, set safety working area.

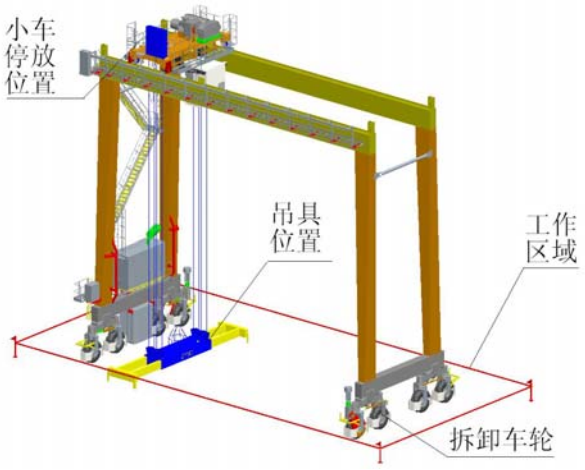


Figure 4-72 step1

Rotate the wheel 90° which need disassembly (if the tyre is broken, please overleap this step, and carry out step 3 directly.)

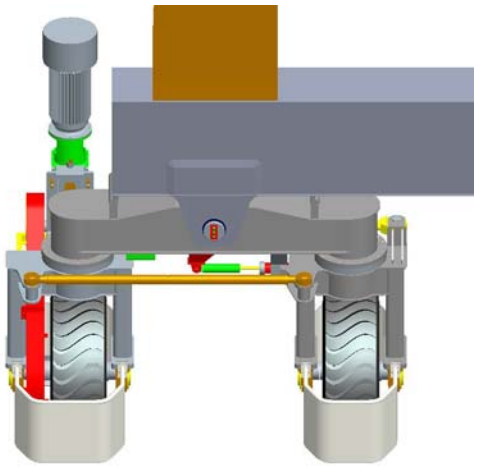


Figure4-73 step2

Remove the bolts from idle wheel shield

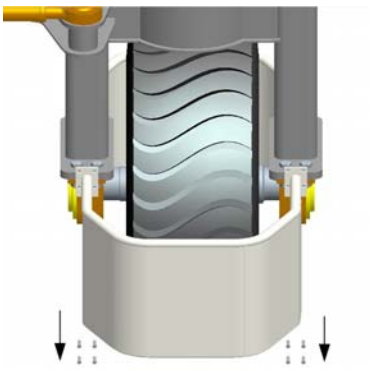


Figure4-74: step3

Take down idle wheel shield

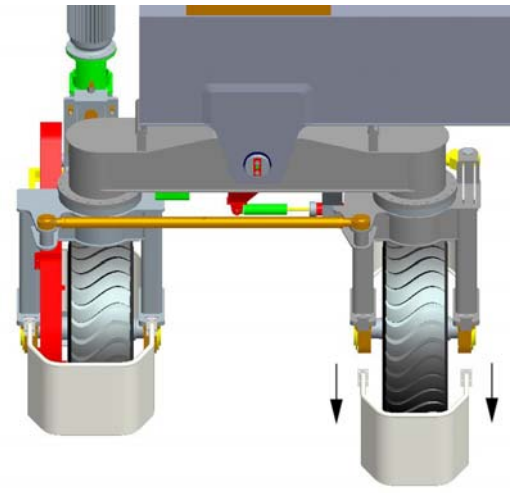


Figure4-75 step4

Remove the ZP3 shear block between idle wheel frame and idle wheel bearing block

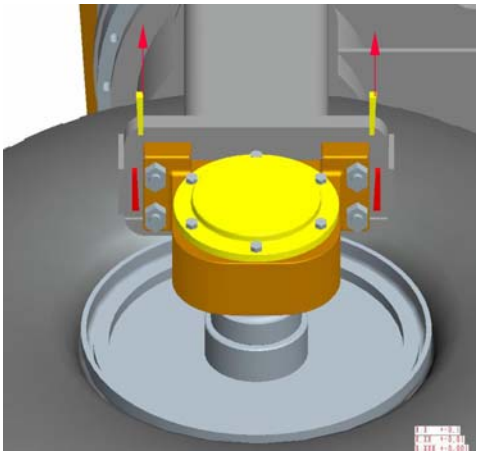
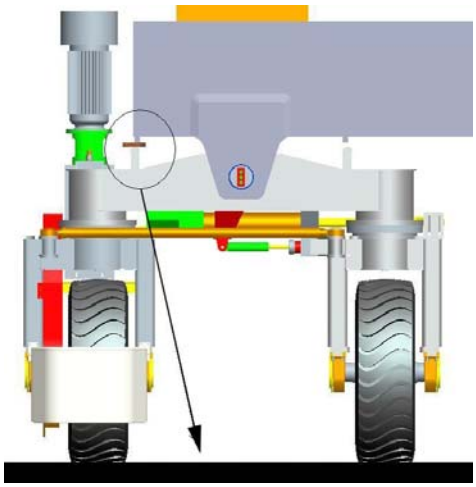


Figure4-76 step5

Insert a 33mm wood plate into the gap between sill beam and equalizer beam.



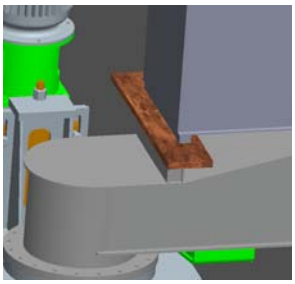
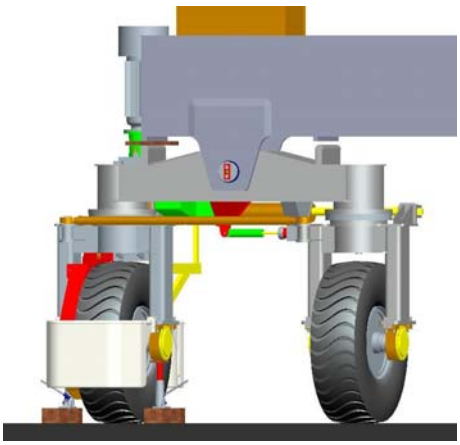


Figure4-77 step6

Set 2 50t hydraulic jacks right under the 2 bearing blocks, insert a wood sheet between bearing block and jack for protection. As bearing block is high, if the jack is not long enough an iron or wood block could be used, don't jack up the wheel this moment.



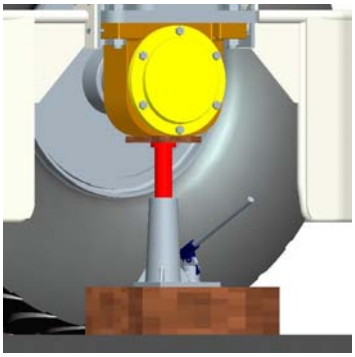


Figure4-78 step7

Remove bolts between wheel carriage and bearing block.



Figure4-79 step8

Jack up the 2 hydraulic jacks simultaneously, make the idle wheel out from ground(about up 50 mm from ground surface), because the bolts and shear blocks have been removed at the time, so the wheel will drop away by self weight, for safety, some person should hold the wheel for fear turnover.

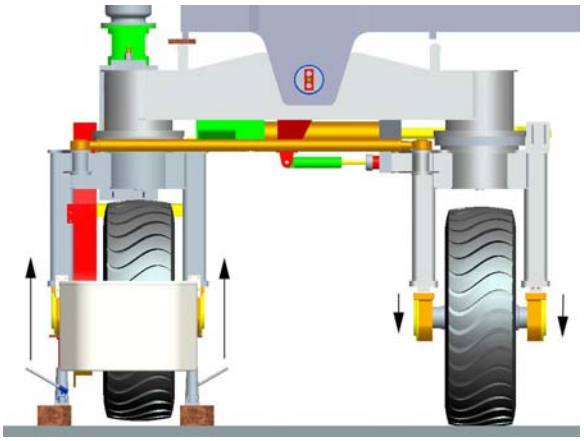


Figure4-80 step9

Take down the idle wheel assembly, then move it to specified position by forklift.(If the wheels don't rotate 90° , because of carriage frame, fork lift can't fork the wheel easily, so let the wheel roll out from the frame first)

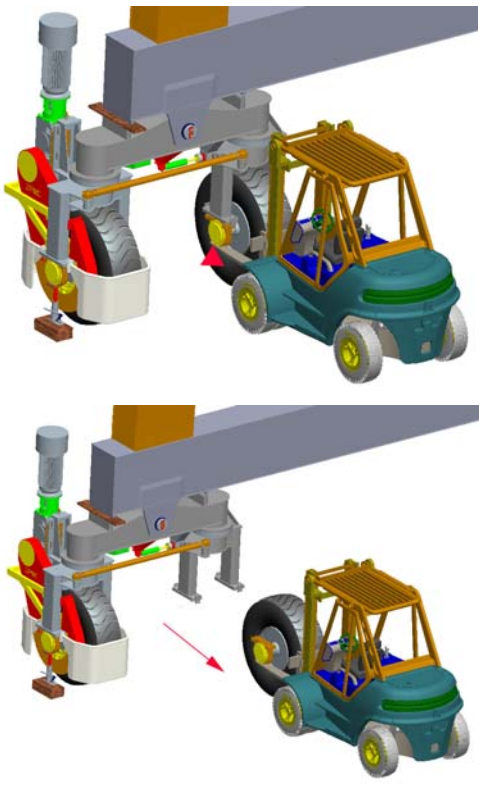


Figure4-81 step10

Note: Assembly procedure is just the reversal of disassembly procedure of idle wheel

4.6.3 Drive wheel disassembly of 8 wheels gantry traveling

Stop trolley to right position, set safety working area.

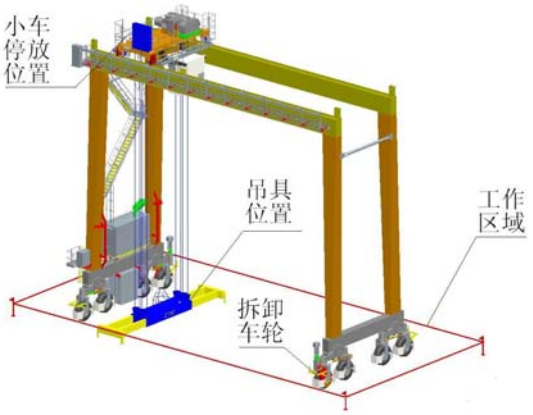
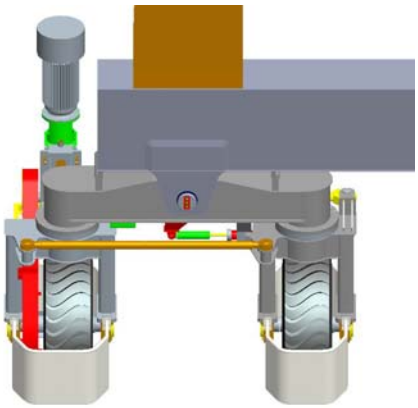


Figure4-82 step1

Rotate the wheel 90° which need disassembly (if the tyre is broken, please overleap this step, and carry out step 3 directly.)



Figre4-83 step2

Remove the bolts from drive protection frame



Figure4-84 step3

Take down drive protection frame.

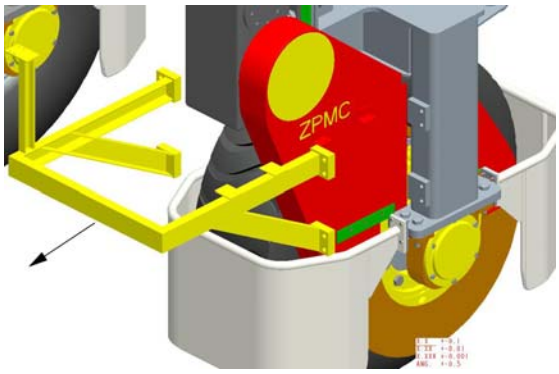


Figure4-85 step4

Remove the bolts from drive wheel shield, then remove the shield.

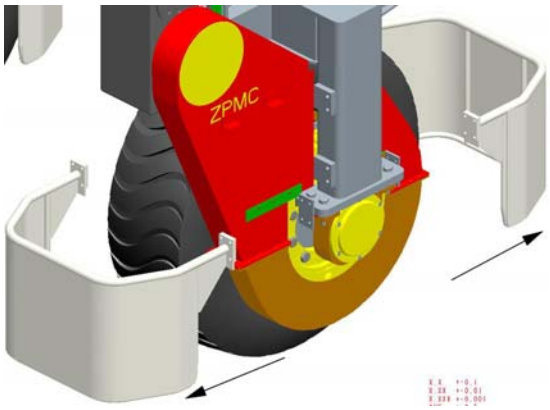


Figure4-86 step5

Remove the bolts of chain cover.

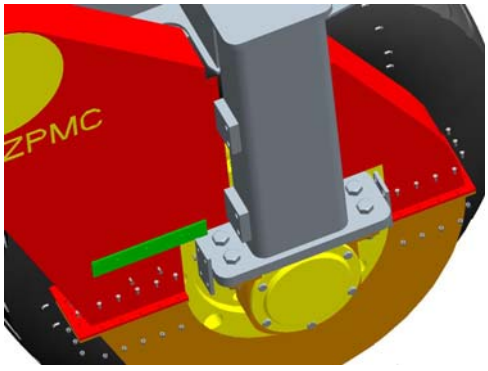


Figure4-87 step6

Remove chain cover.

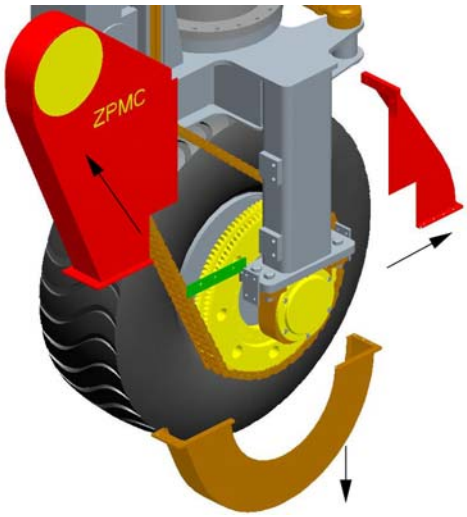


Figure4-88 step7

Loose the bolts between motor and carriage(only loose, no need take down)

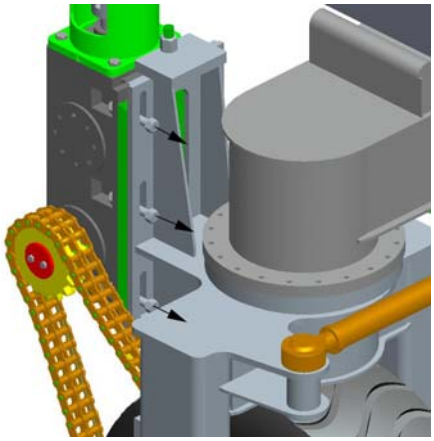


Figure 4-89 step8

Loose the 2 adjusting bolts under the reducer, let the reducer have the room to release the chain for the convenience of its disassembly (If the reducer is already at the lowest position, the chain can't be loosen any more, then disassemble the chain according to the following explanation.)

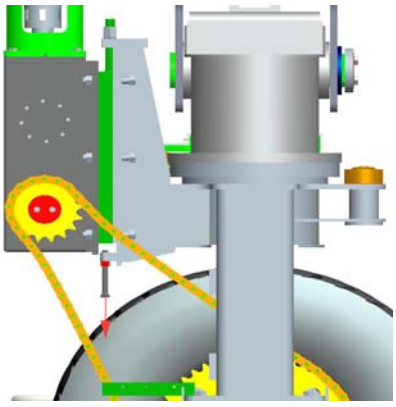


Figure4-90 step9

Adjust the M36 nut on the top of the reducer, lower the reducer to loose the chain for the following disassembly.

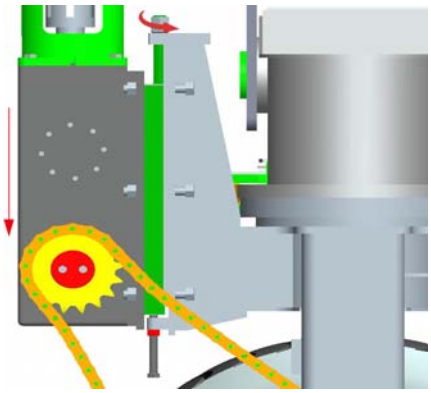


Figure 4-91 step10

First find the joint pin which connect the ends of chain.

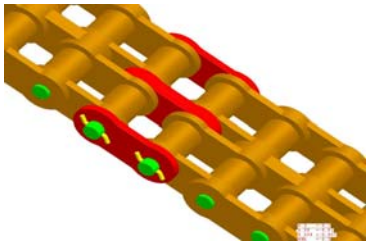


Figure4-92 step11

Pull out the lock pin, remove the joint plates, then disconnect the chain's two ends, take out the chain.

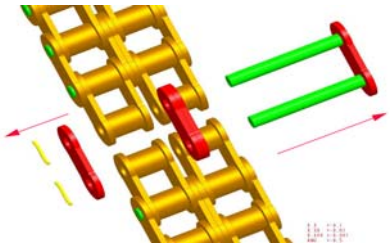


Figure4-93 step12

Remove the ZP3 shear block between drive wheel frame and drive wheel bearing block

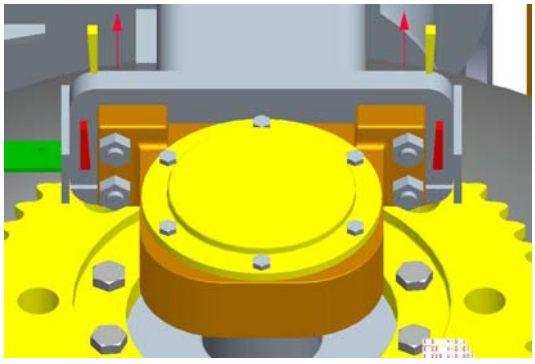


Figure4-94 step13

Insert a 33mm wood plate into the gap between sill beam and equalizer beam.

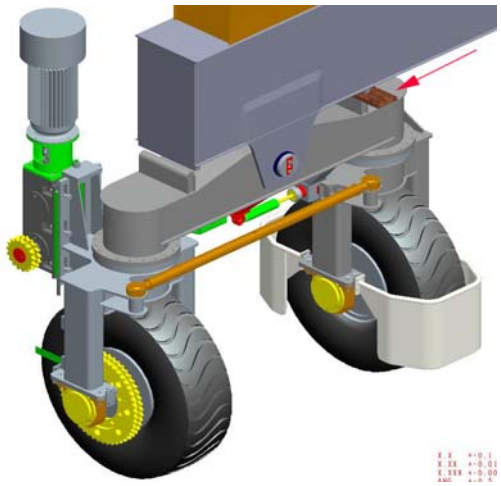


Figure4-95 step 14

Set 2 50t hydraulic jacks right under the 2 bearing blocks, insert a wood sheet between bearing block and jack for protection. As bearing block is high, if the jack is not long enough an iron or wood block could be used, don't jack up the wheel this moment.

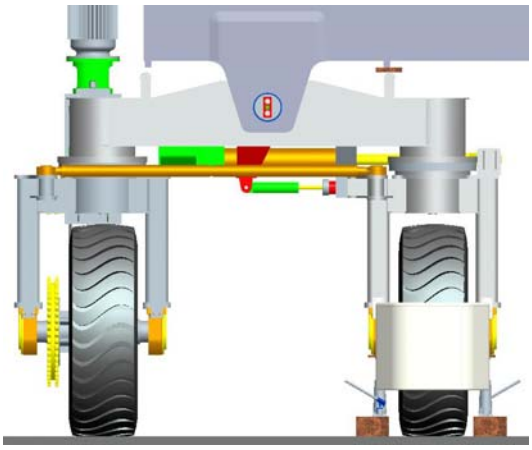


Figure4-96 step15

Remove bolts between wheel carriage and bearing block

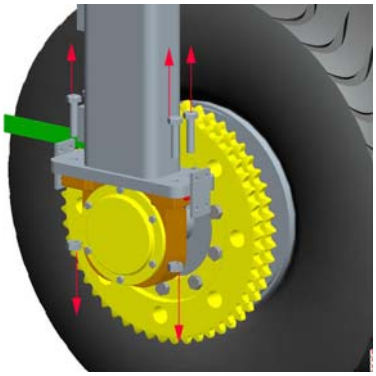


Figure 4-97 step16

Jack up the 2 hydraulic jacks simultaneously, make the drive wheel out from ground(about up 50 mm from ground surface), because the bolts and shear blocks have been removed at the time, so the wheel will drop away by self weight, for safety, some person should hold the wheel for fear turnover.

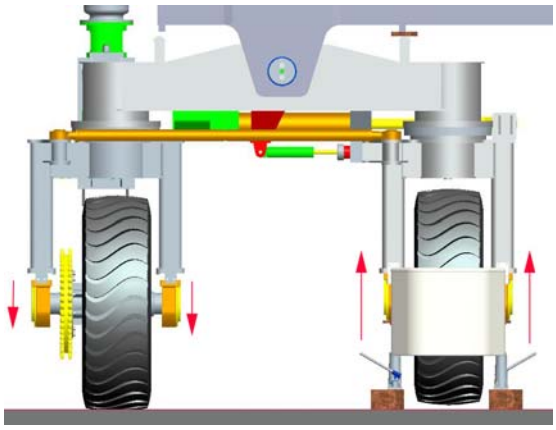


Figure4-98 step17

Take down the drive wheel assembly, then move it to specified position by forklift.(If the wheels don't rotate 90° , because of carriage frame, fork lift can't fork the wheel easily, so let the wheel roll out from the frame first)

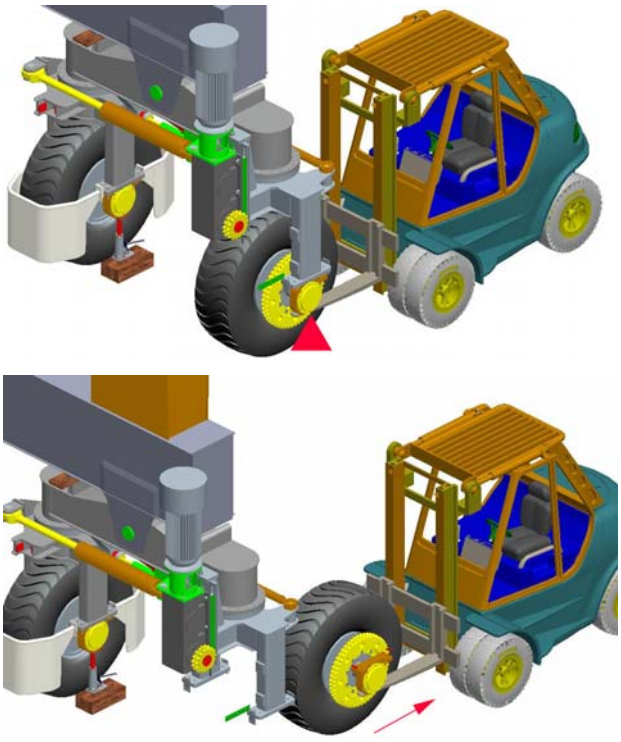


Figure4-99 step18

Note: Assembly procedure is just the reversal of disassembly procedure of idle wheel

4.6.4 Replacement of tyre

8 wheels type gantry traveling device have 2 types of wheel assembly which are drive wheel assembly and idle wheel assembly, shown as figure 4-112. 16 wheels type gantry traveling only has one type of wheel assembly as shown in Figure 4-115.

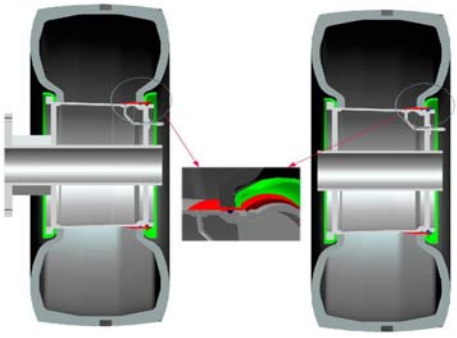
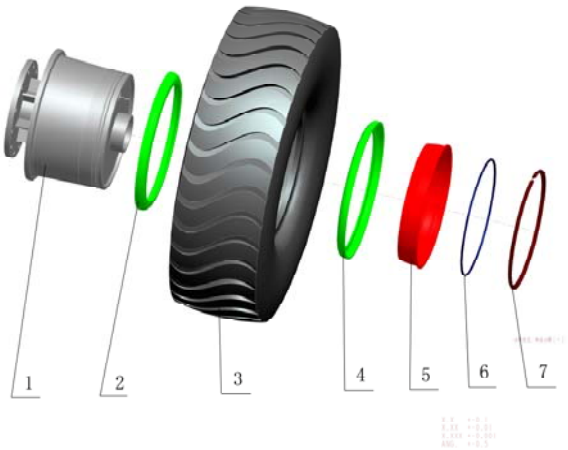


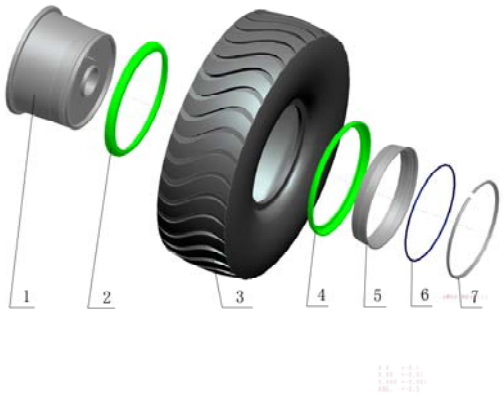
Figure 4-112 8 wheels gantry traveling type



1 drive rim 2 baffle ring 3 tyre 4 baffle ring

5 tyre race 6 seal 7 lock ring

drive wheel assembly of 8 wheel gantry traveling



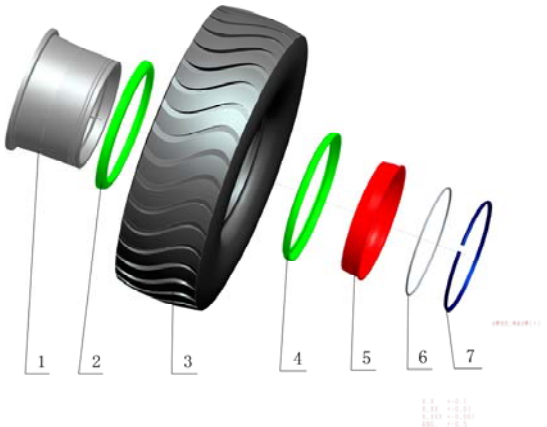
1 idle rim 2 baffle ring 3 tyre 4 baffle ring

5 race tyre 6 seal 7 lock ring

Figure 4-114 idle wheel assembly of 8 wheel gantry traveling



Figure4-115 16 wheels gantry traveling



1 rim 2 baffle ring 3 tyre 4 baffle ring

5 race tyre 6 seal 7 lock ring

Figure 4-116 wheel assembly of 16 wheels gantry traveling

	Installation and maintain tyre and wheel rim is a little dangerous, the work should performed by professional person. Incorrect operation may cause tyres or rims located wrongly, even cause explosion while bearing external force.

1） Preparations

Tools: manual, hydraulic, automatic tyre dismounting machine. Figure4-117 is normal hydraulic tools for dismounting tyre:



A) hydraulic dismounting tyre tool set





B) oil tube C) pedal switch

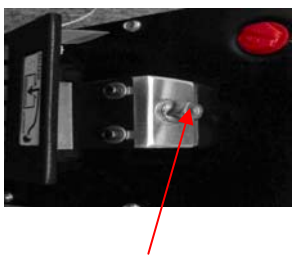




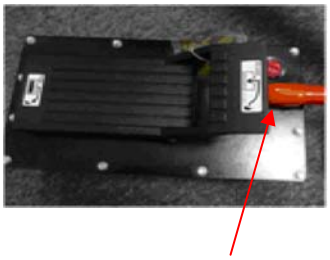
D) oil tube, air tube quick joint E) working cylinder

Figure4-117 hydraulic dismounting tools

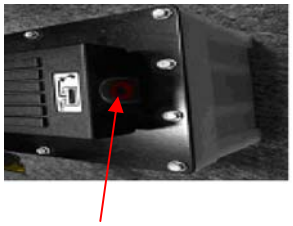
Connection of dismounting tools: refer to figure 4-118



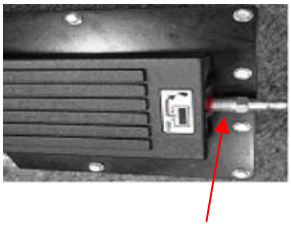
A) remove plug



B) install oil tube



C) pull out plug



D) install air tube joint



E）installation finish

Figure 4-118 Connection of dismounting tools

	Note: This equipment involves high pressure air & fluid, please confirm the joint and the tube work well before using it.

- Make sure the tyre is released entirely before removing any parts such as nut, lock and so on.(Under pressure air broken rim easily get out and cause damage.)
- Pull out valve core to release the tyre entirely.(while removing stud of tyre the components may jump out if there is pressure air inside tyre)
- Inserting a wire into valve core and turning several rounds for fear the blocking up.(Air blowing may caused by blocking up at core, air bleeding may caused by water frozen)
- Operation of tools should be far away from butt weld.
-

2）Disassembly of tyre

A） Take down the lock ring, result refer to Figure4-119



Figure 4-119 step A

B） Take down the rubber seal. Shown in Figure4-120



Figure4-120 step B

C） Disassembly of tyre race, pry the tyre race a little out with tyre lever, insert small nuts, then push working cylinder into the gap. See Figure4-121



Figure 4-121 step C

D） After insert cylinder into gap, adjust bolt at the end of cylinder, hold the rim and tyre race with a flat bar, make the two parts contact tightly. Figure 4-122



Figure 4-122 step D

E） Start pedal switch. Shown in figure4-123



Figure4-123 Step E

F） Pull out the tyre race. Shown in Figure 4-124



Figure 4-124 step F

G） Take down tyre race. Shown in figure4-125



Figure4-125 step G

H） Take down the baffle ring on the other side. Shown in figure 4-126



Figure4-126 step H

I） Take down the rim and the baffle ring on the other side. Shown in figure4-127



Figure4-127 step I

	While prying the rim with dismounting tools, be sure stand at one side of tyre and apply force with one hand. Because the falling tools may jump out toughly to cause damage.

3）Cleanness

- Clean the rim and repaint the worn part to resist erosion, and be useful for tyre maintenance and replacement. The rust at lock ring and groove should be cleaned entirely, or else that will affect accurate assembly. Air filter could resist rust effectively. And that should be checked periodically. (All parts should be cleaned especially grooves at lock ring)

4）Tyre

- Tyre assembly procedure is just the reversal of the disassembly procedure as above description.
- Before assembly, check all the components, a new one should be replaced if any cracking, broken, serious abrasion & rust are found.(cracking, broken and rust could lead to worse performance, distorting and returning may not be used)
- In any condition don't try to re-machine, forge, heat, even brazing. Should be replaced by a new one or well one with same brand, dimensions.
- Clean the rust, ash and foreign matter at tyre lip and rim groove. Clean the inner room of tyre.
- To the tyre which has air leakage or no enough pressure, if you doubt the possibility of broken of wheel component, please don't charge again.(Air leakage and no enough pressure may cause wheel broken or part shifting)

5）air charging

- While charging the tyre, knocking lock ring or rim is prohibited.
- Before air charging, make sure all the parts are installed rightly.
- Don't charge the tyre if the installation is not finished. Put tyre to the fence or other containers. First charge the pressure to 5psi/0.35bar and inspect the installation of all parts, the condition of O shape seal. If there is any problem, should recharge the tyre. If the there is no problem at 5psi/0.35bar, then charge tyre continuously.
- While charging tyre standing on it is prohibited, suggest using extended tube type indicator to let the personnel out from the ejecting area of the tyre. Install indicator or pressure adjustor at inlet valve, pre-set wanted pressure value. If the tyre is in fence or other containers, don't lean on it (for feat the jumping out of tyres).
- Don't perform welding or heating work around tyre.(The heat will cause pressure change, then lead to explosion, the storehouse could catch fire)
- Check the leakage after charging. Shown in below:

6）storage



Figure 4-128 tyre storage

- Normally, no matter tyre is new or old (tube is taken down from rim and get cleaned), needed to be stored at cool & dry condition, should be laid uprightly. Being laid horizontally is prohibited, to make the clearance between lips is proper. Keep away external force for feat of distorting.
- The following conditions should be avoided
  - direct sunshine exposure
  - ozone
  - oil or other contamination
  - high temperature and dampness
- If the tyre is stored open air, cover by canvas. If the tyre is installed to some equipment , a bracket should be used to avoid the load to be directly laid on tyre. If there is no choice, then rotate the tyre monthly for fear of deformation.

4.7 Replacement and adjusting of trolley wheel

4.7.1 Preparations

Trolley wheels are part of the trolley assembly (1 wheel per corner, 4 wheels in total). For RTG trolley, wheels have 2 type which are driving wheels and driven wheels.This device is driven by universal shaft, the reducer driving type please refer to other drawings.The constitution of wheel assembly is shown in Figure 4-130.Before replacement understand the drawing for the trolley wheel structure (错误！未找到引用源。). Personnel and tools requirements needed are show in 错误！未找到引用源。.

Table 4-10 wheel drawings for refer

trolley traveling wheel	JL2200601，JL220060101

Table 4-11 personnel and tools for wheel replacement

personnel/quantity		tools/quantity	
lifting worker	2	jack	4 (15t)
mechanic worker	2	hand chain block	4 (3t)
		torque wrench	1
		other mechanic tools	some

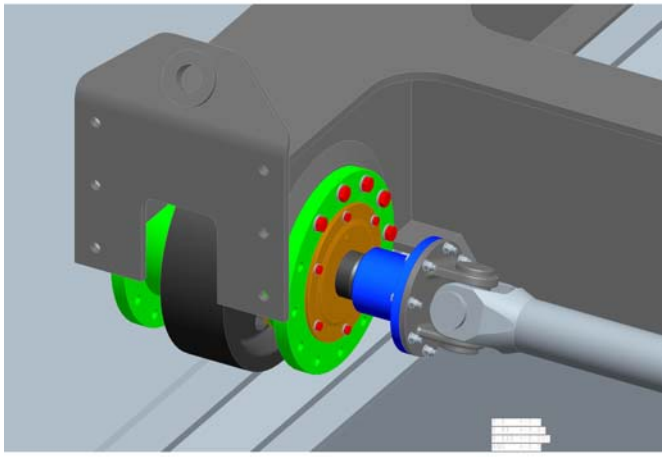
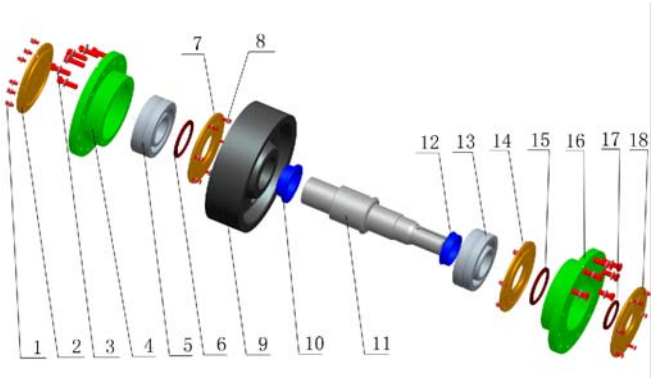


Figure4-129 trolley wheel



1Blind cover bolt 2Blind cover 3bearing block bolt 4bearing block

5bearing 6oil seal 7through cover 8through cover bolt 9wheel

10shaft sleeve 11wheel shaft 12shaft sleeve 13bearing 14through cover

15oil seal 16bearing block 17oil seal 18through cover

Figure4-130 disassembly drawing of trolley wheel

4.7.2 Replacement procedure

	If noise or metal dust is evident when the wheels are running. Stop at once and check them. Take appropriate action to correct the problems.
	During the running in time for new cranes (150~300h), checking is very important. When build up stress is released, a trolley frame can deform appreciably. Trolley wheels must be checked and adjusted.
	Once adjustment is complete, conduct a trial run with an empty load. If these results are successful, conduct a trial run with a rated load. During this period, if these is any problem, adjust the wheel eccentric sleeve seat until performance is acceptable

Trolley wheel replacement requires considerable technology. Workers must read the ZPMC instructions carefully, arrange the procedure consider conditions

around, but safety must be the most important.

Move the trolley to rest position, while sunshine with wind speed less than 5m/s.(The room below the trolley is enclosed by fenders, no crossing of vehicles or person), personnel mush wear safety belt , make sure the jack up point on trolley frame.

Lay the spreader on ground, slack 600mm wire rope at least (To assure the trolley frame is contacted to crane after being jacked up 350mm)

1) Lock out crane motions which affect safety by proceeding lockout, tag-out procedure. Shown in figure4-131

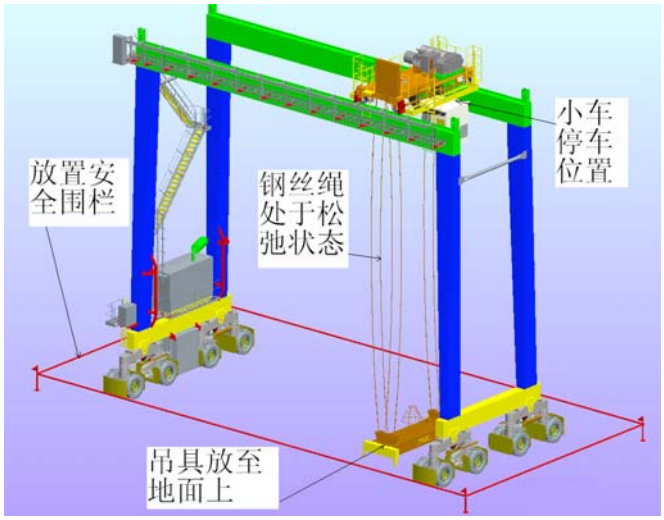


Figure4-131 step1~3

Choose the wheel to be dismounted.

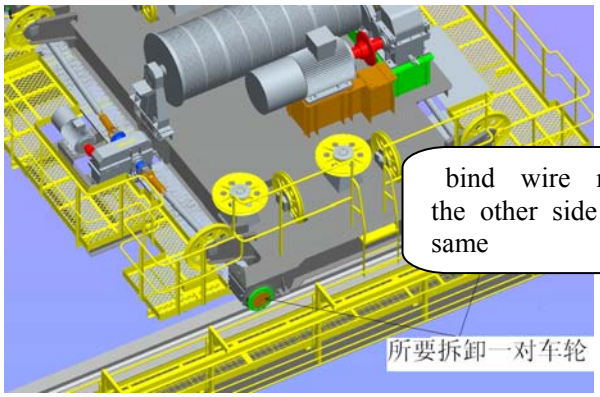


Figure4-132 step4

Remove horizontal trolley wheel and its support or end anti-collision plate, the removed parts should be fixed reliably.

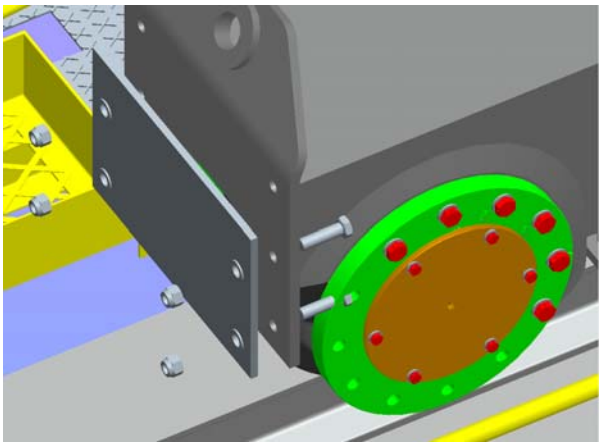


Figure4-133 step 5

If the energy chain system is ZPMC, the upper parts above the bearing of floating head need to be dismounted. Release the cable fixed to leading structure of trolley frame, set aside enough cable to meet the requirement of jacking up 300mm. Avoid the cable being extended too much to cause broken, shown as the following figure. If the energy chain system is IGUS or WAMPFLER, the above operation is no needed.

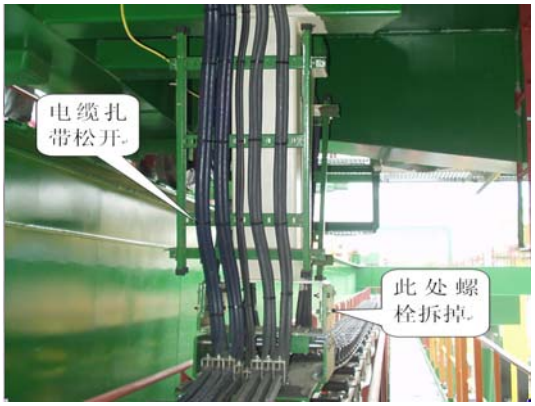


Figure 4-134 step 6

Get the wire rope or chain block through the lug at front & back ends of trolley, then tie the trolley down to the girder bottom(set aside some allowance to let the trolley jacked up smoothly) no unexpected horizontal force is allowed.

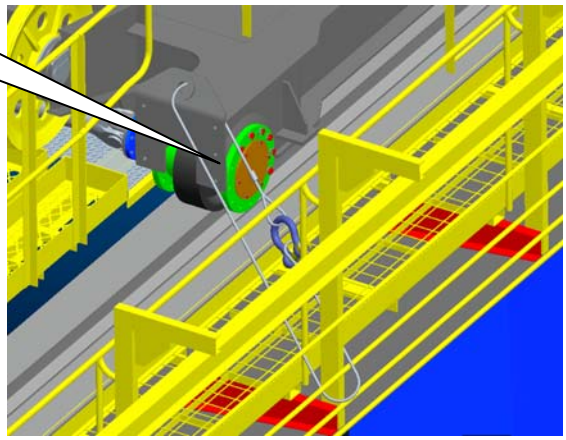


Figure4-135 step 7

Remove the bolts at universal shaft side, adjusting universal shaft, get wheel and universal shaft disconnect, universal is laid down on the platform. Remove the grease tube.

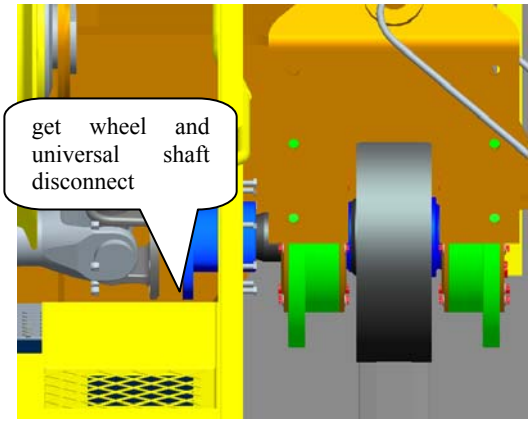


Figure4-136 step 8

Lay the jacks at the position shown in figure, make sure the jack is completely upright.(Totally 4 jacks)

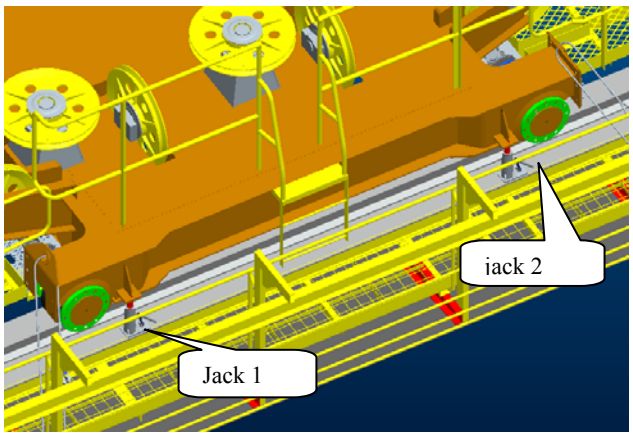


Figure 4-137 step 9

Get wire rope through trolley wheel, fixed to lug at trolley ends.

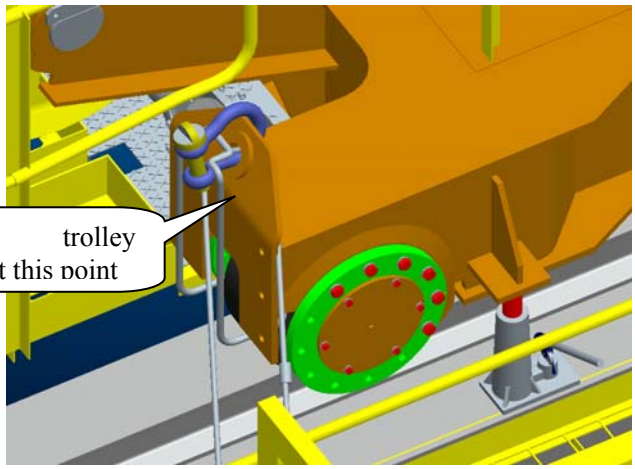


Figure 4-138 step 10

Remove bolts between eccentric sleeve and trolley frame, each side 6.

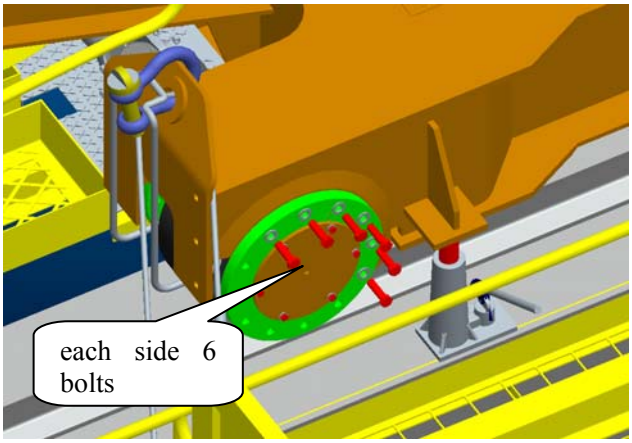


Figure4-139 step11

4 jacks jack up simultaneously with person holding the wheel. If there is enough room eccentric sleeves could be pulled out from outside direction. If no enough room, jack the trolley up until the gap between wheel and rail is about 160mm, then the eccentric sleeves may be pulled out after the assembly position are marked. Be careful while jacking up, 4 jacks should jack up simultaneously, the trolley should be kept horizontal and moved slowly.

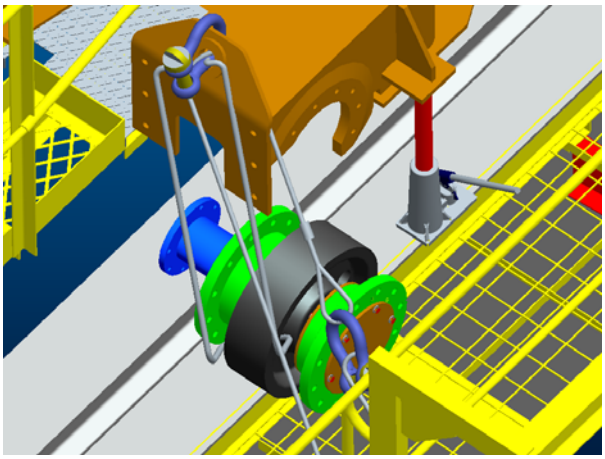


Figure4-140 step12

	Mark the assembly position before operation, the eccentric sleeve should be assembled like used to be.
	After jacking up trolley, insert wood blocks between trolley and rail.

Lift the wheel to ground with lifting crane or service crane. Move it to shop.



Figure4-141 step13

	The repair work or replacement of wheel or shaft should be done at shop.

At shop, first remove end cover.

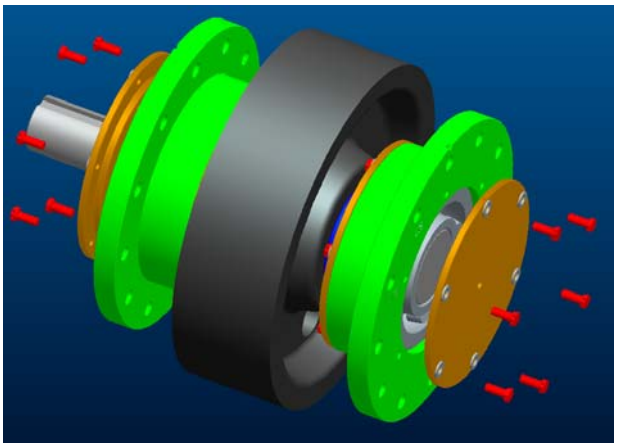


Figure 4-142 step14

Pull out bearings with puller, move them away.

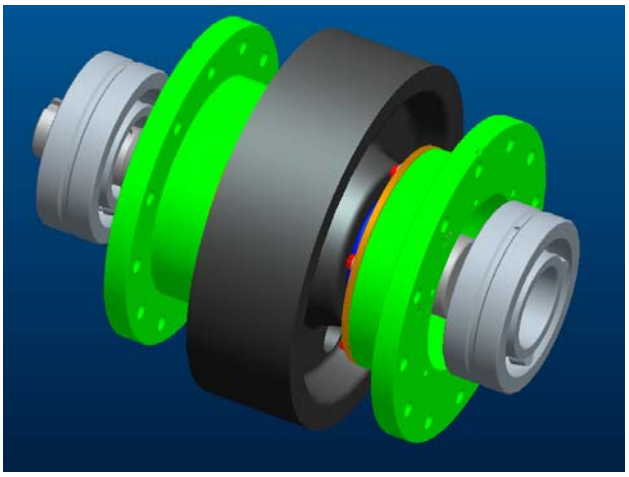


Figure4-143 step 15

The fit between wheel and shaft is tight, so the wheel and shaft should be replaced together to ensure the proper fit tight enough to deliver required torque.

Use new wheel , shaft or new bearing to take place of old ones, and assemble the wheel reversely according to the above procedure. Mount cover, fasten bolts and input the wire.

Have a test run of trolley. If no problem the replacement of wheel and bearing is finished.

4.7.3 Adjusting of trolley wheel

To reduce self-weight, RTG frame normally is designed to be light and relatively soft. A new RTG's trolley wheel will run deviation after being used for a while, which may be also seen after changing wheel. This time adjusting for wheel is needed.

Each trolley wheel is supported by two anti-friction bearings which are housed inside two eccentric bearing sleeves., Both eccentric bearing sleeves are 1mm

adjustable. Please refer to the **错误！未找到引用源。**. Eccentric structure of wheel refer to the Figure4-145

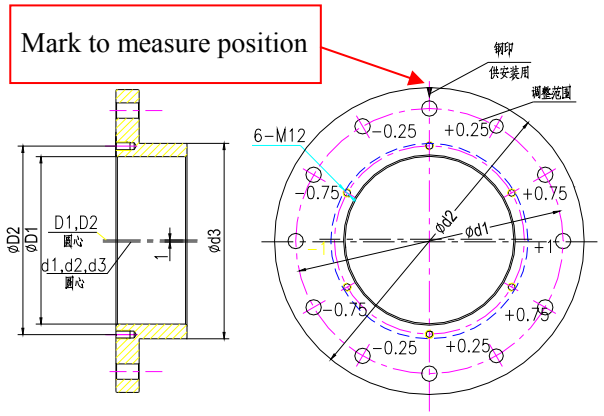
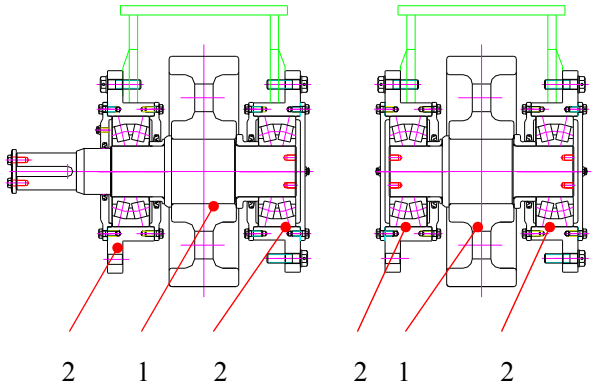


Figure4-144 Mark at the eccentric sleeve



1 trolley wheel 2 eccentric sleeve also as bearing block

Figure4-145 assembly of eccentric wheel

Through different turning combinations of these two eccentric sleeves, the trolley wheels can be re-aligned in three conditions. 1) Horizontally Deviated; 2) Vertically leaned; 3) Vertical Lift-off.

Because side rigidity is not good, so should pay more attention to horizontal deviation. If the deviation between wheel and girder is over 30mm, the trolley wheel should be adjusted.

As rigidity of trolley frame is no good, and gantry is soft, the vertical leaning and deviation could be ignored.

Initial position of trolley wheel

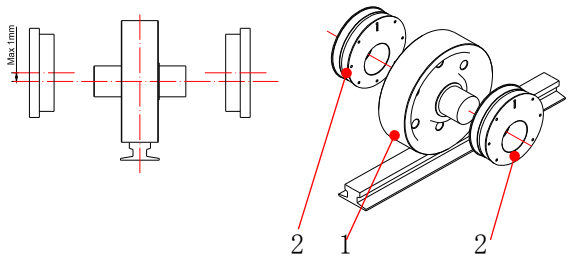


Figure 4-146 Initial position of trolley wheel

1 wheel 2 mark of eccentric sleeve before adjusting

To express the question, Figure 4-146 suppose the initial position of trolley wheel. But the eccentric of wheel is exaggerated(not according to real scale). Attention! Mark the initial position is very important. It is very useful for adjusting trolley wheel, if the wheel can't be adjusted to good condition, the wheel can be resumed easily by using the mark.

wheel horizontal deviation

图 4- shows how to adjust the wheel deviated horizontally. Turn the left sleeve 90 degree clockwise, and turn the right sleeve 90-degree anti-clockwise. The wheel can be deviated horizontally to the left by 0.41° . If you turn the two sleeves in the opposite direction, the wheel will be deviated to the other direction.

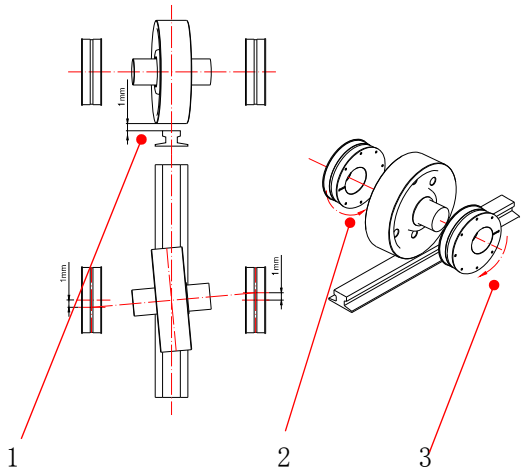


图 4-147 Figure4-147 wheel horizontal deviation adjusting

1 wheel is lifted <=1mm 2 rotate 90 ° clockwise

3 rotate 90° anti-clockwise

When doing this adjustment, please gradually turn the eccentric sleeve. Every time, a 30 degree angle turning is recommended, then see if the trolley running condition has been improved. If necessary,

	turn another 30 degree. Confirm the right direction, this decide which wheels need adjusting and which direction is right. If the running condition gets worse, then turn reversely.

Relationship between wheel adjusting and eccentric sleeve rotating is shown in Table 4-12

Table 4-12 relationship between wheel adjusting and eccentric sleeve rotating

	combination 1	combination 2
right eccentric sleeve	90º	no rotating
left eccentric sleeve	90º	180º
wheel adjusting result		
horizontal deviation	√	
vertical leaning		√
vertical deviation	√	√
	combination 3	combination 4
right eccentric sleeve	180º	no rotating
left eccentric sleeve	180º	90º
wheel adjusting result		
horizontal deviation		√
vertical leaning		√
vertical deviation	√	√

according to above table, only combination affect horizontal deviation

The next is a example of adjusting wheel:

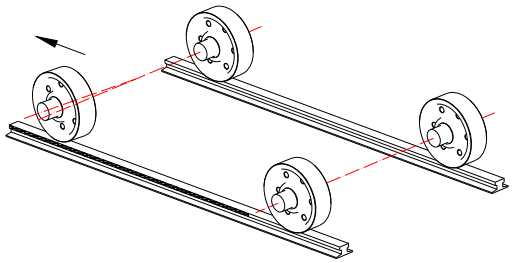


Figure4-148 example of adjusting wheel horizontal deviation

This figure illustrates that the left wheel at the front of trolley have a inner horizontal deviation when the trolley run towards front. The left horizontal trolley wheel will press the left surface of trolley rail. The following method could be adopted to fix the problem.

1) At the trolley horizontal wheel side if trolley wheel has a small inner horizontal deviation, then turn the outer eccentric sleeve of the trolley wheel clockwise an angle.

2) If the wheel deviation is large, turn the outer eccentric sleeve of the wheel at the other side without horizontal trolley wheel. Turn the sleeve an angle clockwise, and observe the running condition.

3) If the running condition gets better after adjusting turn the eccentric sleeves at same direction. If not, turn the eccentric sleeves at reverse direction.

4) Sometimes, both two wheels have inner horizontal deviation is a better condition than two parallel wheels just like the toe-in of car structure. This arrangement make the trolley runs along a straight line more easily.

5) Normally only outer eccentric sleeves need adjusting.

6) Normally, only one wheel needs adjusting at a time, sometimes only two wheels need adjusting. Only in special condition all 4 wheels need adjusting.

7) Clear up all tools. Have a test run of trolley, inspect the result.

	Appling the method to RTG trolley wheel properly may need a lot attempt before doing well.

4.8 Replacement of main hoist sheave

4.8.1 Preparations

Sheaves are part of main hoist mechanism, have 2 arrangements which are 4 ropes and 8 ropes. Sheave constitution is shown in Figure 4-149, 4-150, before replacement understand the drawing for the sheave composition (错误！未找到引用源。). Personnel and tools requirements needed are show in 错误！未找到引用源。.

Table 4-123 sheave drawings for refer

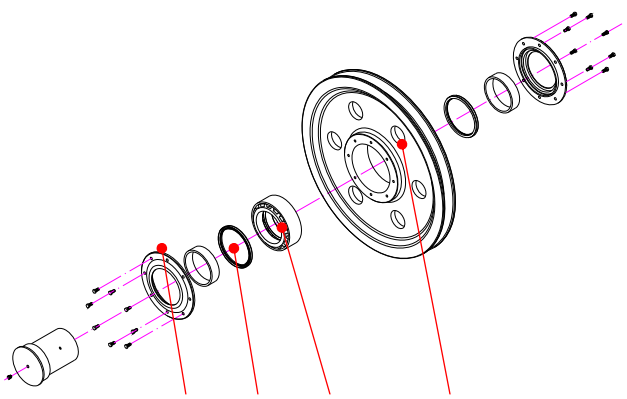
sheaves on trolley	JL2200603，ZMHL760
sheaves on spreader	ZMHL760

Table 4-14 personnel and tools for sheave replacement

personnel/quantity		tools/quantity	
lifting worker	1	hand chain block	2 (0.5t)
mechanic worker	2	other mechanic tools	some
electric worker	1	torque wrench	1



Figure4-149 actual picture of sheave



1 2 3 4

Figure4-150 assembly drawing of sheave

1end cover 2seal 3 bearing 4 sheave

4.8.2 Replacement procedure

For 4 ropes type sheaves shown in Figure 4-151, there are 4 sheaves on trolley, and 4 sheaves on spreader. The spreader could be laid down to ground, it is easy to replace sheaves.



Figure4-151 4 rope type trolley sheaves

For 8 rope type trolley sheave, totally 12 sheaves are all arranged on trolley, shown in Figure4-152. There are no sheaves on spreader.



Figure4-122 8 rope type sheaves

Replacement procedure of sheave on trolley:

Stop trolley at rest position.

Confirm the sheaves and sheave shafts need replacement, move the lift crane to the side of RTG, or move RTG to the

underside of maintenance crane, to get the sheave down to ground.

Lay the spreader to ground and slack the main hoist rope. Take out wire rope from groove of sheave and fixed it.

Lock out crane motions which affect safety by proceeding lockout, tag-out procedure. Shown in figure4-131

Get wire rope from the sheave need replacement with 2 chain blocks.

Take down sheave assembly including shaft. Here are the methods according to different base types:

- Horizontal sheave of 4 rope type. Remove the 2 covers on the top, remove 2 stop plates, then take out the sheave as a whole assembly including shaft. Refer to Figure 4-153

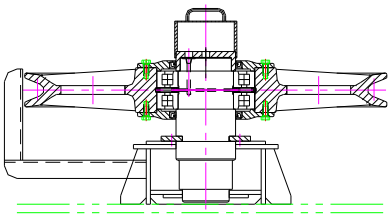


Figure4-153 horizontal sheave of 4 rope type

- Vertical sheave of 4 rope type. Remove 4 bolts at both side, remove upper half support, then take out sheave as a whole assembly including shaft.

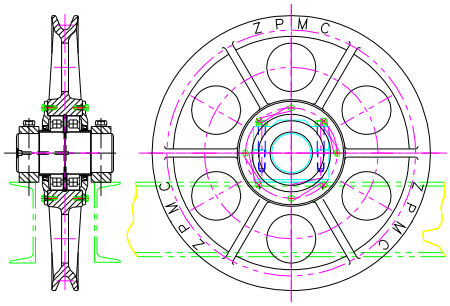


Figure4-154 vertical sheave of 4 rope type

- Horizontal sheave of 8 rope type. Limited by circumstance, part of sheave may be under motor. So in this condition the sheave can't be lifted upright directly, the bolts between sheave support and trolley structure should be removed firstly, then take out the sheave together with support. The next is removing 2 stop plates, then take out sheave as a whole. Refer to figure 4-155.

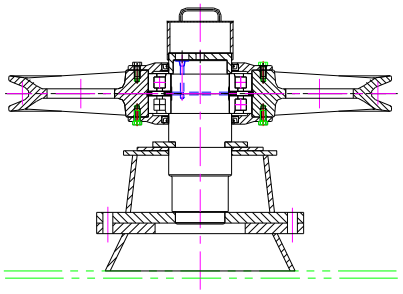


Figure4-155 horizontal sheave of 8 rope type

- Leaning sheave of 8 rope type. Horizontal sheave of 4 rope type. Remove the 2 covers on the top, remove 2 stop plates, then take out the sheave as a whole assembly including shaft. Because the sheave is leaning, side force should be applied during operation. Refer to Figure 4-156

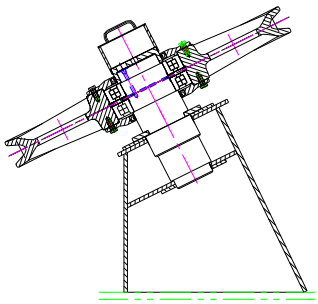


Figure4-156 leaning sheave of 4 rope type

- Vertical overhang sheave of 8 rope type. Remove 2 stop plates, then take out the sheave as a whole assembly including shaft. Shown in Figure4-157.

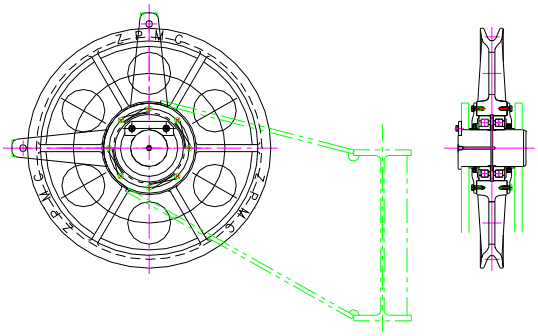


Figure4-157 vertical overhang sheave of 8 rope type

Lay the sheave assembly on the ground, repair it in shop. Figure4-158



Figure4-158 step7

Knock out wheel shaft, be careful not to break the grease hole.



Figure4-159 step8

Take down sheave assembly. If there are problems, remove the cover bolts.



Figure4-161 step9

Pull out the upper half bearing with puller. Puller can be tied to bolt holes of sheave cover, its claw could hook the inner ring of bearing.



Figure4-162 step10

Take out inner spacer ring, outer spacer ring and so on.



Figure4-163 step 11

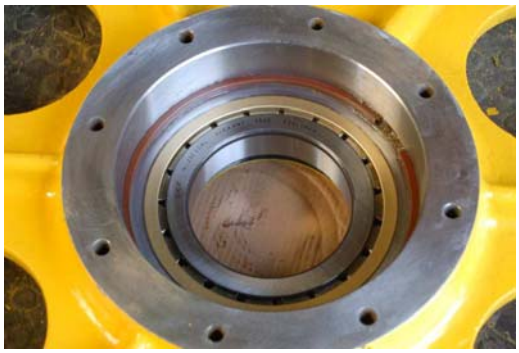


Figure4-164 outer spacer ring(brown part)

Turn over the sheave, take out the other half bearing by same way.

Assemble the sheave to initial condition reversely according to above method.

Recover the wire rope. Have a test run, then finish the replacement of sheave.

4.9 Replacement and adjusting of horizontal trolley wheel

To avoid rail gnawing, trolley wheel normally has no rim. Horizontal trolley wheel take over the leading function.

Each horizontal trolley wheel is composed by 1 bracket, 2 eccentric shafts, 2 wheels and bearings and etc. They are fixed to trolley frame by bolts. Assembly drawing is shown in Figure 4-194 & Figure 4-195.

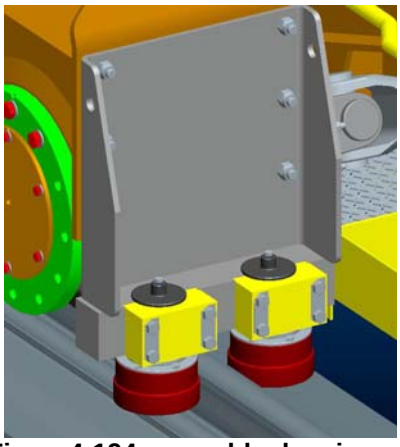


Figure4-194 assembly drawing of horizontal trolley sheave

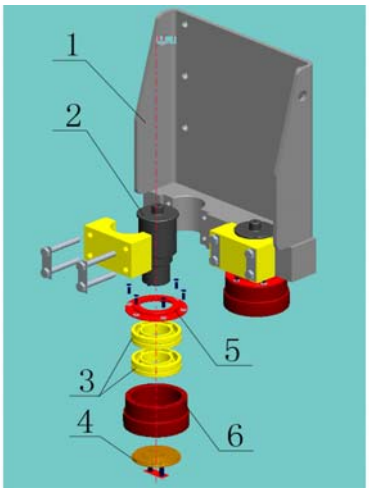


Figure 4-195 assembly of horizontal trolley wheel

1bracket	2eccentric shaft	3bearing	4spacer ring
5through cover	6horizontal wheel		

4.9.1 Preparations

Keep crane no load, stop the trolley to a position with both sides between horizontal wheels and rail have clearance.

Personnel and tools for horizontal wheel replacement is shown as Table 4-

Table 4-15 wheel drawings for refer

horizontal trolley wheel	JL2020612

Table 4-16 personnel and tools for horizontal wheel replacement

personnel/quantity		tools/quantity	
lifting worker	1	other mechanic tools	some
mechanica worker	2	Torque wrench	1

4.9.2 Disassembly of horizontal trolley wheel

Stop trolley at rest position, kept no load, lay the spreader on the ground.

Remove bolts of wheel shaft.

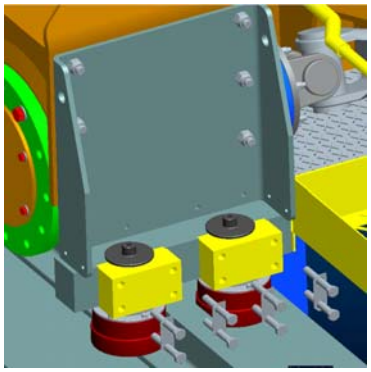


Figure 4-196 step32

Remove the locating block.

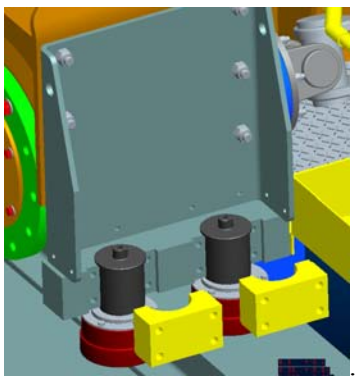


Figure4-197 step33

Take out horizontal wheel assembly.

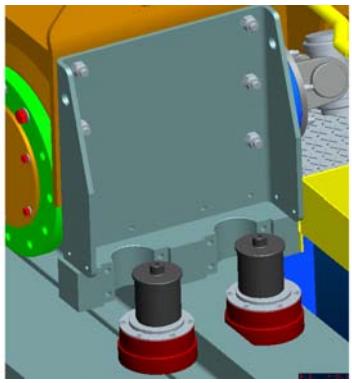


Figure4-198 step34

Put wheel assembly at safe area for disassembling.

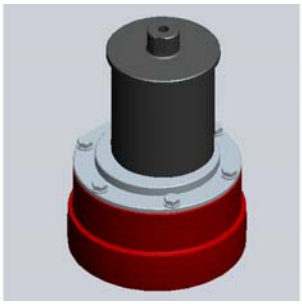
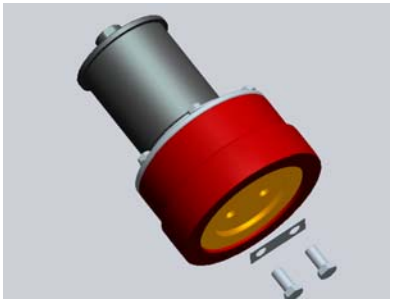


Figure 4-199 step 35

Remove the bolts of stop ring at bottom



Figre4-200 step 36

Remove stop ring.

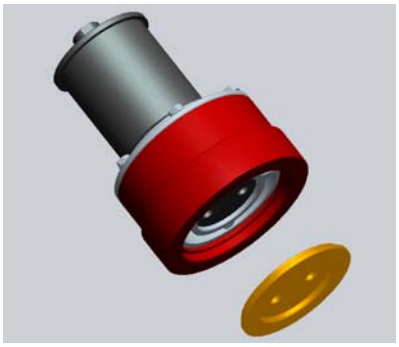


Figure4-201 step37

Remove the bolts of through cover at top



Figure4-202 step33

Pull out wheel from shaft with puller.

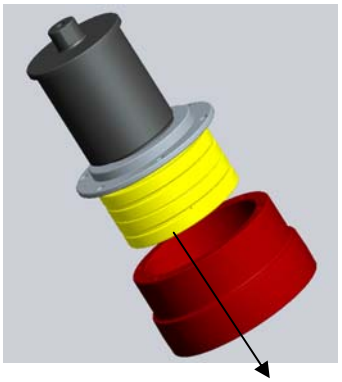


Figure4-203 step39

Pull out 2 bearings from shaft with puller.

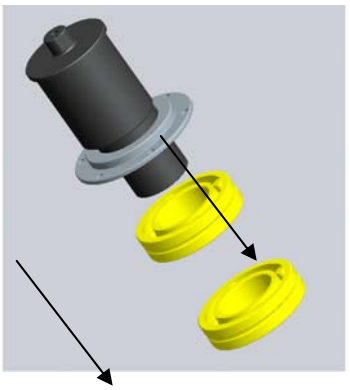


Figure 4-204 step40

Take out upper through cover.



Figure4-205 step41

4.9.3 Adjusting of horizontal trolley wheel

A new RTG's trolley wheel will run deviation after being used for a while, which may be also seen after changing wheel. At this time adjusting for horizontal trolley wheel is needed.The adjusting of horizontal wheel is mainly to assist trolley wheel adjusting. The aim is to make trolley run along straight line, with even clearance at both sides between rail and horizontal wheels.While trolley running, only one side contacting of horizontal trolley wheel.

Eccentric distance of wheel shaft is 5 mm. Shown in Figure 4-206. Adjusting clearance by rotating eccentric shaft.

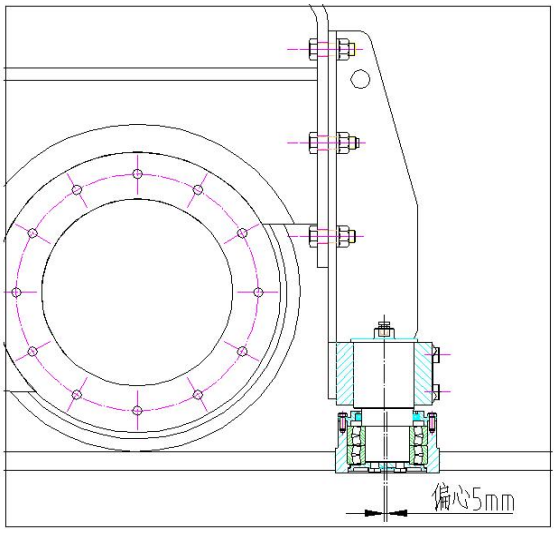
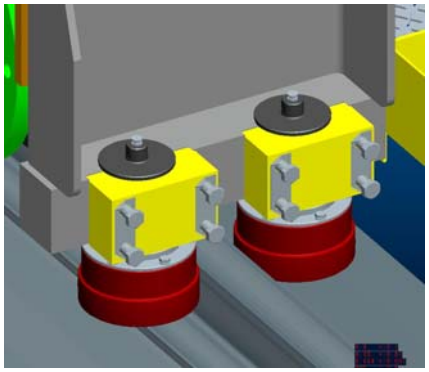


Figure4-206 eccentric structure of horizontal trolley wheel

Adjusting procedure:

Loose bolts which fix wheel shaft, create clearance between locating block and base. So, horizontal wheel can rotate freely.



Hold shaft head with spanner, then turn it as below method to proper position.

A） Turn spanner as red arrow direction, the clearance between wheel and rail gets smaller.

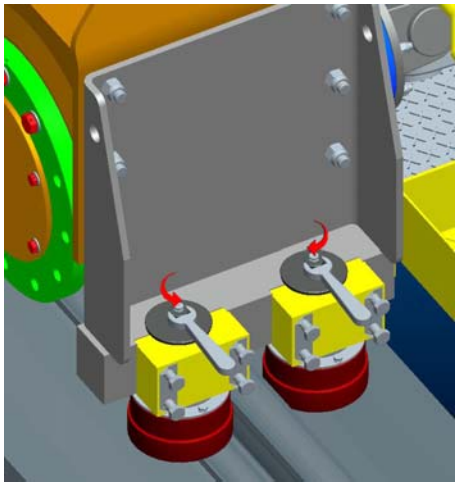


Figure4-208 step 2

B） Turn spanner as red arrow direction, the clearance between wheel and rail gets larger.

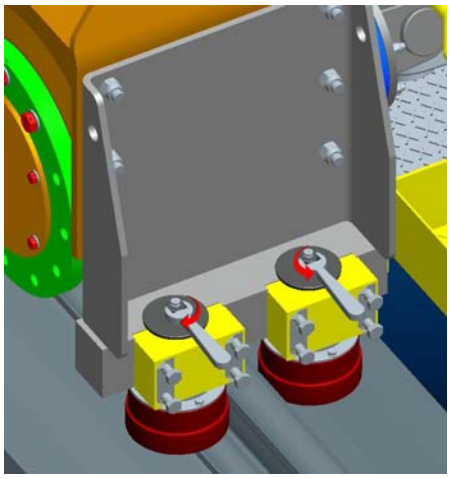


Figure4-209 step3

Then fasten the bolts of locating block.

CHAPTER 5 WIRE ROPE

5.1 BASIC KNOWLEDGE OF QIRE ROPE
5.1.1 construction
5.1.2 core 2
5.1.3 type of laying 2
5.2 OPERATION
5.2.1 transportation and storage
5.2.2 unwind and wind
5.3 TERMINATION OF WIRE ROPE END
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5.3.2 wire rope cutting
5.4 WIRE ROPE INSPECTION
5.4.1 factors influence the service life of wire rope
5.4.2 wire rope inspection method
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5.4.4 inspection content
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5.4.6 procedure of wire rope inspection
5.5 WIRE ROPE LUBRICATION
5.5.1 importance of lubrication
5.5.2 selection and requirement of lubricant
5.5.3 lubrication methods of wire rope
5.5.4 lubrication of main hoist wire ropes
5.6 ROPE RE-REEVING
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Chapter 5 wire rope

Summary

Wire rope is the easily worn-out component, and not include in the guarantee scope.

Usually wire rope brand is decided by consumers according to their experience, and the new wire rope is also purchase by consumers. As one of the most important component, ZPMC has pay a lot of attention to the wire rope, so we make this chapter.

This chapter describe the common knowledge, storage, transport method and the criteria of wire rope replacement; and the inspection, lubrication, adjustment and exchange wire rope are the key points of this chapter.

This part introduces the common knowledge of usage and maintenance. Refer to the manual provided by the supplier first. Meanwhile, codes and standards of maintenance, replacement wire rope for different countries need to be strictly obeyed.

5.1 Basic knowledge of wire rope

5.1.1 Construction

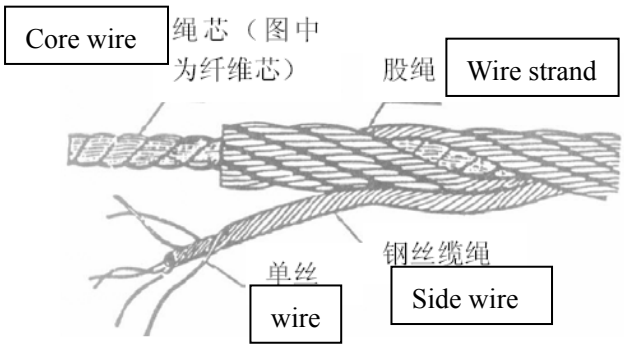
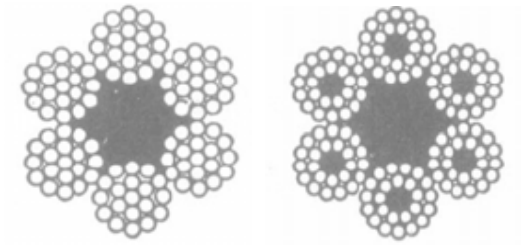


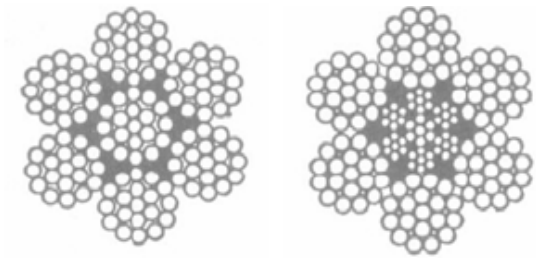
Figure 5-1：Wire rope appearance (example)

5.1.2 Core

Figure 5-2 Common core wire for RTG



(a) fiber core and fiber strand core



(c) (IWSC)　　(d) (IWRC)

图 0-1 core wire martial

Figure 5-2：Core of wire rope

5.1.3 Type of laying

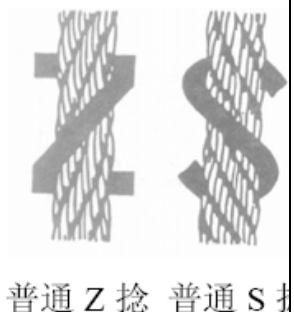
1) direction of lay

Right hand lay and left hand lay, we usually use left hand lay.

2) method of lay

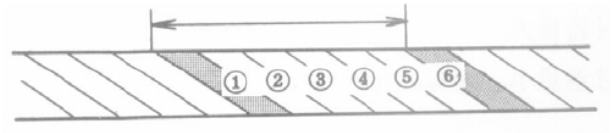
- Lang lay: side wire lay is the same direction of strand lay
- Ordinary lay: side wire lay is the opposite direction of strand lay

wire rope type and method of lay

	advantage	disadvantage
 顺 Z 捻　顺 S 捻 Right hand lang lay left hand lang lay	Wires on the surface are longer, so this type is well wear resistant, soft and flexible, well fatigue resistance.	Side wire will surfer bigger torque.
 普通 Z 捻　普通 S 捻 Ordinary right hand lay Ordinary left hand lay	This type is not easy to twist and loosening, and easy to handle.	Only shorter wires on the surface are strong, this type is not good at the wear resistant.

3) length of lay

Usually, length of one lay is 6.2~8.2 times of the wire rope diameter. The way to text the length is show as follow (use 6 strand wire rope as an example):



measure way of trlength of lay

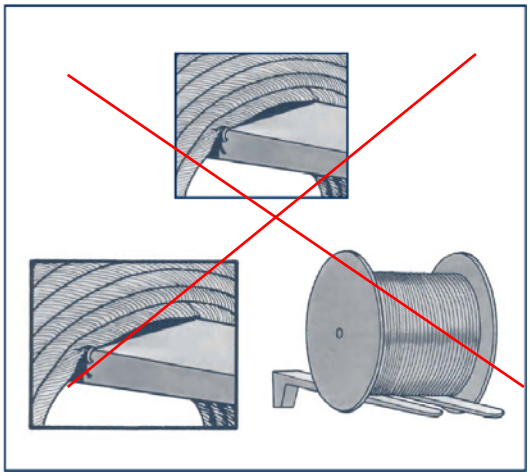
5.2 Operation

5.2.1 transportation and storage

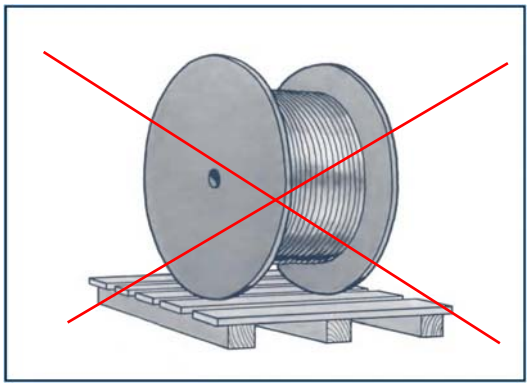
To prolong their life, the wire rope must be properly transported, stored, cleaned, inspected and installed. Wire ropes are easily affected by external damage. Therefore, they must be handled with utmost care during transport and unloading. All wire ropes should be stored clean, dry and cool and soil contact must be avoided anyway. Wire ropes must always be handled with equipment that will not damage the rope.

Reels of wire rope should be stored in an area where they will not be damaged by objects falling, chemical spills, excessive heat, open flames, welding, and excessive heat. Reels should be stored vertically on their flanges and covered to prevent dirt and debris from embedding into the wire. When storing reels, ensure that one reel will not roll into another. This will prevent one reel flange from hitting the wire rope surface of another reel. If necessary, reel flanges should be choked to prevent movement. Figure 5-4 shows the improper methods of wire rope transportation which must be avoided, and figure 5-5 shows the proper methods.

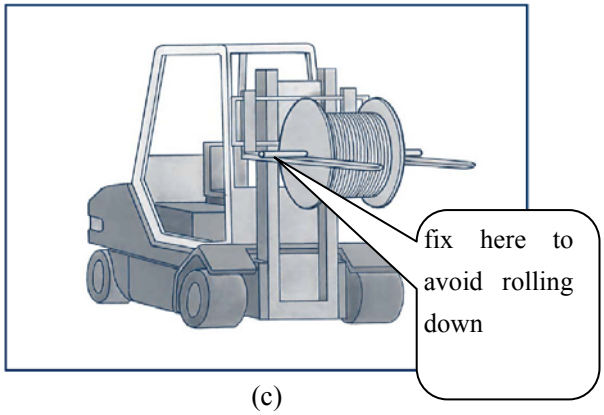
	**Do not store wire rope in atmosphere of volatile chemical or moisture. Wire rope will rust.**
	**Do not use any detergent oil to clean wire rope. Detergent oil soaks into and dissolves the lubricant inside the rope and will cause internal corrosion. Clean with a wire brush.**



(a)

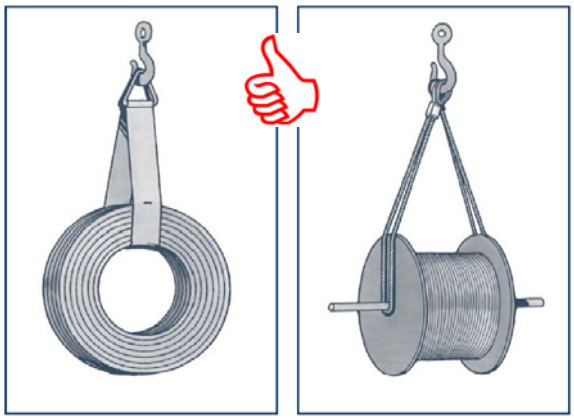


(b)

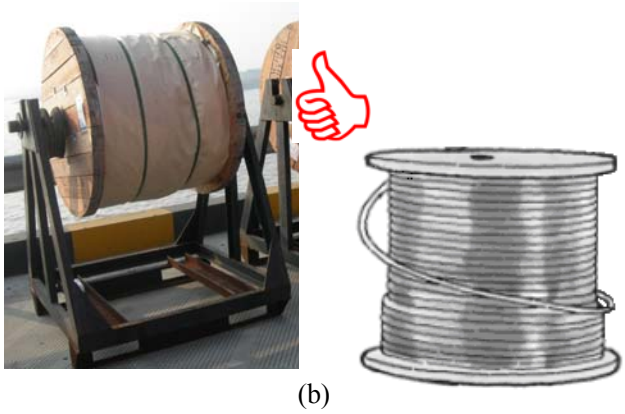


(c)

图 0-2 improper methods

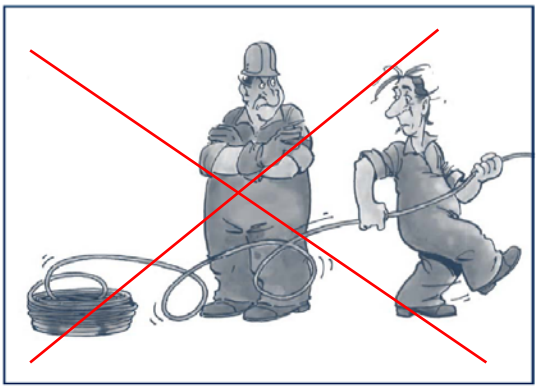


(a)



(b)

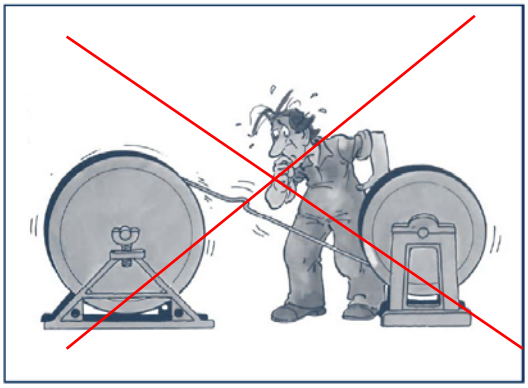
proper methods



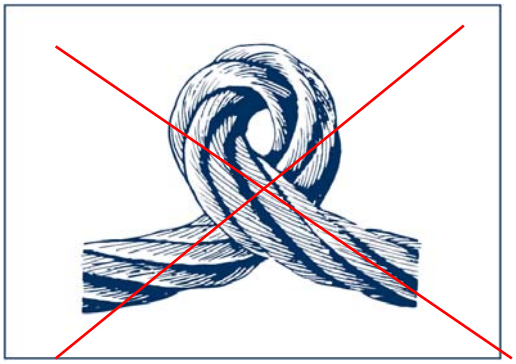
（a）



（b）



（c）

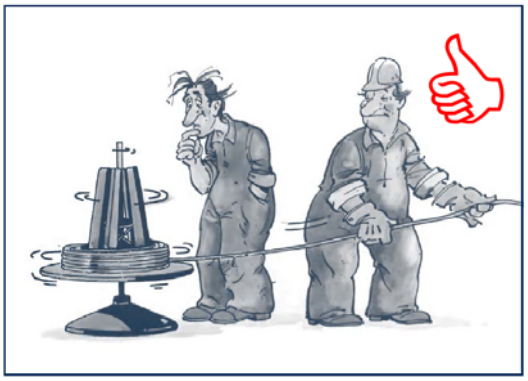


（d）

improper unwind and wind methods

5.2.2 unwind and wind

Figure 5-6 and 5-7 show the proper/improper methods of unwinding a wire rope from a reel. Considerable care must be taken when unwinding a wire rope. Their operational performance is substantially influenced by the way they are handled. Bending or twisting wire rope can cause internal damage that will adversely affect the operational life of the wire rope. Avoid unwinding wire rope on the ground to protect the wire rope from debris. If there is no other option, make sure the ground is clean or covered with boards or plastic sheets.



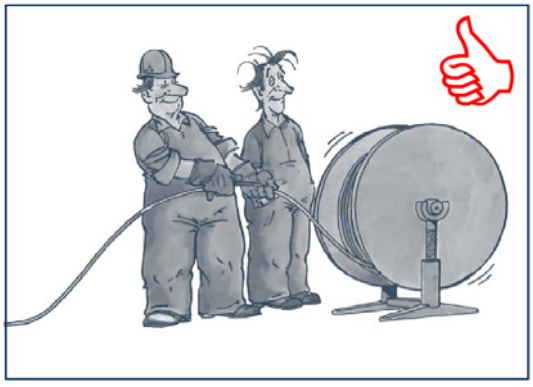
（a）



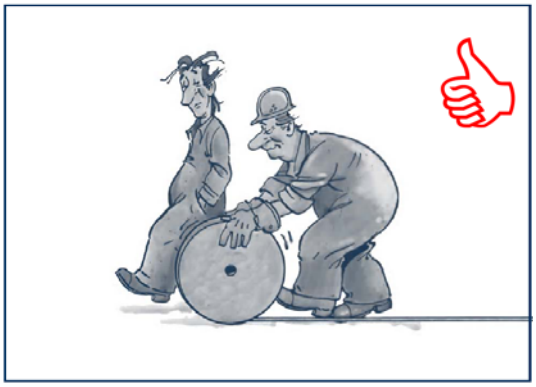
（b）



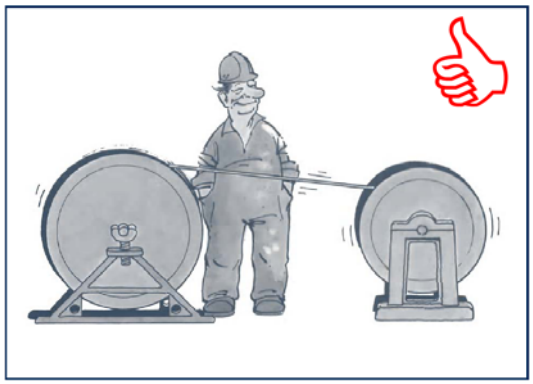
（c）



（d）



（e）



（f）

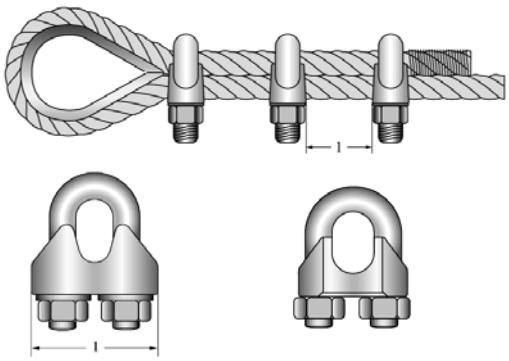
proper unwind and wind methods

Forbid to unwind the wire rope with the reel fixed, wire rope will be tangled and will cause kink or deformation.

5.3 termination of wire rope end

5.3.1 termination of wire rope end

5.3.1.1 clip termination



clamps fixation distance

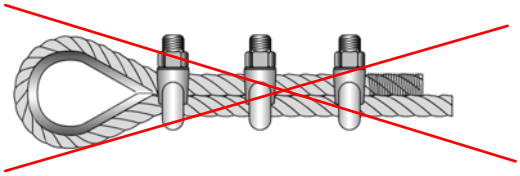
Usually clip termination method use the clips, and the loop back method uses a rope grip and the loop should be lashed to the live part of rope by a soft wire serving or tape to prevent loosening of the rope in service.

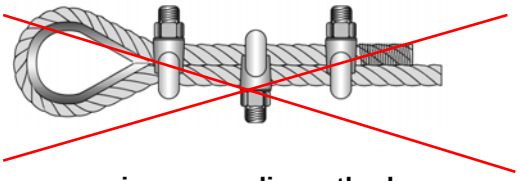
The method of looping back should not be used if there is a possibility of interference of the loop with the mechanism or structure.

Pay attention to the distance of each clips, see figure 5-8, the clips cannot be less than min quantity, and the bolts of the clips need to tighten, see table 5-2. Figure 5-9 show the proper/improper methods of using the clips, improper wire rope termination methods will lessen the wire rope end's strength, which will make the accident occur.

clips torques and clips min quantity

diameter	Nm torque	Min quantity
5	2.0	3
6.5	3.5	3
8	6.0	4
10	9.0	4
13	33	4
16	49	4
19	67.7	4
22	107	5
26	147	5
30	212	6
34	296	6
40	363	6



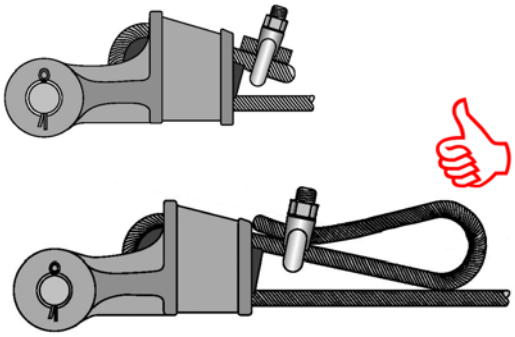


improper clip methods

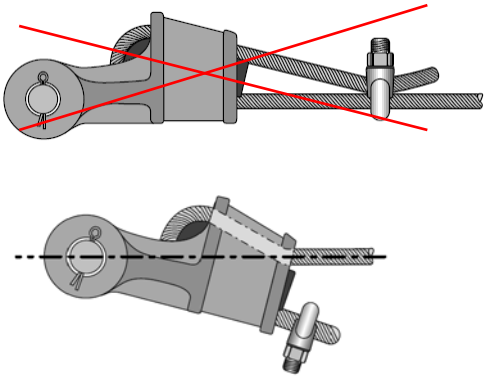
5.3.1.2 wedge termination

When terminating a rope end with a wedge socket, ensure that the rope tail cannot withdraw through the socket by securing a clamp to the tail or by following the manufacturer's instructions. The tail length should be a minimum of 20 rope diameters for all Rotation Resistant wire rope and a minimum of 6 rope diameters for 6 and 8 strand ropes.

Figure 5-10 shows two recommended methods of securing the rope tail of a wedge socket termination.



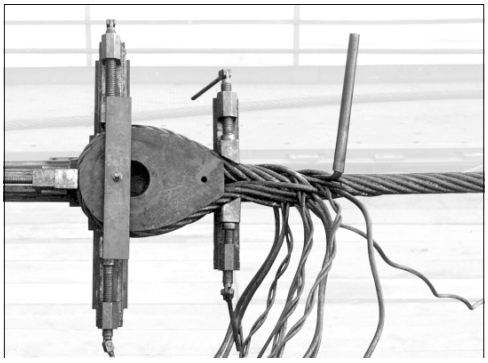
recommended methods



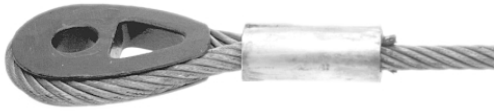
improper methods

5.3.1.3 other termination method

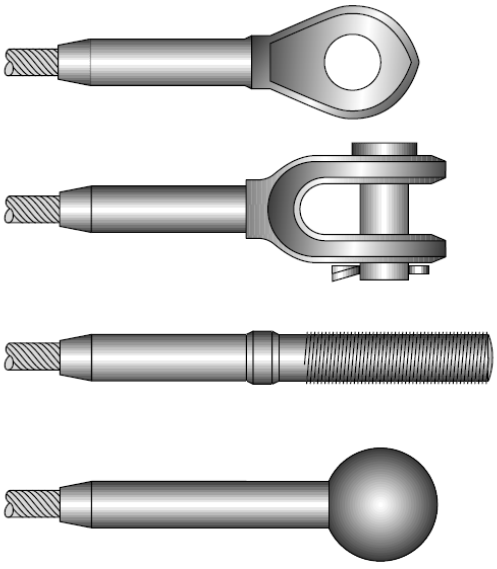
Figure 5-12 (a) shows the weave termination method, which is an old method and rarely used on RTG, and only use for some unimportant wire rope with small diameter. (b) shows the compressed metal splice termination method, which need the special equipment to finish it. (c) shows the cast termination method.



(a)



(b)



(c

other termination methods

5.3.2 wire rope cutting

Before cutting the wire rope, the two end should be lashed with small- gauge wire, which can prevent the loosening of the wire rope. After lashing, the wire rope can be cut. Usually, each cutting end with one small-gauge wire is enough, but for counter- rotation wire rope, at least two small- gauge wire should be used, and at 6 times the rope diameter apart. The procedures are as follows:

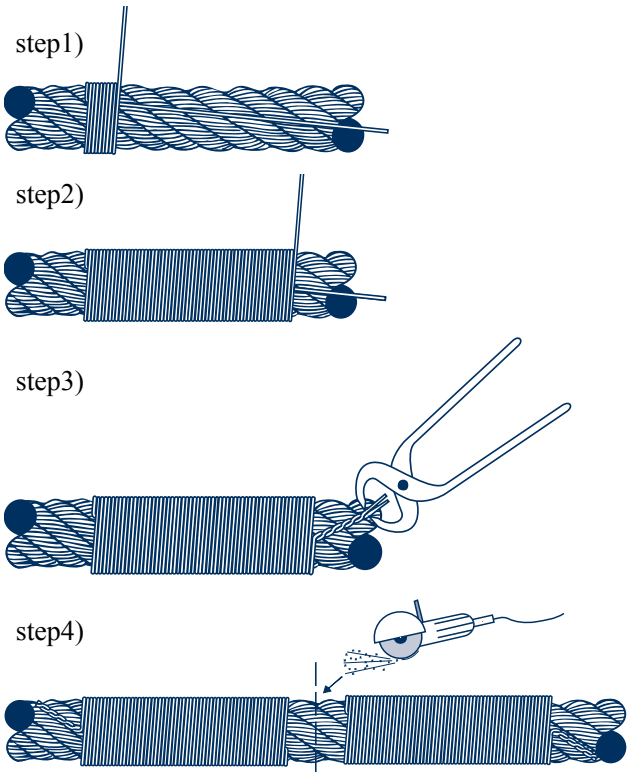
1) Reel the small- gauge wire on the wire rope, and keep reeling parallel, closely and tightly.

2) Using the pliers to make it tightly.

3) After reeling one wire strand long, using the pliers to tighten the two ends and twist to knot. Gently knock the knot end to make it cling to the wire rope.

4) After finishing one end, do the same thing at another end. And using the cutting machine to cut the wire rope after finish both.

Upon procedures refer to figure 5-13



cutting procedures

To avoid wire strand loosening, most of the wire ropes ends are lashed. Figure 5-14(a) shows the usual way of the end handing method. In some particular cases,

other methods also be taken, show as in figure 5-14 (b), (c). Weld the end of the wire strands, and polish the end to taper or crop shape. A ring can be welded to convenient traction wire rope.



（a）lashed end



（b）taper type end



(c) lashed and welded end



(d) lashed and ring end welded

wire rope end

5.4 wire rope inspection

Wire rope service life is limited and influences by many factors such as it quality, overload, environment, and the maintenance. Different damage types will occur as signs before it is useless. So regular inspection need to be taken to find these signs and hidden dangers and analysis the main reason to lead to the useless. Chang the wire rope before it is useless and eliminate the adverse factors in the future use to extend the service life.



The performances will drop quickly before it's useless. When the wire rope close it's must exchange standard, more inspection need to be taken to make sure the safety use.

5.4.1 factors influence the service life of wire rope

wire rope service life influence by many factors, include the working condition, maintenance, working environment and ect. For RTG wire rope, the main factors are listed as follows:

1) Wire rope bear variational load constantly and at the same time it need to pass through drums and sheaves, which make the wire rope buckling repeatedly, and is easy to cause the wire rope's fatigue.

2) Water, salt atmosphere make the wire rope corrosion: the moisture mixed the soluble salt, dissolved saline ions will accelerate the diffusion of the oxygen ions in the air, which result in the wire rope with the air oxidation reaction speed fold increase.

3) The impact of the embedded foreign matter in the wire rope: because terminal operations are out of doors, hostile working environment and with big wind. Some objects such as grits, sands are easy to be involved in the sheaves, drums, wire ropes and even embedded in the center of the wire ropes which will cause local stress concentration and result in abnormal wear and surface wire broken.

4) The impact of wire rope frequently operation: lubrication coat the wire as thin film. When the wire rope be tightened, some of the lubrication will be extruded out to the surface of the rope, while some will be absorbed back to interior of the rope. Repeated extrude out and absorb back will lead to the release of the interior lubrication and accelerate the oxidation of the lubrication.

there are lots of other factors that influence the wire rope service life, not only factors that involve in the design and usage, but also other abnormal factors, such as follows:

1) Multi-layers of the rope on the drum can result in severe distortion in the underlying layers.

2) Bad reeving (due to excessive fleet angles or slack winding) can result in mechanical damage, shown as severe crushing, and may cause shock loading during operation.

3) Small diameter sheaves or drums can result in severe bending stress of the rope, and will certainly lead to early wire breaks.

4) Oversize grooves offer insufficient support to the rope leading to increased localized pressure, flattening of the rope and premature wire fractures. Grooves are deemed to be oversize when the groove diameter exceeds the nominal rope diameter by more than 15%

5) Undersize grooves in sheaves will crush and deform the rope, often leading to two clear patterns of wear and associated wore breaks.

6) Excessive angle of fleet can result in severe wear of the rope due to scrubbing against adjacent laps on the drum. An excessive angle of fleet can also induce rotation causing torsional imbalance.

7) Using the wire rope of incorrect size, construction, or grade can result in wire rope breaking;

8) Dragging rope over obstacles can result in severe friction and wire rope breaking;

9) insufficiency lubrication

10) drums or sheaves have concave- convex surfaces;

11) Wire rope jumping off the sheaves can result in serious accident

12) Subjecting it to acid fumes

13) mounting error attachment can result in overload local stress at the joint, which make the rope easy to break.

14) Subjecting it to severe or continuing overload and fierce impact such as emergency stop can result in different degrees damage.

5.4.2 wire rope inspection method

The wire rope inspection must be done by the person who is trained and authorized.

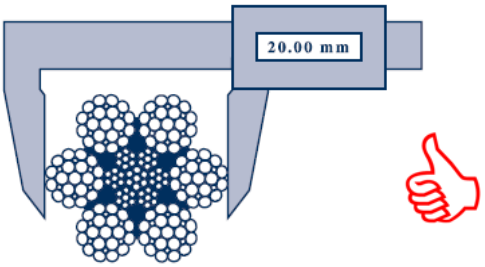
1) wire rope diameter measurement

The size of a wire rope is designated by its diameter. The diameter of a wire rope should be measured with a set of vernier calipers.

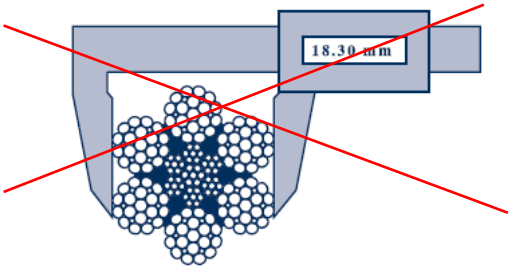
The proper method is showed in figure 5-15(a), and 5-15(b) is the improper method.

To ensure an accurate measurement of the diameter of a wire rope, always measure the rope at three places and at least 1.5m apart. Use the average of the three measurements as the diameter of the rope.

The nominal diameter of a new wire rope must be measured and recorded before it is installed on the crane. The results of every subsequent check should also be recorded in the journal for that wire rope.



（a）proper method

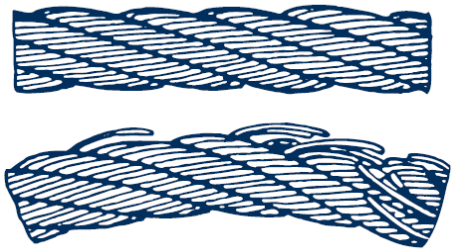


（b）improper method

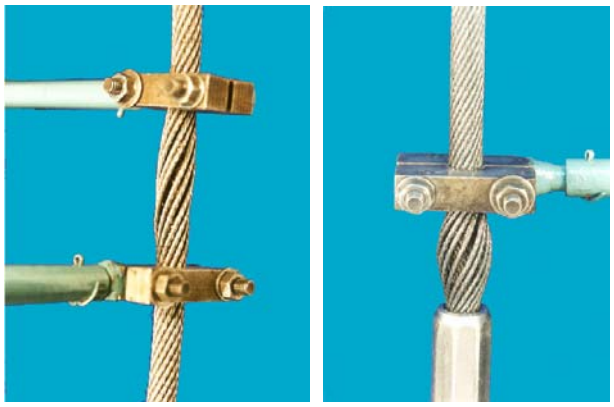
measurement

2) ocular estimate

Ocular estimate is the common method. Removing dirt can ease the flaws' observation before using this method. Bending the wire rope to find the broken wire when it is too tight, see figure 5-16. Interior flaws can be found when use the methods showed in figure 5-17. Open part of the rope to check if there is any corrosion, broken wire phenomenon inside.



bend the wire rope to check the broken wire



(a) (b)

check wire rope interior

3) hand feeling method

Hand feeling method is holding the wire rope, and makes the wire rope pass through the hand slowly to feel the flaw on the wire rope surface. But this way has dangers, which need to be carefully used. Any sharp on the surface of the wire rope make cause hurt on the hand. To avoid the hurt, can use two pieces of semicircle wood tube which has the same inner diameter to feel the heave when it passes through the half tube. When hand feels the wood tube slightly jump, then take away the hands. Another method is cover the wire rope with one layer cotton yam, not only surface dirt can be cleared, but also the cotton yam will be grappled away when any edge heave is pass through.

4) （NDT）

There are lots of instruments on the market, which use the NDT to inspect the surface and inner flaw of the wire rope. The advantage of these instruments is they can easily inspect the inner part of the wire rope and find the flaw which is not easy to find. And the growing trend can be told to the inspector through the computer analysis. But the reliability needs to be verified through the fact. Other information refers to the NDT instrument manual.

NDT using electromagnetic means may be used to detect broken wires and loss in metallic area. This method complements the visual examination but does not replace it.

5.4.3 inspection part and period

The whole length of the wire rope needs to be inspected in principle, and the easy damage parts need to pay more attention. For instance, wire rope parts which pass through the sheaves frequently, dead end, rope clamp position, roll in and roll out drum and sheaves, pass through the equalized sheaves are easily be ignored, because these parts of the wire rope are still. But the experience is that these parts have flexural fatigue for the wire rope's springing and swaying. Wire rope parts which have been covered by drum or sheave grooves should be pay more attention too.

Ocular estimate must be done before everyday work, especially the connecting part and dead end part.

Wire rope should be carefully checked within one week, after first usage, and inspected within one week, after wire break occur.

Every mouth regular inspection should be done after the first inspection.

Wire rope should be comprehensive inspected after accident which can cause the damage of the wire rope or

equipment occurs, and decreases the inspection interval after that.

Wire rope also should decrease the inspection interval before it close to the replacement period.



According to the usage situation, inspection period should be shortened when necessary.

5.4.4 Inspection content

To find the wire rope wear and break as early as possible, schedule should be made according to it actual situation. A whole inspection procedure should include the follows:

1) Inspect wire rope diameter to see if it is decreasing, and how about the decrease rate.
2) If there is any corrosion, and how about the lubrication
3) The exterior and interior wear situation.
4) If there is any visible break, and the property of the break. Wire rope must be changed if the break reaches specified degree.
5) If there is any phenomenon of crushing, flattening, bird cage flattening, kinking, or other distortion, kinking wire rope and other kinds of severe deformation wire rope cannot be repaired and used.
6) Inspect the components which are connected or contacted with the wire rope, broken joint must be changed immediately; the surface situation of the sheave and drum grooves, if they are reach the changeable degree.

The result of the inspection should be recorded in the work diary or special log.



The inspection result should be recorded for following.

Refer to ISO 4309:2004(E) Annex B which lists the recommend format for single inspection and working record. The working record can be used to predict the wire rope lifetime and do more exact inspection period, it also can be used to analyze and take measures which can increase the lifetime of the wire rope.

5.4.5 Discard criteria

Wire rope's safety usage is qualified by the following criteria:

1) Broken wires: property and quantity, wire break at the joint position, concentrate wire break and the increase rate of the broken wires. Usually wire break is the main damage form in RTG hoist, trolley, tray trolley.
2) Wire strand breakage
3) Reduced wire rope diameter, and the wire core's degeneration.
4) Weakening of the elasticity
5) Exterior and interior wear
6) Distortion: waviness, cage flatten, wire core or wire strand protrusion or distortion, wire protrusion, local increase in wire rope diameter, flattening, kinking, bending.
7) Damage cause by electric arc or heating
8) Rate of permanency elongation

Not only consider the each aspect impacts individually, but also consider the comprehensive impacts to judge if the wire rope reaches the discard criteria.

The discard criteria can be found in GB/T5972-2006, ISO 4309:2004（E），DIN15020, DIN3088 and etc. The local criteria shall be taken priority of those criteria. Due to a variety of factors, there is no precise rule deciding when the wire rope must be discarded. According to the judgment of qualified personnel, it may be allowed to use to the end of the shift even if the wire rope meet one of the discard criteria. The wire rope should be discarded after the shift, at the end of the day or before the

beginning of next shift.

The content of discard criteria of GB/T5972-2006 and ISO 4309:2004（E） is as below.

5.4.5.1 Nature and number of broken wires

In the case of 6- and 8-strand ropes, broken wires usually occur at the external surface. In the case of rotation-resistant ropes, there is a probability that the majority of broken wires will occur internally and are "non-visible" fractures. Detail criteria are given in **错误！未找到引用源。**. While the number of broken wires is not less than that in the table, the wire rope must be discarded. Picture 1 and Pictuer 2 in Table 5-4 are two typical broken wires.

5.4.5.2 Broken wires at termination

If there are broken wires at, or adjacent to the termination, the cause of this deterioration shall be investigated and, where possible, the termination shall be remade, shortening the rope if sufficient length remains for further use, otherwise the rope shall be discarded.

5.4.5.3 Localized grouping of broken wires

Where broken wires are very close together, constituting a localized grouping of such breaks, the rope shall be discarded. If the grouping of such breaks occurs in a length less than 6*d* or is concentrated in any one strand, it may be necessary to discard the rope even if the number of wire breaks is smaller than the maximum number shown in **错误！未找到引用源。**. Pictures 3 in **错误！未找到引用源。** show the typical localized grouping of broken wires. They are usually the main damages to the main hoist and anti-shaking device of RTG.

5.4.5.4 Rate of increase of broken wires

In applications where the predominant cause of rope deterioration is fatigue, broken wires will appear after a certain period of use, and the number of breaks will progressively increase over time. In these cases, it is recommended that careful periodic examination and recording of the number of broken wires be undertaken, with a view to establishing the rate of increase in the number of breaks. This premise may be used to propose a future date for rope discard.

5.4.5.5 Fracture of strands

If a complete strand fracture occurs, the rope shall be immediately discarded.

5.4.5.6 Reduction of rope diameter resulting from core deterioration

Reduction of rope diameter resulting from deterioration of the core can be caused by:

1) Internal wear and reduction of wire diameter .
2) Internal wear caused by friction between individual strands and wires in the rope.
3) Deterioration of a fiber core;
4) Fracture of a steel core.
5) Fracture of internal layers in a rotation-resistant rope.

If these factors cause the actual rope diameter to decrease by 3 % of the nominal rope diameter for rotation resistant ropes, or by 10 % for other ropes, the rope must be discarded.

5.4.5.7 Decreased elasticity

Under certain circumstances elasticity decreasing is usually associated with the working environment, a rope can sustain a substantial decrease in elasticity and is thus unsafe for further use.

Decreased elasticity is difficult to detect. However, it is usually associated with the following:

1) Reduction in rope diameter but different from which caused by wear.

2) Elongation of the rope lay length
3) Lack of clearance between individual wires and between strands, caused by the compression of the component parts against each other.
4) Increased stiffness

If any above phenomenon are observed or examiner has any doubt, advice shall be obtained from a specialist. Decreased elasticity of a rope can lead to abrupt failure under dynamic loading and is sufficient justification for immediate discard.

5.4.5.8 External wear

See Picture 13 in **错误！未找到引用源。** for the reference. Wear is promoted by lack of lubrication, or incorrect lubrication, and also by the presence of dust and grit. Wear reduces the strength of ropes by reducing the cross-sectional area of the steel strands. If, due to external wear, the actual rope diameter has decreased by 7 % or more of the nominal rope diameter, the rope shall be discarded even if no wire breaks are visible.

5.4.5.9 External and internal corrosion

Corrosion occurs particularly in marine and polluted industrial atmospheres. It will diminish the breaking strength of the rope by reducing the metallic cross-sectional area, and it will accelerate fatigue by causing surface irregularities which lead to stress cracking. Severe corrosion can cause decreased elasticity of the rope.

Corrosion of the outer wires can often be detected visually. See Picture 14 in **错误！未找到引用源。**.

See Picture 14 in **错误！未找到引用源。** for the internal corrosion. It is more difficult to detect than the external corrosion which frequently accompanies it, but the following indications can be recognized:

1) Variation in rope diameter; in locations where the rope bends around sheaves, a reduction in diameter usually occurs. However, in stationary ropes it is uncommon for variation in diameter to occur.
2) Loss of clearance between the strands in the outer layer of the rope, frequently combined with wire breaks between or within the strands.

If there is any indication of internal corrosion, the rope should be subjected to internal examination as indicated in Clause 5.4.2. Confirmation of severe internal corrosion is justification for immediate rope discard.

5.4.5.10 Waviness

See picture 11 in **错误！未找到引用源。** for reference. In the case of waviness (see Figure 5-18), the wire rope shall be discarded if, under any load condition on a straight portion of rope that does not bend around a sheave or drum, the following condition is met: d1>4d/3 or d1>1.1d.

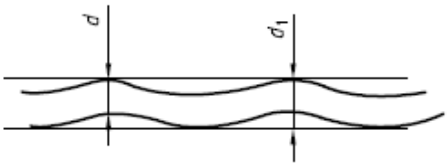


Figure 0-3 Waviness

5.4.5.11 Basket or lantern deformation

See picture 12 in **错误！未找到引用源。**. Basket or lantern deformation, is a result of a difference in length between the rope core and the outer layer of strands. Different mechanisms such as excessive fleet angle that cause the rope touch the flange or the rope running over a “tight sheave”. The rope shall be discarded if the basket or lantern deformation occurs.

5.4.5.12 Core or strand protrusion/distortion

See picture 5, 7 in **错误！未找到引用源。**. Rope with core or strand protrusion/distortion shall be immediately changed.

5.4.5.13 Wire protrusion

See picture 4 in **错误！未找到引用源。**. In wire

protrusion, certain wires or groups of wires rise up, on the side of the rope opposite to the sheave groove, in the form of loops. Rope with wire protrusion shall be immediately discarded.

5.4.5.14 Local increase in diameter of rope

See picture 17 in **错误！未找到引用源。**. This condition usually relates to a deformation of the core such as in particular environments, a fiber core can swell up owing to the effect of moisture. If this condition causes the actual rope diameter to increase by 5 % or more, the rope shall be immediately discarded.

5.4.5.15 Flattened portions

See picture 19 in **错误！未找到引用源。**. Flattened portions of rope which pass through a sheave will quickly deteriorate and may damage the sheave. In such cases the rope shall be discarded immediately. Flattened portions of rope in standing rigging can be exposed to accelerated corrosion, and shall be subject to inspection at a prescribed shortened frequency if retained in service.

5.4.5.16 Kinks or tightened loops

See picture 18 in **错误！未找到引用源。**. Kink or tightened will cause imbalance of lay length and reduction in strength which will cause excessive wear, and in severe cases the rope will be so distorted that it will have only a small proportion of its strength remaining. Rope with a kink or tightened loop shall be immediately discarded.

5.4.5.17 Bending

Bends are angular deformations of the rope caused by external influence. Rope with a severe bend will suffer similarly to flattened portions of rope and should be treated as in Clause **错误！未找到引用源。**.

5.4.5.18 Damage due to heat or electric arcing

Ropes that have been subjected to heat or electric arcing effects, externally recognized by the colours produced in the rope, shall be immediately discarded.

Table 0-1 Number, reached or exceeded, of visible broken wires in single-layer and parallel-closed Ropes working in steel sheaves, signaling mandatory discard of rope[1]

Rope category Number RCN[2]	Number of load –bearing wires in all outer strands of the rope[3] (n)	Number of visible broken wires[4]			
		Mechanism group of M1~M4		Mechanism group of M5~M8	
		in rope over a length of [5]		in rope over a length of	
		6d	30d	6d	30d
01	n ≤ 50	2	4	4	8
02	51≤n ≤ 75	3	6	6	12
03	76 ≤ n ≤ 100	4	8	8	8
04	101 ≤ n ≤ 120	5	10	10	20
05	121 ≤ n ≤ 140	6	11	11	22
06	141 ≤ n ≤ 160	6	13	13	26
07	161 ≤ n ≤ 180	7	14	14	29
08	181 ≤ n ≤ 200	8	16	16	32
09	201 ≤ n ≤ 220	9	18	18	36
10	221 ≤ n ≤ 240	10	19	19	38
11	241 ≤ n ≤ 260	10	21	21	42
12	261 ≤ n ≤ 280	11	22	22	44
13	281 ≤ n ≤ 300	12	24	24	48
21	4-strand ropes, n ≤100	2	4	4	8
22	3 or 4- strand ropes，n ⩾100	2	4	4	8
23	At least 10 outer strands	2	4	4	8

1: Above values usually apply to ropes working in sheaves that are made of steel with multi-layer coiling. They do not apply to ropes working in sheaves that are made of nylon with single-layer coiling

2: Detail rope category number RCN. Where:

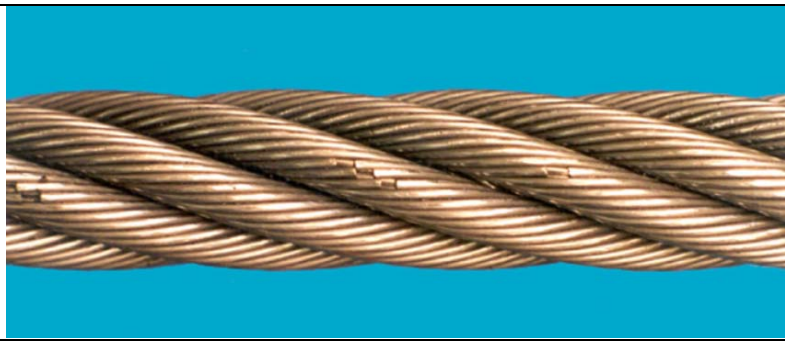
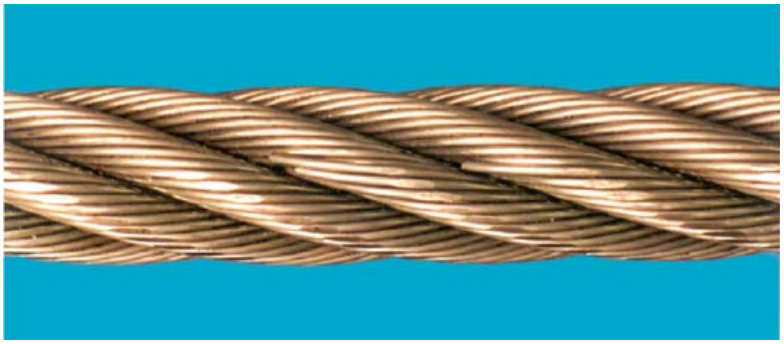
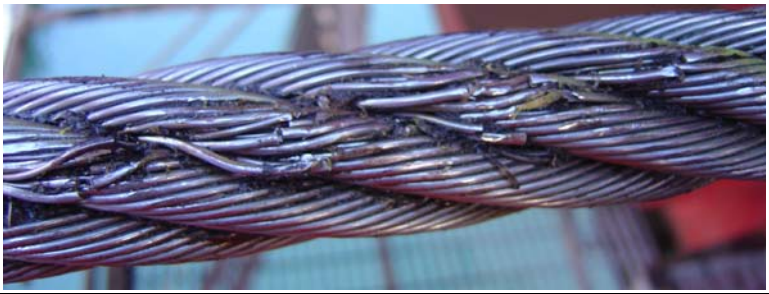
05-Including rope of 6XFi(29)-IWRC, 07-Including rope of 6X36WS-IWRC usually used for main hoist. 23-Rotation-resistant rope usually used for maintenance vehicle.

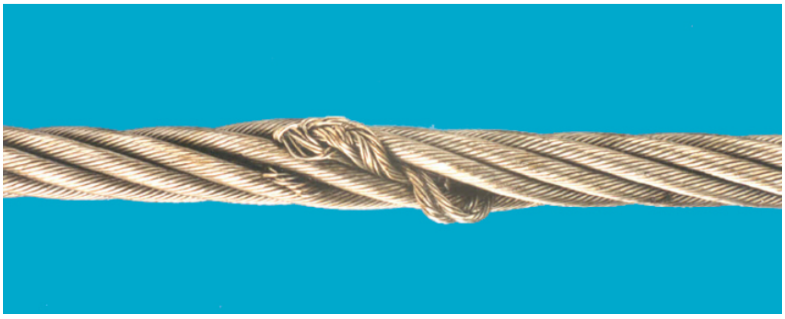
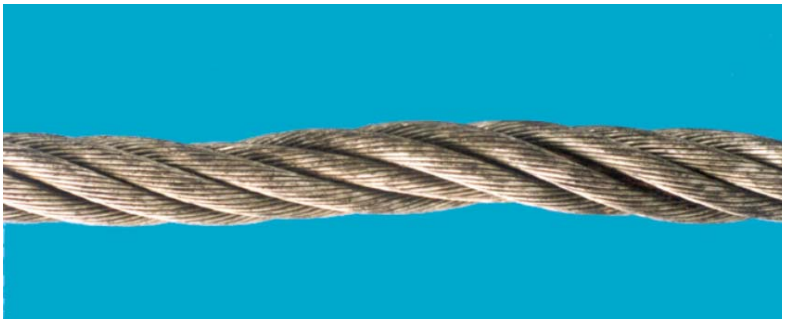
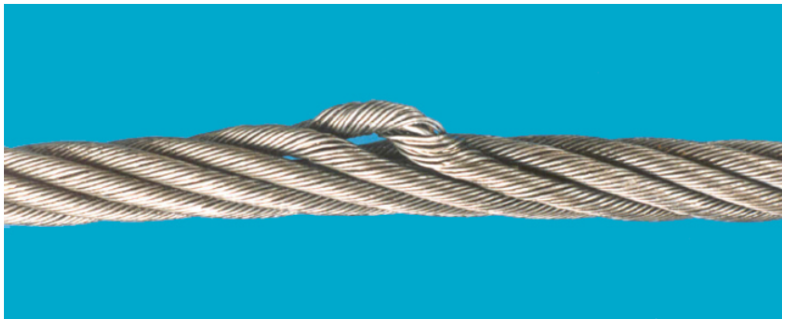
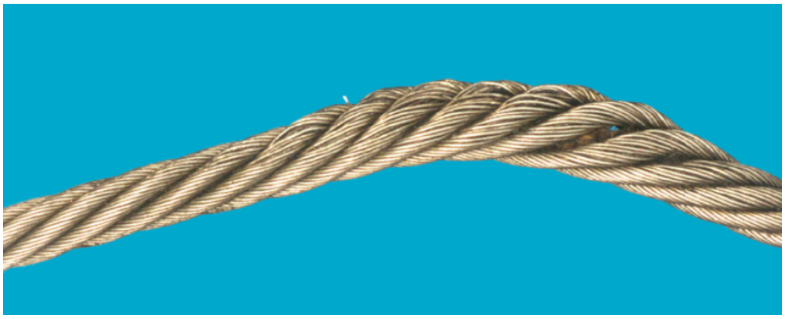
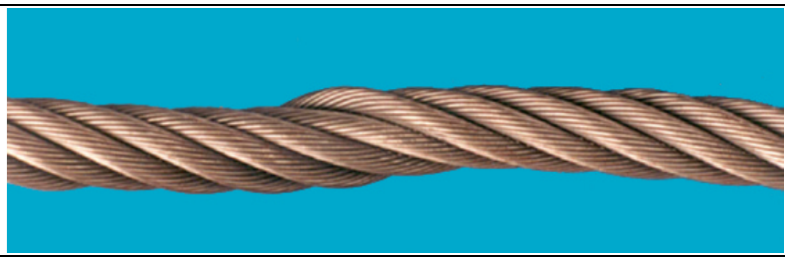
3: Filler wires are not regarded as load-bearing wires and are not included in the value of n. In ropes having a number of layers of strands, only the visible outer layer is considered in the value of n. In ropes having a steel core, this is regarded as an internal strand and is not considered in the value of n.

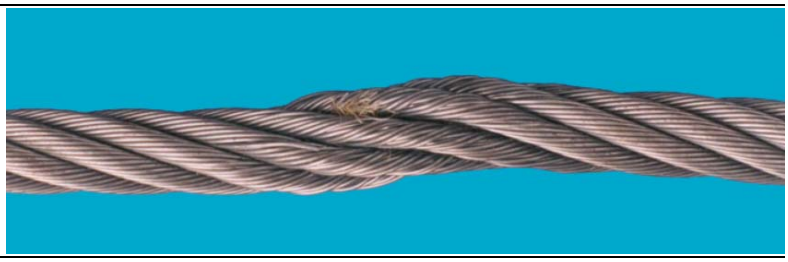
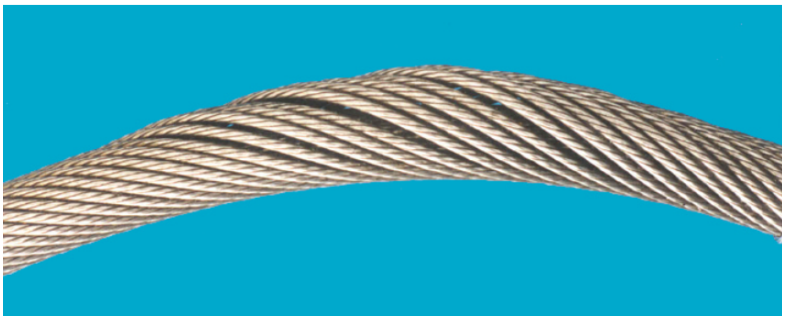
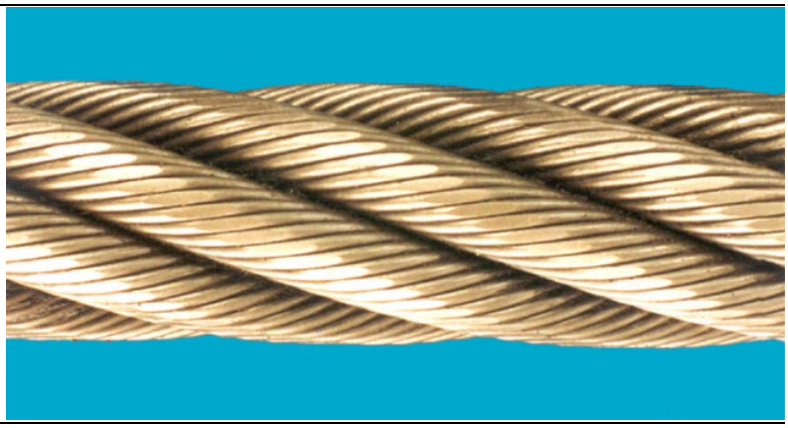
4: A broken wire may have two ends.

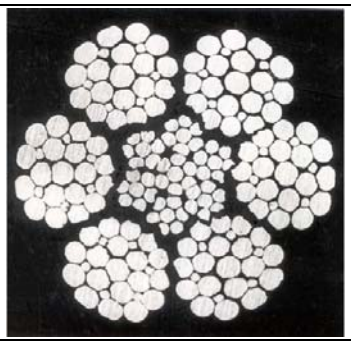
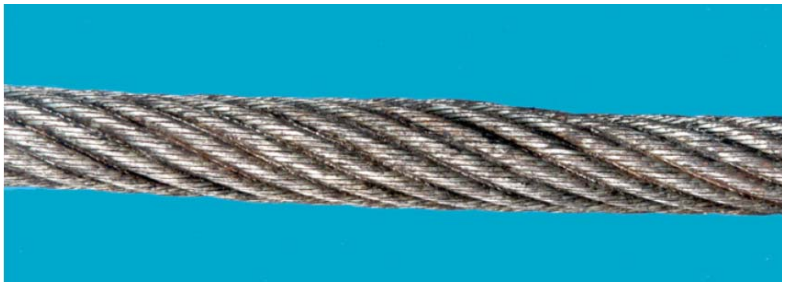
5: d=nominal diameter of rope.

Table 0-2 Examples of Damaged Wire Rope

No	Description	Picture
1.	[Crown wire breaks] Usually caused by fatigue	
2.	[Valley wire breaks]	
3.	[Localized grouping of broken wires]	 
4.	[Wire protrusion]	

No	Description	Picture
5.	[Core protrusion-single-layer rope]	
6.	[Local reduction in rope diameter-sunken strand]	
7.	[Strand protrusion/distortion]	
8.	[Flattened portion]	
9.	[Kink(Positive)]	

No	Description	Picture
10.	[Kink(Negative)]	
11.	[Waviness]	
12.	[Basket deformation] Caused by torque imbalance	
13.	[External wear]	
14.	[External corrosion]	

No	Description	Picture
15.	Internal corrosion	
16.	[Protrusion of inner rope of rotation-resistant rope]	
17.	[Local increase in rope diameter due to core distortion]	
18.	[Kink]	

No	Description	Picture
19.	[Flattened Portion]	

In case of wire rope abnormalities, the maker of the wire rope should be determined. A colored tracer thread is embedded in the core of the rope. The manufacturer can be found by this tracer thread.

Use the right hand lay wire rope both on right hand and left hand drum



Left hand lay



Right hand lay

5.4.6 Procedure of wire rope inspection

5.4.6.1 Inspection procedure of main hoist rope

Table 0-3 Manpower and tools for the inspection of main hoist wire rope

Description		Quantity	Notes
Manpower	Operator	1	With radio
	Examiner	1	With radio
	Supervisor	1	With radio, must care examiner at any time
Tools	Duster cloth	Some	For cleaning
	Vernier caliper	1	For measurement
	Screwdriver	1	Forchecking internal of rope
	Chalk/tape	1	For marking
	Inspecting form	1	For recording
	Pen	1	
	Others		Other special tool for rope inspection.

The homologous number of personnel and tool for the inspection are given in 错误！未找到引用源。. The RTG shall be switch to maintenance mode before inspection. Taking “change rope” mode or using the emergency drive to move wire rope.

Examination may be according to following steps:

1) Move the trolley to the parking position. Hoist and lower the spreader at slow speed for several times. Make the spreader as close to the ground surface as possible when lowering it. Examiner can stay on both sides of the main hoist drum to check the wire ropes. This step aims to check the wire ropes wrapped on the drum. Pay close attention to the moving drum while inspecting. When the spreader is lowered to the ground surface, rings on the drum shall be checked.
2) Examiner can stand on the platform outside of sheave. Hoist and lower the spreader at slow speed for several times. Make the spreader as high as possible when hoisting it. This step aims to check the wire ropes of main working parts. Pay close attention to the moving sheave while inspecting.
3) Lowering the spreader as low as possible, examiner can stand at the headblock to check the wire ropes nearby.
4) The problematic area should be checked carefully.
5) In 4-rope hoisting system, examiner should stand on the platform around the trolley,

checking the end of wise ropes on both sides of the rocking beam.

6) In 4-rope hoisting system, examiner should stand on the maintenance platform under the trolley, checking the end of wise ropes under the trolley frame.

Once the above-mentioned procedure is completed, the full length of main hoist wire rope has been inspected.

	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**
	**The RTG must be in maintenance mode for this task. If hands have to be on rope for inspection, they must always be away from the pinch point.。 An additional person must act as a spotter between the operator and the mechanic(s) inspecting the rope. All safety rules listed in this manual applies to the maintenance.**

5.5 Wire rope lubrication

	**It has been proven that a well lubricated wire rope will last four times as many working hours as a non-lubricated wire rope.**
	**Always use wire rope manufacturer's recommended lubricant!**

5.5.1 Importance of lubrication

The importance of periodic lubrication will never be overstressed. The effects are as follows:

1) Each time a rope bends or straightens, the wires in the strands of the rope must slide on each other. Thus, a film of lubrication should be present on each moving part. When exposed to the same test condition, a well-greased rope showed four times as many working cycles as a non-greased rope.

2) A second important reason for lubricating the wire rope is that it keeps away moisture which can cause interior corrosion under simultaneous bacterial proliferation in the fiber core. The lubricant also prevents deterioration of the fiber core. A rusty rope is a liability, since there is no means of visual inspection, which can determine the remaining strength of a corroded rope.

5.5.2 Selection and requirement of lubricant

The main lubricants for wire ropes used in a container terminal are as below:

1) Grease: Mainly is lithium soap base grease. Sometimes are mixed with mineral oil for better penetrability.

2) Special lubricant for rope such as : ESSO SK(semi-liquid), FUCHS Ropelife (Grease), Yizhi601(Oil).

3) Special synthetic base oil for rope such as: Ultrachem Chemlube 812(synthetic ester base

oil), synthetic ester base oil supplied by PAO.

The grease that is one kind of protection of coving to rope may be most wildly used for the lubrication of the rope traditionally because of convenience for using, economical efficiency and efficiency also good effectiveness. Compared to the lubrication oil that is liquid and has better penetrability, the grease are more easily to be tainted with foreign grains in larger size such as dust, slag or grits. Those hard grains may also be pushed inside the wire rope and cause direct damages to rope. On the other hand, Oil lubrication also has better performance in resist the moisture, salt spray for the interior of rope than grease lubrication. Additionally grease lubrication is also making the visual inspection more difficult.

Below are some main requirements to the lubricant of the wire rope used in RTG:

1) Strong antioxidant capacity.

2) The capacity to insulate the wires from moisture and salinity. That means the lubrication shall have a strong capacity of adsorbing and binding with the wires.

Usually ester oil has a better performance in Iron-binding capacity, permeability, film strength and water resistant. The user may determine the lubrication according to the actual condition of different terminal.

5.5.3 Lubrication methods of wire rope

Lubricant may be applied with an oil can, brush or any effective method. The objective is to apply a uniform coating to the entire length of the wire rope. Sometimes multipoint lubrication program should be used to make sure all the part of the wire rope can be lubricated thoroughly. Wire rope lubrication include manual lubrication and auto lubrication, auto lubrication is not frequently used on the RTG, and the manual lubrication methods are showed in figure 5-19.

Frequent and not too excessive lubrication of the wire ropes produces better results than only occasional but rich lubrication. As soon as the grease film vanishes in some areas of the wire rope, it has to be re-greased.

Greasing has to be done only when the wire rope is completely dry. Otherwise, the existing moisture will be contained the rope and the corrosion be increased from the inside. Wire rope should be cleaned with a wire brush to remove any accumulation of dirt, grit, rust, or other foreign matter. The purpose is to remove all old lubricant and foreign matter from the valleys between the strands and from the spaces between the outer wires. This gives newly applied lubricant ready penetrate into the rope.

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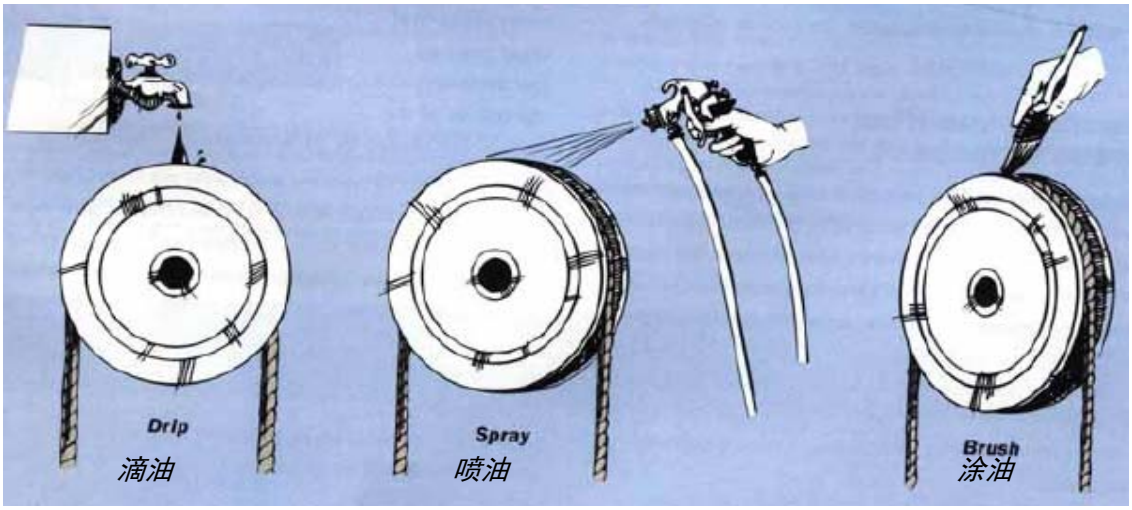


Figure 0-4 Manual lubrication methods

5.5.4 Lubrication of main hoist wire ropes

During the whole procedure, the RTG must be in maintenance mode.

The necessary manpower and tools are shown in 错误！未找到引用源。.

Table 0-4 Manpower and tools for the lubrication of wire rope

Description		Quantity	Notes
Manpower	Operator	1	With radio
	Examiner	1～2	With radio
	Supervisor	1	With radio, must care examiner at any time
Tools	Duster cloth	Some	For cleaning
	Vernier caliper	1	For measurement
	Chalk/tape	1	For marking
	Note book	1	
	Pen	1	
	Lubrication tool	1	Provided by dock

1) Ensure that the RTG is in Maintenance Mode.

2) Move the trolley to the parking position.

3) Select a proper position to start the lubricating such as drum. Most parts of the rope can be lubricated by rotating the drum slowly. There are always some dead ropes that could not reach by rotating the drum. You have to manually lubricate it. Anyway, the full length of the rope shall be lubricated. During the procedure, the drum and sheave should be rotated slowly. The whole procedure is similar to the inspection of wire rope.

5.6 Rope re-reeving

The ropes reach the discard criteria shall be replaced by new one. The manpower and tools needed for the rope re-reeving are given in 错误！未找到引用源。. The wire rope type and size are given in 错误！未找到引用源。. Reference drawings are also given in 错误！未找到引用源。.

Table 0-5 Manpower and tools needed

Description		Quantity	Notes
Man power	Mechanics	3	With radios. At least one person is qualified as an operator.
	Supervisor	1	With radio
Tools	Duster cloth	Some	For cleaning
	Chisel	1	
	Torque wrench	1	
	Wrench	As required	M16,M20,M24,M30
	Hemp rope	Some	According to the length of rope re-reeving. Dia: ~20mm
	1 ～ 2t chain block	2	
	Mat board	1	Circular, Dia: 80mm
	Rubbers, wood plate, wear-resistant block, etc.	Some	

	Steel wire、Iron plate, etc.	Some	Temporarily fixing rubbers, wood plate, wear-resistant block and binding the end of rope
	Grinder or rope cutter	1	Cut wire rope
	Pincers	1	To colligate rope end
	Rope connection device	1	
	Electric welding machine	1	Welding temporary base or wire rope
	Chalk/ punched tape	1	For marking
	Note book and pen	1	For recording
	Drawings and instructions of re-reeving	1 set	
	Forklift or Ladder truck	1	

Table 0-6 Rope type and length

Description	Notes
Main hoist rope	Type: DSR IWRC 6xFi(29), Grade: 1770N/mm^2， right lay， multi-layer coiling. Dia：26mm Min Break load：434KN Length：64+62+65+68＝259m, total 1/reach RTG

Table 0-7 Reference drawings

Drawing No	Description
JR2200604	Spreader rotation system
JR2200603	Hoisting system

	**Barricade the area under the RTG where falling parts or rope may harm person and vehicle. Don't allow the person other than concerned to enter the procedure area.**
	**Keep away from suspended loads!**
	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**

5.6.1 Procedure of main hoist rope re-reeving in 8-rope reeving system

The 8-rope reeving system is shown in Figure 5-36.

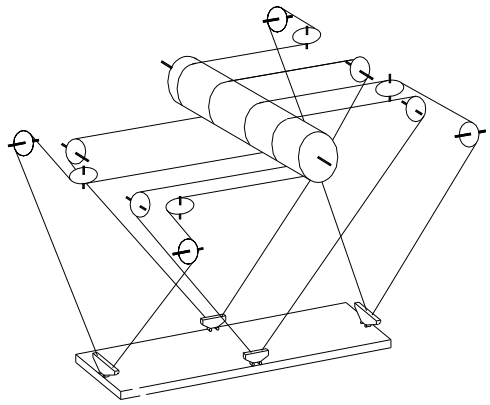


Figure 0-36 8-rope reeving system

Procedure of rope re-reeving is as follows:

1) First of all, prepare some new wire ropes according to the length and quantity in the drawings. One end of the pressed joints should be done if pressed joints are used. Prepare a few hemp ropes whose diameter is about 20mm and steel wire for binding.

1) Move the trolley to the parking position.

2) Ensure that the RTG is in Maintenance Mode. All other operations affecting the safety are banned.

3) Rotate the spreader to zero position. If having tilting or translation function, the spreader also needs to return to zero position.

4) Lower the spreader to the ground and slacken the main hoist ropes until they are on the spreader headblock. Continue to slacken the ropes until about 1 loop to the securing bolts of rope clamps. After the ropes relax, close the operation system and diesel generator sets.

5) Tie the 1 or 2~4 old ropes temporarily with hemp ropes or chain hoist (To prevent the ropes slipping off).

6) Loosen the securing bolts of rope clamps on drum and take off the rope clamps. Then take off the rope from the drum. Bind the hemp rope with the end of the rope with mild steel wire. 3~4 parts need to be bound, each part should be bound 10 loops and about 15cm apart. Connect the old rope end with the hemp rope by rope connecting device (Refer to Figure 5-21).

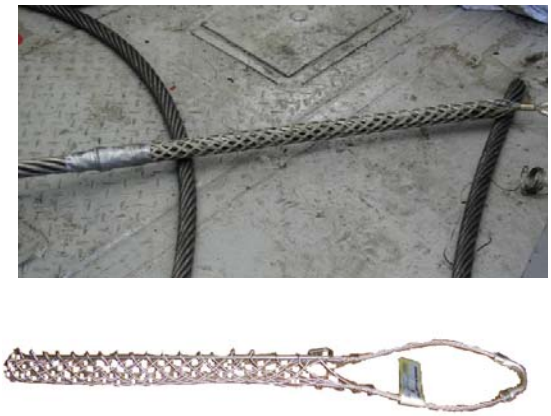


Figure 0-21 Rope connecting device

7) Tighten the hemp rope, lower the old rope to the ground slowly. Pay attention to observe the speed of the rope and whether the rope wears or hooks the structure. And whether the rope runs out of the sheave should be observed. On the headblock, if the rope wears the structure, rubbers, wood plates, and wear-resistant blocks should be fixed.

8) Bind the hemp rope with the end of the old rope, take off the pin connecting the end of the rope (Usually called pressed joints) and the adjusting screw rod according to Figure 5-37. When taking off the pin, the end of the rope should be pulled (To prevent the rope end touching the headblock
paint).

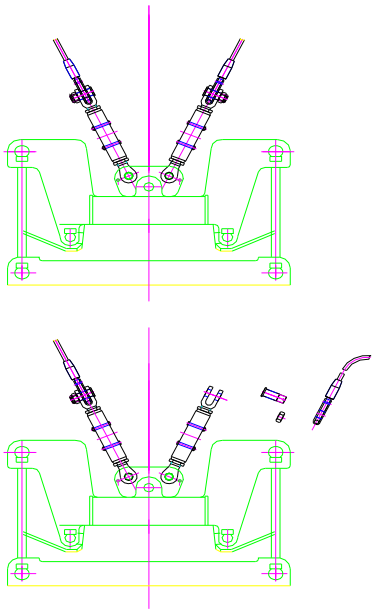


Figure 0-37 Taking off the pressed joints from the adjusting screw rod

9) Connect the end of the new rope with the turnbuckle by the pin.

10) Pull the new rope end to the main hoisting drum and winding the wire ropes onto the drum by 3 loops (Note the loop number should be consistent to which left on the drum). Then fasten that ropes with bolts primarily.

11) Replace the whole 8 ropes according to the above methods in turn.

12) Adjust the lengths of the ropes preliminarily.

13) Hoist the spreader slowly until the spreader is 200~300mm to the ground. Check the condition of the rope and the headblock. Loose the clamp bolts and tighten the rope. Cut the remaining rope and tighten all the bolts to make sure the rope suspension basically consistent.

14) When the spreader hoist up to max, check if there are about 0.5~1 empty rope grooves on the drum. Or the ropes need to be adjusted.

15) The suspended lengths of the 8 ropes must be the same in 8-rope hoisting system. So the ropes must be adjusted after replaced.

16) Cut the remaining rope and tighten all the bolts according to **错误！未找到引用源。**10. Due to the rope is elastic, the tightening torque of the bolts is less than the normal value.

17) Run the main hoist up and down slowly and check if the control system is working properly such as: the position of Cam. Correct every abnormal phenomenon immediately.

18) Lubricate the ropes. The procedure can refer to the steps listed above.

Table 5-10 Tighten torque of bolts

Bolts	Tighten torque
M24	430N m
M30	750 N.m

During the procedure of rope re-reeving, one examiner is needed on the trolley and one is needed on the ground to observe the operators and the ropes.

5.6.2 Re-adjusting rope lengths

During reeving of the wire ropes, the lengths of the 8 ropes must be the same. If not, the stress of the 8 ropes will vary greatly and the spreader slewing position will be not correct. Re-adjusting rope lengths in 8-rope system is a complicated problem and needs to study carefully. The manpower and tools needed for the rope adjustment are given in Table 5-11.

Table 0-8 **Manpower and tools for the rope adjustment**

Description		Quantity	Notes
Man power	Mechanical	3	With radios and one act as operator
	Supervisor	1	With radio
Tools	Wrench	1	M12
	Pliers	1	
	Chalk/ Punched tape	1	For marking

The procedure of the 8-rope adjustment is as follows:

1) Ensure that the RTG is in Maintenance Mode. All other operations affecting the safety are banned.

2) Rotate the spreader to zero position. If having tilting or translation function, the spreader also needs to return to zero position.

2) Move the trolley to the parking position. Lower the spreader until it is 200~500mm to the ground to make sure it is convenient to observe.

3) Observe whether the rotating beam is in a level. If not, adjust it by the adjusting screw. If the rotating beam is in a level, the spreader has been adjusted. Refer to Figure 5-38.

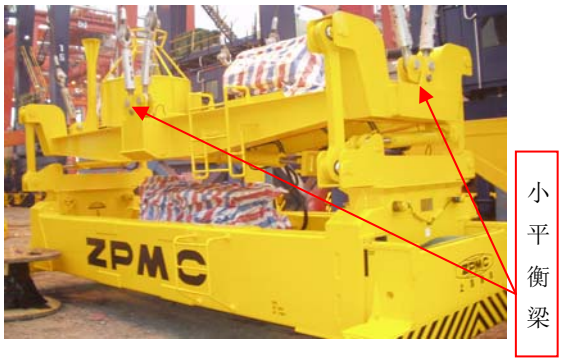


Figure 0-38 he adjusting screw on the rotating beam

4) Hoist the spreader until it is 40 feet to the ground and check the height that the four corners of the spreader are to the ground. Refer to Figure 5-39 and
record.

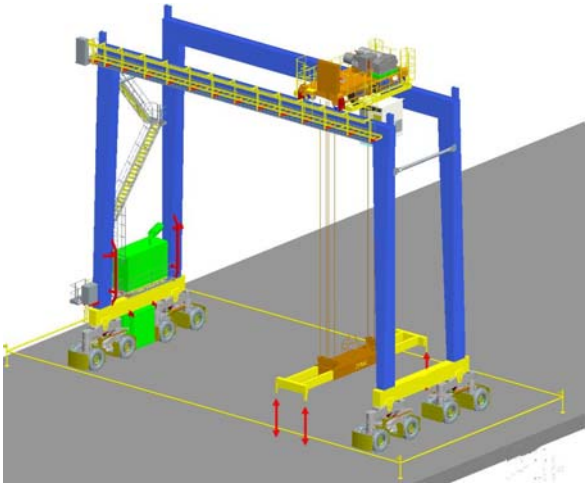


Figure 0-39 The height of the spreader inspection

5) Move the trolley to opposite the parking position. Check the height that the four corners of the spreader are to the ground and record.

6) Compare the two sets of values above and determine the ground which is more smooth be measuring datum.

7) When the spreader is 40 feet to the ground, measure the distance from the two closer corners of the spreader to the gantry. The two measuring points on the spreader and the gantry shall be

symmetrical. Record the values and compare them. Refer to Figure 5-40.

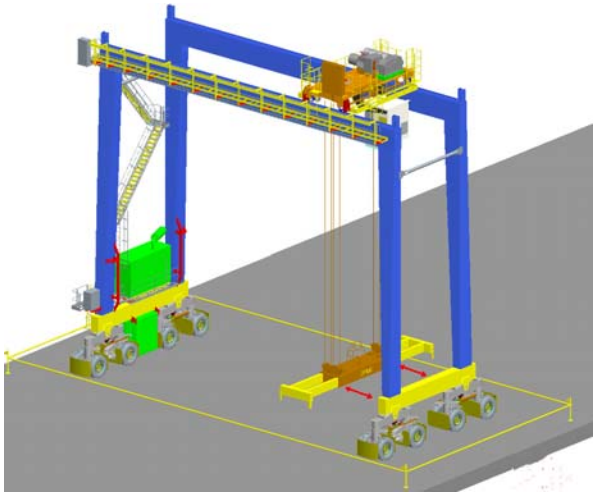


Figure 0-40 The level of the spreader inspection

8) Compare the distance from the two closer corners of the spreader to the gantry. If the left one is 50mm larger than the right one, adjust as follows:

   A) Tighten up the left front screw rod for 1/4 loops.

   B) Slacken the left back screw rod for 1/4 loops.

   C) Slacken the right front screw rod for 1/4 loops.

   D) Tighten up the left back screw rod for 1/4 loops.

9) Compare the distance from the four corners of the spreader to the ground. If the left one is 30mm less than the right one, adjust as follows:

   A) Tighten up the left front screw rod for 1/2 loops.

   B) Tighten up the left back screw rod for 1/2 loops.

   C) Slacken the right front screw rod for 1/2 loops.

   D) Slacken the right back screw rod for 1/2 loops.

10) Compare the distance from the four corners of the spreader to the ground. If the front one is 20mm larger than the back one, adjust as follows:

   A) Tighten up the left front screw rod for 1/2 loops.

   B) Tighten up the right front screw rod for 1/2 loops.

   C) Slacken the left back screw rod for 1/2 loops.

   D) Slacken the right back screw rod for 1/2 loops.

11) Hoist and lower the spreader for several times and observe whether the beam is in a level. The adjustment may need many times to complete. Usually the distance from left to right shall be less than 30mm and the height difference shall be less than 20mm. The goal of the adjustment is to make it convenient for the driver to aim the box.

12) Observe the difference of the four weight sensors in electrical room. And the difference shall be controlled within 30%.

13) Various factors shall be considered due to the difference in all directions. The values above are for reference only. The adjustment may need many times to complete.

14) Finally try to hoist and lower the heavy box for several times. Repeat the steps above until controlling the difference within specified range.

15) Put the locking bolts into the screw rods, tighten the nuts and pins. Now the ropes have been adjusted.

Various factors shall be considered when adjusting the rope, the

adjustment may need many times to complete.

CHAPTER 6 APPLICATION, REPAIREMENT, AND INSPECTION OF PAINTING

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6.1.1 suitable scope
6.1.2 standards and codes
6.1.3 surface treatment
6.1.4 painting application
6.1.5 painting regulation
6.1.6 inspection
6.1.7 colors
6.2 APPLICATION AND REPAIREMENT TECHNOLOGY
6.2.1 introduction
6.2.2 surface treatment
6.2.3 shop primer
6.2.4 paint application
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6.3 INSPECTION AND JUDGMENT FOR REPAINTING
6.3.1 aging or deterioration of coating and judging standard for repair
6.4 PAINTING SYSTEM
6.5 EUROPEAN RUST SCALE

Chapter 6 Application, repairement, and inspection of painting

Summary

Corrosion of metals exist everywhere and make great harm. There are 1/10 steel loss for rust or other kinds of corrosion every year according to statistics.

RTG usually work outdoor beside the coast, with great salt atmospheric, sometimes even suffer with corrosivity cargo (for instance, beside the bulk- cargo dock). Because the exist of the corrosion, the structure anticorrosive painting lose its effect, which not only impact the outlook of the crane, but also make the metal construction corrosion and decrease it’s loading ability, even broken the structure.

Consumer should do the structure routing care and maintainence, arrange the renew of the painting and recover the anticorrosive fuction of the damaged painting, while ZPMC make sure the painting have the suitable anticorrosive ability and lifetime.

Consumer should reinforce the management of the anticorrosion, enlength the anticorrosive lifetime of the painting, and it has great meaning for the crane’s safety normal work, enlength the lifetime of the crane and reduce the maintenance charge of the crane.

Painting not only embellishes the crane, but also has great impact on protecting the structure from the corrosion. Can crane structure reach the max design lifetime is mostly decided by the anticorrosion care and maintenance work of the crane.

This chapter will descript the surface treatment, painting appilication, repairing technology, inspection and judgement standard when crane manufacture.

6.1 general principles

6.1.1 **suitable scope**

The procedure covers the surface preparation, materials and coating application requirements for coating the fabricated steel, equipment and appurtenances of the container crane structures.

The color of finish coating is to be in accordance with the owner's requirement, while that of the primer and intermediate coating shall be up to the manufacturer's availability.

6.1.2 **standards and codes**

The latest edition of the ASTM, the Steel Structures Painting Council Surface Preparation (SSPC-SP) and the International standard (ISO) written Specifications are part of this Specification.

international standard

standard	discription
ASTM A123	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
SSPC-SP-1	Solvent Cleaning
SSPC-SP-2	Hand Tool Cleaning
SSPC-SP-3	Power Tool Cleaning
SSPC-SP-10	Brush-off blast cleaning
SSPC-SP-COM	Surface Preparation Commentary
SSPC-PA2	Measurement of DFT with M. Gauge
SSPC-SP-11	Power Tool Cleaning to Bare Metal
ISO 8501	Visual assessment of surface cleanliness
ISO 8502	Tests for the assessment of the surface cleanliness
ISO 8502-6	Sampling soluble impurities on surface to be painted
ISO 8503	Surface roughness characteristics of blast-cleaned steel surface substrates.
ISO 8504	Preparation of steel substrates before application of paints and related products

6.1.3 **surface treatment**

1) All surfaces to be coated shall be clean of all contaminants that would affect the integrity of the coating system. Prior to surface preparation, the surfaces shall be cleaned, as necessary, to remove all oil, grease, dirt, salts or other foreign materials. This cleaning shall be done with appropriate detergent and/or solvent cleaning. Detergent or solvent cleaning shall be performed as outlined in SSPC-SP-1.

2) SSPC-SP-3。When approved, power tool cleaning shall be as per SSPC-SP-3 and SSPC-SP-11. Hand tool cleaning shall be as per SSPC-SP-2.

3) 50mm around the damaged part of the painting need to be thinned to ensure the adhesive force for the original painting and the new painting.

6.1.4 **painting application**

1）Surface need to be clean, dry and dust free before apply ZP842

2）Apply ZP842 within 4 hours after the surface treatment.

3）Coatings shall be applied under the environmental conditions prescribed in the pertinent ZPMC's product data sheets.

4）Coat shall be applied as per ZPMC's recommendations.

5）Thickness requirements shall be met with each coat and total thickness shall not be "made-up" in any one coat.

6）Maximum recoating interval time shall be as per ZPMC's recommendations.

7）All areas around the corners, welding lines, sharp edges, bolts and nuts should be applied with shop primer before painting.

6.1.5 **painting regulation**

1) Structure exterior

Surface treatment according to SSPC-SP-3 power tool cleaning, see 6.1.3

2 × ZP842 (intermedia)

1 × ZP135 (top coat)

2) structure interior

1 × ZP 842 (intermedia)

3) galvanized area

remove all the grease according to 6.1.3

1 × HEMPEL'S 15552 organic epoxy primer

1 × ZP135

6.1.6 **inspection**

1) All work should be subjected to inspection. The inspectors shall be given advance notice prior to the start of surface preparation or coating application.

2) All the inspections should be carried out at each hold point defined below

Completion of surface preparation, prior to primer application.

Completion of application of each coat (primer, intermediate and topcoat).

Final acceptance of completed coating work.

6.1.7 **colors**

painting color codes

components	colors
structure exterior	Maersk Blue
structure interior	RAL 9016
driving gantry	Red RAL 3020
driving gantry guarder	RAL9016/3020
cabin exterior	RAL 9003
cabin interior	RAL 7015
power set exterior	RAL 9003
power set interior	RAL 7015
E-room exterior	RAL 9003
E-room interior	RAL 7015
walkway,handrail	BS Golden Yellow 356
headblock	BS Golden Yellow 356
spreader	BS Golden Yellow 356

6.2 application and repairement technology

6.2.1 **introduction**

1）The procedure covers the surface preparation, materials and coating application requirements for coating the fabricated steel, equipment and appurtenances of the container crane structures.

2）The coating system of this specification is based on:

The International standard ISO 12944 "Protective Paint System", under corrosive category "C4"- for coastal or industrial environment

With high durability more than 15 years of performance reaching level Ri 3 as defined in ISO 4628-3 or the rust ratio Re3 defined in European Rust Scale.

3）The color of finish coating is to be in accordance with the owner's requirement, while that of the primer and intermediate coating shall be up to the manufacturer's availability.

6.2.2 **surface treatment**

The surface shall be prepared in a manner suitable for coatings. Chemical de-rusters or rust converters are not allowed.

1) Detergent / Solvent Cleaning

All surfaces to be coated shall be clean of all contaminants that would affect the integrity of the coating system. Prior to surface preparation, the surfaces shall be cleaned, as necessary, to remove all oil, grease, dirt, salts or other foreign materials. This cleaning shall be done with appropriate detergent and/or solvent cleaning. Detergent or solvent cleaning shall be performed as outlined in SSPC-SP-1.

2) Hand or Power Tool Cleaning

Power tool cleaning is a way used power machine such as sander to clean the rust and paint.

Once finished, no erosioned paint can be seen on the surface, and the edge need to be smooth and equality.

Do not polish the surface, surface roughness can ensure the adhesive force of the new applied paint.

Power tool cleaning shall be as per SSPC-SP-3 and SSPC-SP-11. Hand tool cleaning shall be as per SSPC-SP-2.

The sharp edges, such as those occurring in rolled structural members as well as those resulting from flame cutting, welding, etc., shall be removed by suitable methods such as a soft grinding wheel before sandblasting and the application of any coat of paint. (Refer to SSPC-SP5 and SSPC-SPCOM).

3) Abrasive Blasting Cleaning

Prior to abrasive blast cleaning of the steel, remove oil and grease and other contaminations with a suitable detergent / solvent, according to SSPC-SP1 “Solvent cleaning”.

All welding seams must have a surface finish that ensures that the quality of the paint system will be maintained in all respects. The sharp edges should be smoothed and the radius of the rounding should be approximately 1 mm. Holes in welding seam, undercuts, cracks, residual flux and weld spatter etc. should be avoided. If found, they must be remedied by welding and/or grinding.

Steelwork (including welding, flame-cutting and grinding) must be finished before the surface preparation starts.

Prior to abrasive blasting, sight glasses, nametags, machinery, electrical fixtures, etc. which are subject to damage from either the blast or from the abrasive material remaining shall be protected. If possible, nametags should be installed after painting is complete. If it is not possible to coat under the tags, their edges shall be sealed with silicone to prevent moisture accumulation.

Blasting shall be accomplished so that the previously-painted surfaces will not be contaminated by abrasive or blast wastes.

Only dry steel grits, silica sand and aluminum silicate grit-type abrasives shall be used (refer to ISO 8504). The abrasive shall comply with government regulations. The abrasive shall be used in accordance with ZPMC’s Specifications and shall contain no harmful impurities, such as salt. Beach sand or rounded river sand is not acceptable.

All steel surfaces shall be abrasive blasting to obtain:

Steel Surface Blasting Treatment

item	instrcution
Cleanliness	SSPC SP No.10 “Near White Blast Cleaning” “Sa 2.5” according to ISO 8501 or “Near White Blast Cleaning”- SSPC SP No.10.
Roughness	N9a to N10a (Ra = 6-12.5 micron) - Rugotest No.3, Keane-Tator Surface Compactor, 2-3 mils segments (average maximum peak to valley roughness 50-75 micron).
condition	Should be below No.3 scale as per ISO 8504

Surfaces shall not be abrasive blast cleaned when the ambient or the substrate temperature is less than 3°C above the dew point temperature. Blast cleaned surfaces, which show evidence of rust bloom or has been left uncoated overnight, shall be re-cleaned to the specified degree of cleanliness.

Steel surface slivers, laminations, laps, scabs or seams exposed by blasting shall be repaired on any portion of the steel to be painted prior to coating. Such area shall be re-blasted prior to coating.

No inhibitive washes intended to prevent rusting shall be used after blasting.

Welding undercuts and porosities shall be repaired by welding or grinding. No organic filler materials are allowed.

When the abrasive blasting is completed, remove all grits and dust.

Surfaces, which are impractical to blast clean may be power tool or hand tool cleaned. Power tool cleaning shall be in accordance with SSPC-SP 11 or 3, listed in order of preference. Hand tool cleaning shall be as per SSPC-SP-2.

4) Galvanized Surface

Generally， galvanized surface noneed to be painted.

For better rust prevention, parts of the crane galvanized components will be painted, such handrail.

Using detergent or solvent cleaning to clean the surface dirts, no to break the galvanized surface when using hand or power cleaning, otherwise zinc lay need to be repaired.

6.2.3 **shop primer**

1) surface preparation

All steel plates, bars etc., which are to be coated with shop-primer, must be abrasive blasted to Sa2.5 and surface profile equivalent to Rugotest No3 N9a to N10a. Also refer to Paragraphs 6.1.3

When the abrasive blasting is completed, remove all grits and dust.

2) When the treated surface is dry and free from oil, dust, etc. ZP01704 should be applied by airless spraying machine immediately on the automatic line.

6.2.4 **PAINT APPLICATION**

6.2.4.1 GENERAL APPLICATION REQUIREMENTS

1) Surfaces to be coated shall be clean, dust-free, and dry before the application of any coat of paint and shall meet the requirements for applicable hand, power tool or abrasive blasting surface preparation before priming.

2) The specified coating system primer shall be applied within 4 hours from the moment the surfaces are cleaned, and before rusting, discoloration, or surface contamination occurs.

3) Painting done outdoors shall be done in daylight hours and completed at least one hour prior to sunset. Indoor painting is allowed 24 hours a day if the specified metal and the air temperature and relative humidity requirements are met inside the building at all times during preparation, painting and curing, and sufficient lighting is available.

4) Coatings shall be applied under the environmental conditions prescribed in the pertinent ZPMC's product data sheets.

5) No coating shall be applied:

On wet surfaces

When the surface temperature is less than 3°C above the dew point.

When the ambient or surface temperatures are below the ZPMC's recommended temperature for application and curing.

When the wind speed exceeds 7 meters per second

Humidity is 80% or more

6) Coating materials shall be furnished in the original packing, clearly identifiable, and shall be kept covered, clean and protected. Materials older than ZPMC's recommended shelf life shall not be used.

7) All components shall be thoroughly mixed for the time recommended in the ZPMC's data sheets. All mixing shall be done in clean containers, free from traces of grease, other paints or contaminants. Containers shall be kept covered to prevent contamination by dust, dirt or stain. Paint shall be mechanically mixed in full batches. Partial mixes are not acceptable.

8) Recommended pot life shall not be exceeded. When this limit is reached, the spray pot, hoses, pumps shall be emptied and cleaned, and new material should be used.

9) Care shall be exercised to prevent overspray, spillage or application of coatings to surfaces for which they are not intended. Skips, sags, and drips are to be repaired.

10) Coatings shall be applied in not less than the number of coats specified and shall have individual coat dry film thickness not less than the minimum thickness specified. Wet film thickness needs to be verified at the start of work (until the painter is familiar with the paint and work) with a wet film thickness gauge. Thickness requirements shall be met with each coat and total thickness shall not be "made-up" in any one coat.

11) All coating applied shall be free of defects such as runs, sags, pinholes, voids and bubbles.

12) All coatings shall be allowed to dry for at least the minimum time recommended, considering the temperature and the humidity, before applying the succeeding coats.

13) Maximum recoating interval time shall be as per ZPMC's recommendations. When the maximum overcoat time has been exceeded, it needs to perform additional surface preparation. This operation includes but not limited to hand sanding, pressure washing, mist coat application, degreasing or re-blasting.

14) Each coat (primer, intermediate and topcoat) shall be inspected by the yard's Q.C.

6.2.4.2 SPRAY APPLICATION

1) The applicators shall be skilled in the proper application technique for each coating. Proper equipment, as per prescription in the ZPMC's data sheets, shall be used.

2) Spray guns, lines, and pressure pots shall be cleaned prior to adding new material.

3) Moisture traps or separators shall be installed between the air supply and the pressure pot. Drain valve must be kept open slightly to permit continuous draining of any condensate during operations.

4) All pressure pots shall be equipped with pressure regulators and gauges.

5) Paint shall be applied in a uniform layer, with 30% overlapping of the previous pass. During application, the spray gun shall be held perpendicular to the surface being painted.

6.2.4.3 BRUSH OR ROLLER APPLICATION

1) All welds, which cannot be properly spray coated shall be brush or stripe coated with the same coating before applying successive spray operation.

2) Brushes shall be of a style and quality that will enable proper application of the coating.

3) Brushing shall be done so that a smooth coat even in thickness will be obtained. When applying HB coatings with a brush, two coats may be required to achieve the specified DFT.

6.2.4.4 APPLICATION OF INORGANIC ZINC PRIMER

1) Inorganic zinc may be applied by either conventional spray application (air spray) or airless spray application. If airless spray is to be used, frequent checks must be taken for the wearing of the parts to ensure proper operation. Inorganic zinc shall be continuously mixed during the application by mechanical agitation.

2) The minimum dry film thickness of inorganic zinc primer shall be 75 microns. In order to prevent mud cracking, do not apply over 130 microns. The defective coating (mud cracking) needs to be removed by blasting and the area recoated to the specified thickness. Frequent checks of the film thickness must be made and immediate action must be taken should low DFT be observed.

3) The applicator shall avoid situations where overlapping of inorganic zinc will be necessary. The blast preparation of the uncoated area shall be brought into the existing inorganic-zinc. If the inorganic-zinc has been exposed to the environment and has formed salts or become contaminated, it should be washed with high-pressure fresh water prior to blast cleaning.

4) Zinc salts as a result of the weathering of inorganic zinc primer should be removed with a fresh water jet before the application of subsequent coatings.

5) When inorganic zinc is over coated with an organic epoxy intermediate, it may be necessary to apply a mist coat of the organic epoxy before applying a full coat. This will help reduce substantially the amount of gassing associated with over coating inorganic zinc with epoxies.

6.2.4.5 APPLICATION OF EXTERIOR INTERMEDIATE COAT

1) When the inorganic zinc coating has been inspected and accepted, carry out general a cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharp edges, hidden areas, etc. by brush application.

3) At the allowed recoating interval, apply the ZP842 by airless spray, using the mist coat spraying technique on top of the inorganic zinc surface and then build up the required dry film thickness of 125 microns as indicated in painting system table 2.

6.2.4.6 APPLICATION OF EXTERIOR TOP COAT

1) When the intermediate coating has been inspected and accepted, carry out a general cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharp edges, hidden areas, etc. by brush application.

3) At the allowed recoating interval, apply the ZP135 by airless spray with a dry film thickness 50 microns as indicated in painting system table 2.

6.2.4.7 APPLICATION OF INTERIOR PRIMER/TOP COAT

1) When the surface preparation has been inspected and accepted, carry out a general cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharp edges, hidden areas, etc. by brush application.

3) Apply the ZP842 by airless spray with dry film thickness 125 microns as indicated in painting system table 3.

6.2.5 **INSPECTION**

1) All work should be subjected to inspection. The inspectors shall be given advance notice prior to the start of surface preparation or coating application so that they can witness the work.

2) Work shall not proceed without the approval of the inspectors of each inspection hold points defined below. The inspector shall ensure that all necessary inspections are carried out at each hold point. The inspection hold points are as follows

Completion of surface preparation, prior to primer application.

Completion of application of each coat (primer, intermediate and topcoat).

Holiday detection and repair.

Final acceptance of completed coating work.

3) Scaffolding - Beam clamps or any other equipment which will interfere with or damage coating shall not be used. Building yard shall provide safe accesses to all areas for inspections by the inspector.

4) Prior to final coating acceptance, the entire painted surface shall be cleaned with fresh water. If necessary, a finish coat may be applied to areas stained and contaminated by grinding, weld spatter, general traffic, etc. Areas contaminated by grinding, weld spatter, and other particulate contamination shall be lightly hand-sanded to dislodge particulate prior to coating.

6.2.6 **OTHERS**

1) External / Internal Galvanized

- Apply HEMUCRUL TI-COAT 16490, Grey 19840 on broken galvanized surface.

ONE coat of HEMPEL'S 15552 organic epoxy primer approved at 50-micron DFT

Apply by airless spray ONE full coats of ZP135 polyurethane finish or similar approved at 50-micron DFT.

2) Stainless Steel

Stainless steel should not be painted

3) Non-Ferrous material

Non-ferrous materials need not to be painted

4) Machined surfaces

Machined surfaces are to be masked prior to painting

5) Faying Surface

Apply ZP2704 at 75-microns DFT.

6.2.7 **REPAIR/TOUCH-UP**

1) External Steelworks

When the coating is damaged down to the steel, spot cleaning is necessary:

For areas larger than 0.25 square meters (if the width of the area is over 0.5 meter), the area shall be isolated, blasted to Sa 2.5 / SSSPC - SP - No. 10 in. After the metal surface has been cleaned, the edge of the surrounding coating shall be feather-edged to remove all the cracked, loose or damaged coatings. Feather-edging shall be accomplished by power or hand-sanding with a grit wheel or sandpaper. The prepared surface shall be free of loose, burnt, or blistered coatings. Carry out touch-up coating procedure with inorganic zinc followed by intermediate and top coat.

For areas smaller than 0.25 square meters, use power tool cleaning and apply ZP842 at 125-micron DFT and followed by top coat.

2) Internal Steelworks

Whenever possible, surface preparation must be restored to the degree as originally specified, or as per hand /power tool standards. Apply touch-up coating with airless spray (large areas) or with brush (small areas) ZP842 at 125-micron DFT. If the application is brush, the paint should be applied in two or more coats to achieve the specified total DFT of 125 microns.

6.3 inspection and judgment for repainting

The following passages describe the inspection and repair procedure for the painting system of the crane.

6.3.1 **aging or deterioration of coating and judging standard for repair**

1) rust spreading throughout the steel surface

In some cases, the coating can be removed with the primer in a wide area. In such a case, the rust will eat the steel to make a hole.

Judgment for repainting: consider repainting the coating when the above mentioned defect is found.

2) peeling of the coating

Peeling of the whole coating due to insufficient primer adhesion to the steel surface.

Peeling of the top coating due to insufficient adhesion between the top coating and the intermediate coating.

Peeling of the coating due to cracks.

Peeling of the coating due to poor preparation for painting.

Judgment for repainting: If a straight line can be

drawn with the tip of a drill bit applied on a coating, the coating is considered to be in good condition. If the coating is peeled off, repainting is required.

3) Deterioration of coating

The coating may be easily removable due to external impacts and internal dissolution of the coating itself.

Judgment for repainting:

If the coating is vertically and slightly struck with a steel hammer of 5-10 mm diameter, it may become easily removable or be actually removed.

If the coating is vertically and slightly struck with a steel hammer of 5-10 mm diameter, the intermediate coating or the top coating may be removed.

Such a coating is considered to be deteriorated and requires repainting.

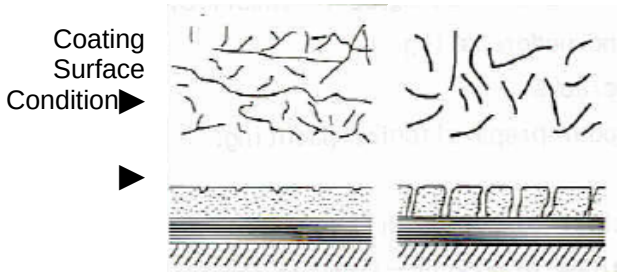
4) Cracks in coating

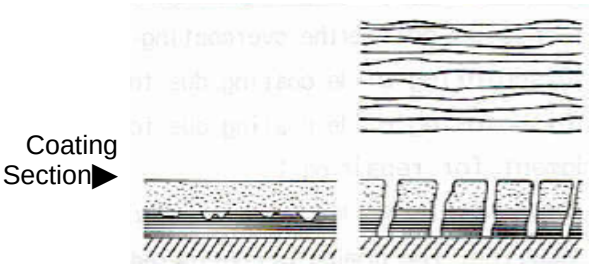
Cracks in a coating occur due to the following causes and in the following ways. Refer to Figure A-1 “Crack in Coating” for diagnosis

If the crack occurs soon after painting, the cause possibly lies in improper painting or poor painting preparation (Applying top coating before the primer dries up etc.)

The cause of cracks possibly lies in the aging of the coating caused by the shrink and the expansion of the painted steel due to temperature change.

In some cases, cracks can occur by sudden temperature change. (Take a special measure for painting in a cold area where temperature changes excessively.)





a: Cracks only on top coating b: Cracks only on top coating c: Cracks only in intermediate coating or primer d: Cracks in both top coating and intermediate/primer

cracks of the coating

Judgment for repainting:

For the State of A and C. No immediate re-painting is required from the view of coating protection. However, the severe cracks may possibly develop into the state D shown in Figure A-1

For the State of B. The top coating is ineffective and requires re-painting. Re-paint it according to the primer protection effect and the rust condition.

For the state of D. In most cases, rust also occurs. Repair the painting as soon as possible.

5) swelling of coating

The causes are considered to be the followings:

In a cold area, the whole coating may be removed in frozen state due to the swelling which once occurred. Refer to Figure 6.2-2 “Surface View of Swelled Coating”. The swelling could be caused due to:

Coating soaked in water

A drop of water heated in high temperature on the coating

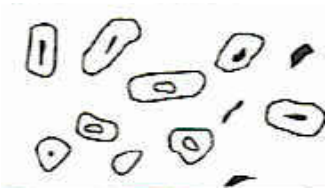
Affected by alkali and acid

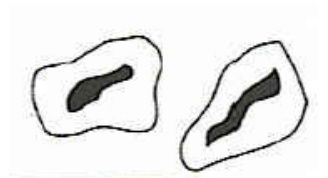
Affected by heat

Judgment for repainting:

If swellings occur in both the top coating and primer and rust may develop, repainting is required.

If the swelling occurs only in the top coating and no rust is found on the steel surface, no immediate repainting is required.





Surface View of Swelled Coating

6) Discoloring of coating:

a. The coating may be suddenly discolored due to internal and external causes after painting.

In most cases, the causes are improper selection of paint for the following conditions or use of inferior paint.

b. The coating gradually becomes discolored as time passes after the paint completion.

Judgment for repainting:

If the coating terribly discolors comparison with the original color, it requires immediate repainting.

In case of slight discoloring, no repainting is required.

7) Blurred coating

In general, the coating loses its gloss by 40% in 3-6 months and 60% in 6-12 months after painting. Therefore the coating will be blurred as time passes.

Judgment for repainting:

Compare the coating with an original sample. If the coating is only blurred, no repainting is required.

6.4 painting system

Table 6-1 shows the primer, intermediate, top coat product number and DFT used on the crane. No need painting parts are also be showed.

All values of film thickness are minimum DFT!

coating system

shop primer

coats	product name	DFT(microns)
shop primer	ZP1704	20
total		20

external area

coats	product name	DFT(microns)
primer	ZP2704	75
intermediate	ZP842	125
top coat	ZP135	50
total		250

internal area

coats	product name	DFT(microns)
top coat	ZP842	125
total		125

galvanized surface(external)

coats	product name	DFT(microns)
primer	Hempel's Wash Primer 15552-11630	50
top coat	ZP135	50

toal	100

galvanized surface (internal)

coats	product name	DFT(microns)
primer	Hempel's Wash Primer 15552-11630	5
top coat	ZP842	75
total		125

repair area (a) external area(big area)

coats	product name	DFT(microns)
primer	ZP2704	50
intermediate	ZP842	125
top coat	ZP135	50
total		250

repair area (b) internal area(small area)

coats	product name	DFT(microns)
intermediate	ZP842	100
intermediate	ZP842	125
top coat	ZP135	50

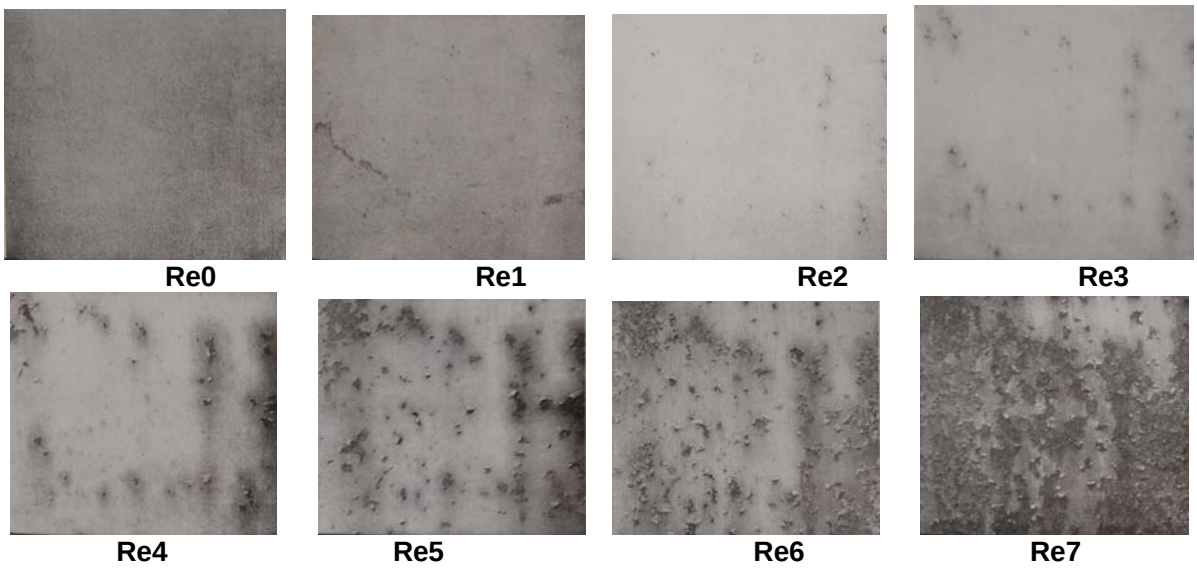
repair area(internal)

coats	product name	DFT(microns)
top coat	ZP842	125
total		125

no painting parts

1	rail head
2	wheel tread
3	groove of wire drum and sheave
4	inside of drum
5	inside of airtight box and pipe
6	stainless steel and hot-dipped galvanized items

6.5 EUROPEAN RUST SCALE



Ten different rusting degrees are in the European rust scale. Figure 6-3 shows 8 stages' description.

The approximate percentages of rusting degree corresponding to the photos are shown as follow. These percentages apply only to visible rust which has broken the paint coating. These percentages figures are intended for the use of those who wish to refer to percentage as well as to the photo-graphs.

Re0	0%	Re4	3%
Re1	0.05%	Re5	8%
Re2	0.5%	Re6	15~20%
Re3	1%	Re7	40~50%

CHAPTER 7 STRUCTURAL INSPECTION AND REPAIR

7.1 GENERAL
7.1.1 routine inspection
7.1.2 periodic annual inspection
7.1.3 metal strcture repair
7.1.4 inspection and record

Chapter 7 Structural inspection and repair

Summary

Metal structure is the frame of the crane, and affects the performance and safety of the crane directly. Cracks need to be found and repair in time. To keep the crane always in good condition and ensure crane's performance and safety, routine inspection and periodic annual inspection are needed.

This chapter describes the procedures of the structural routine inspection and periodic annual inspection, and lists the inspection recording table.

7.1 general

Metal structure is the frame of the crane, and affects the performance and safety of the crane directly. The crane need to be shut down and carefully inspected, when crack is founded. If the crack is not find and repair in time, it will lead to structural damage which will not only broken the crane, but also cause the causalities. So structure needs the routine inspection and periodic annual inspection.

The deformation of the structure part may be caused by the misuse of the crane，crack extension or bolt loosening. Inspection procedures and the routine inspections in the manual are needed to strictly obey.



No attachment or device can be welded on to crane, without the written authorization of the responsible structural engineer. This is the crane's structural integrity requirement.

The inspection ways of the structure include the follows:

1) using the visual sense and experience to observe and judge the structure

- If the connecting part of the structure is stable, flexible and without abnormal noise when operation process.
- If there is any loosening of the bolts.
- If there is any paint wrinkle, peel off or weld crack phenomenon around the connecting part or welds.
- If there is any entire or local deformation, and any bend, fold, twist, damage phenomenon.

2) Using the nondestructive inspection to do the periodic inspection of important welds.

3) Using electrical logging tension method to do the carefully quantitative inspection of the metal structure, which can make a scientific judge to the metal structure's status.

the follows need to be pay more attention

1) Inspect the structure deformation, corrosion, flaw, weld crack, bolt loosening in routine inspection, month and annual periodic inspection.

2) Structure deformation will increase the concentric error of the mechanism which installed on it, increase the operation noise, lead to the bad mesh of the gears.

3) Structure deformation, flaw extension, weld crack, bolt loosening should be report to the maintenance department instantly, and do the repair work under the guide of the structural engineer.

4) The jam drainage holes need to be cleaned to avoid the corrosion of the structure. Repaint the rusty place. Corrosion will decrease the thickness of the structure, enlarge the stress, and may lead to the crack. To

inspect the crack, these areas should remove the rust and paint thoroughly, and inspect according to the illustration.

7.1.1 routine inspection

1) Frequently inspect the welded and related part of the main structure member, especially the supporting part, once found the abnormal thing, the structure need to be repaired.

2) If there is obviously concave- convex phenomenon on the plate of the box structure, crane need to be shut down to find out the cause and only can be restart after eliminating the hidden danger.

3) Frequently check the bolts of the trolley rail clips, or welded rail's welding line and if the bumper support at the girder end has been impacted to deformation.

4) Regular check the deformation and welding line of the trolley frame. Pay attention to the working condition of the trolley wheel axle. Once found the abnormal status, measures should be taken.

5) Pay attention to the tightness of the bolts.

6) Broken paint part of the structure should be removed the rust instantly and repaint.

7.1.2 periodic annual inspection

1) Fatigue cracks may occur after the metal structure has been used for several years. Except the routine inspection, the structure also needs to do the periodic annual inspection.

2) Interval of the periodic annual inspection is 3~5 years.

3) The critical inspection points are those with variation stress and concentration stress points.

4) Inspection method: VT, MT, UT

5) Record the text, and solve the problem when it come out.

7.1.3 metal structure repair

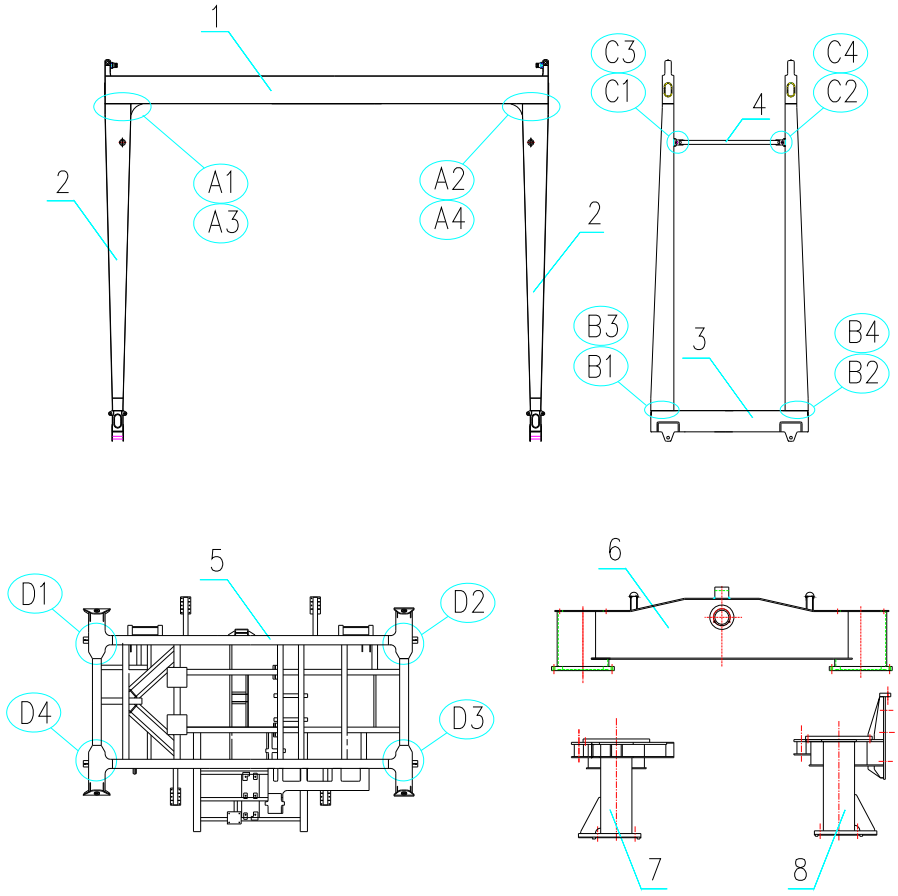
Repair methods should be taken, when flaw or crack turn out. Repair methods should according to the situation of the specific location. Repairmen of the structure is very complicated, consumers cannot judge it by themselves. Connecting with ZPMC after- sales service department and design department will help consumers to get the correct methods.

7.1.4 inspection and record

VT should be used to inspect the structure usually, which can see if there's any abnormal deformation,

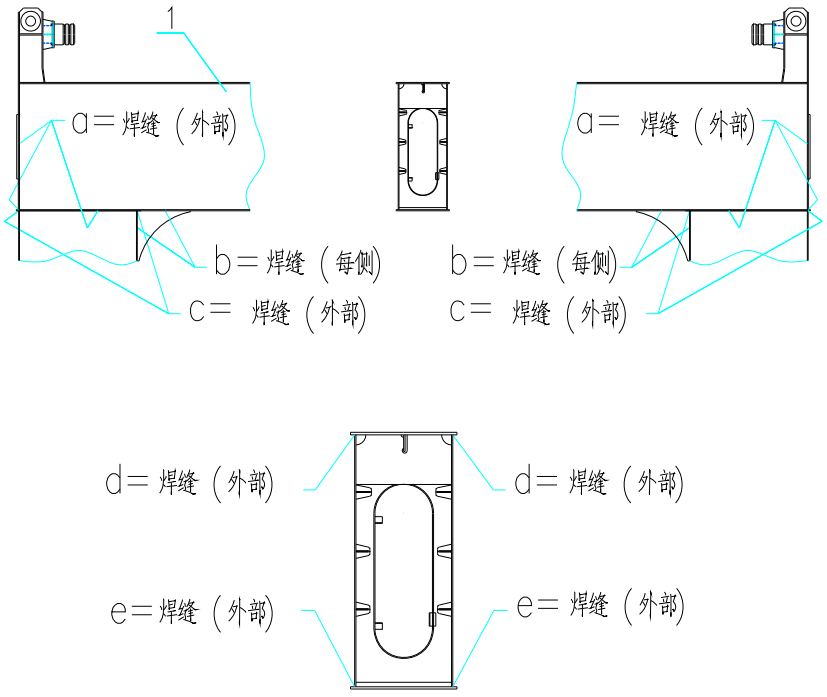
abnormal noise, if the positions of the mechanisms are stable and if there's any part fall off. The follows list the typical position of the structure's inspection and record. Refer to the figure 1~9.

ZPMC	structure inspection record table	page:
project: *** RTG	inspection locations	drawing No (1)



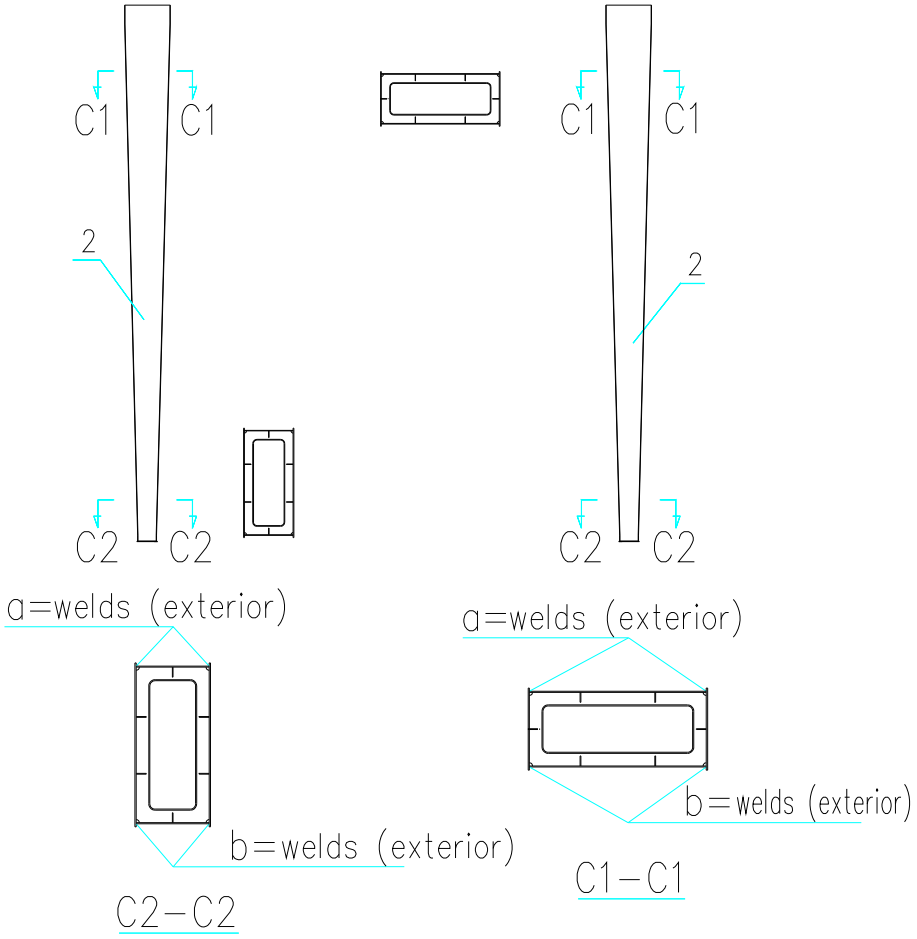
No	component name	No	connection position name
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2	门腿eg	B1-B4	门腿与鞍梁eg to sill beam
3	鞍梁sill beam	C1-C4	撑杆与门腿tennsion bar to leg
4	撑杆	D1-D4	小车架主梁与端梁trolley girder to end beam
5	小车架bar system		
6	平衡梁equilizer		
7	车架（从动）driven bogie		
8	车架（主动）driving bogie		

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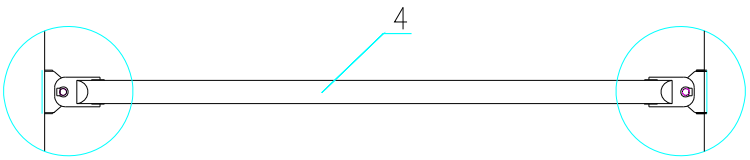
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b (右)	○							
c (右)	○	○						
d (右)	○							
e (右)	○							
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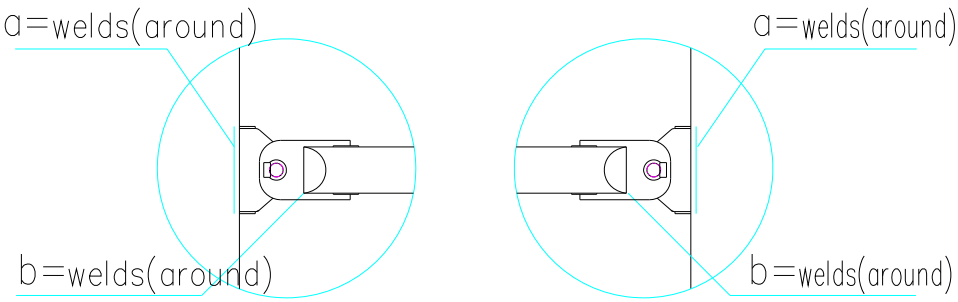
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项目: *** RTG	检查部位 门腿	图号 : (3)



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b (R)	○							
a (L)	○							
b (L)	○							

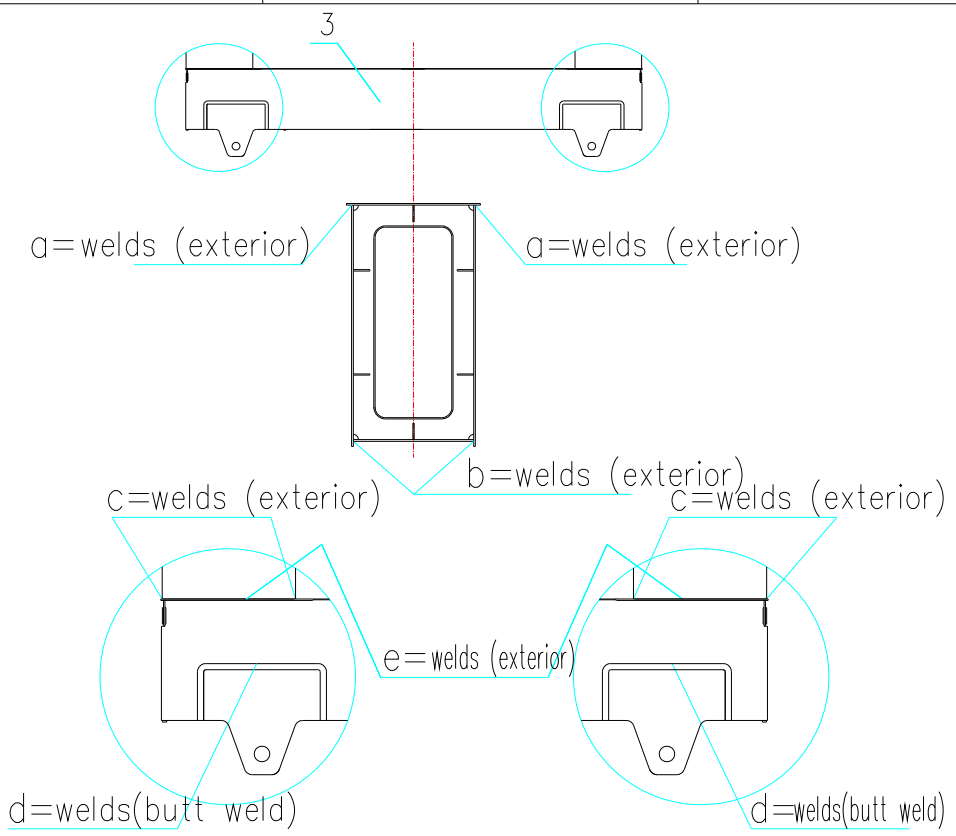
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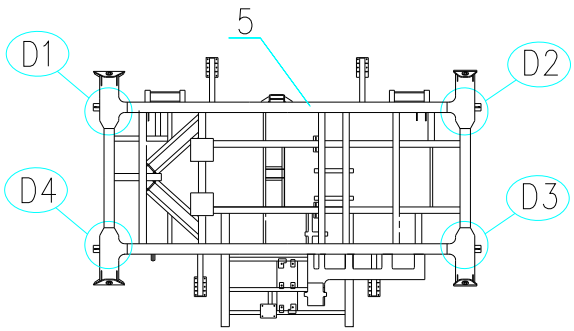
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b (R)	○							
a (L)	○							
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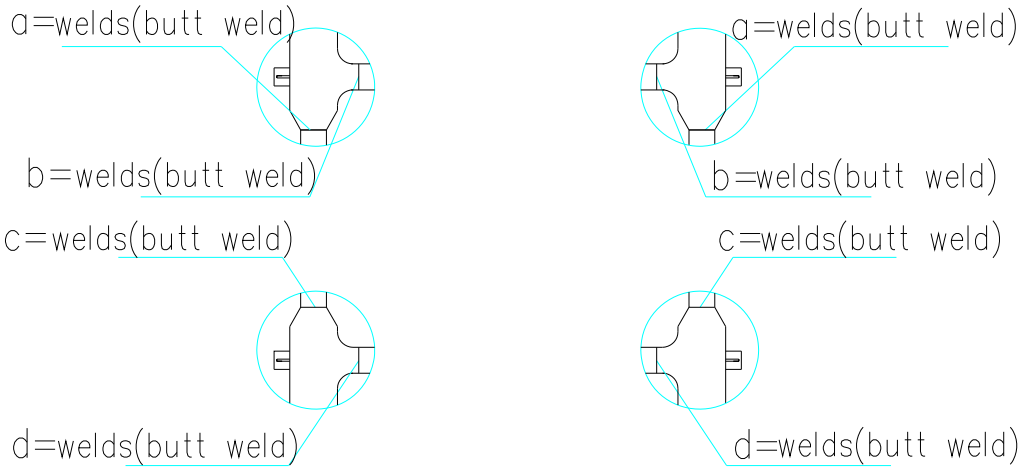
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项目: *** RTG	检查部位 鞍梁	图号 : (5)



ITEM	TEXT			ACCEPT	REJECT	COMMENTS	BY	DATE
	VT	MT	UT					
a (R)	○							
b (R)	○							
c (R)	○	○						
d (R)	○							
e (R)	○							
a (L)	○							
b (L)	○							
c (L)	○	○						
d (L)	○							
e (L)	○							

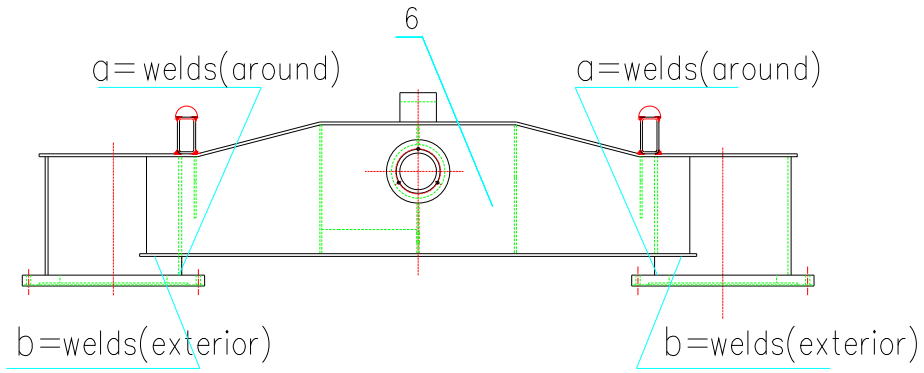
ZPMC	金属结构检查记录	页码 :
项目: *** RTG	检查部位 小车架	图号 : (6)

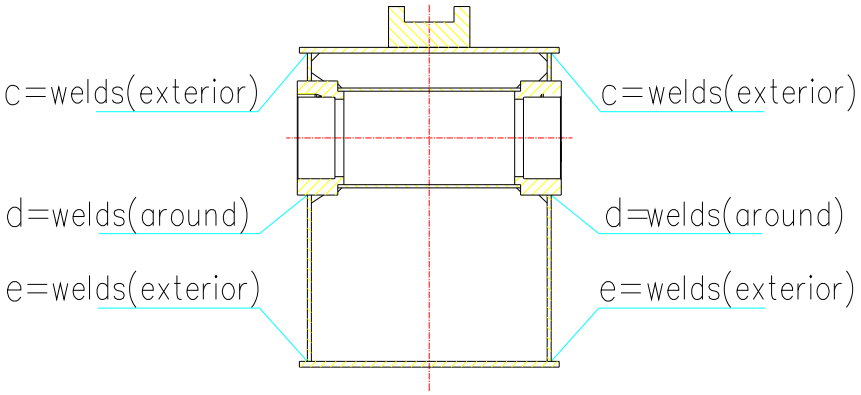




ITEM	TEXT			ACCEPT	REJECT	COMMENTS	BY	DATE
	VT	MT	UT					
a (R)	○							
b (R)	○							
c (R)	○							
d (R)	○							
a (L)	○							
b (L)	○							
c (L)	○							
d (L)	○							

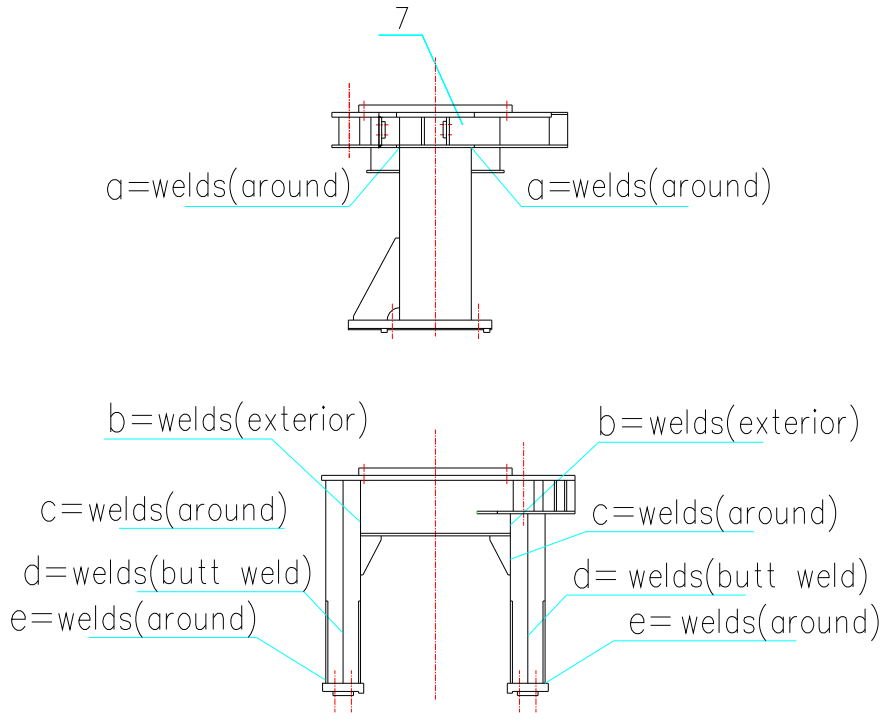
ZPMC	金属结构检查记录	页码 :
项目: *** RTG	检查部位 平衡梁	图号 : (7)





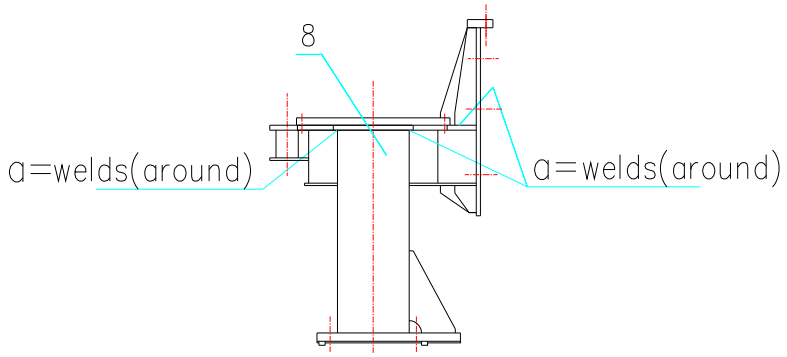
ITEM	TEXT			ACCEPT	REJECT	COMMENTS	BY	DATE
	VT	MT	UT					
a (R)	○							
b (R)	○							
c (R)	○							
d (R)	○							
e (R)	○							
a (L)	○							
b (L)	○							
c (L)	○							
d (L)	○							
e (L)	○							

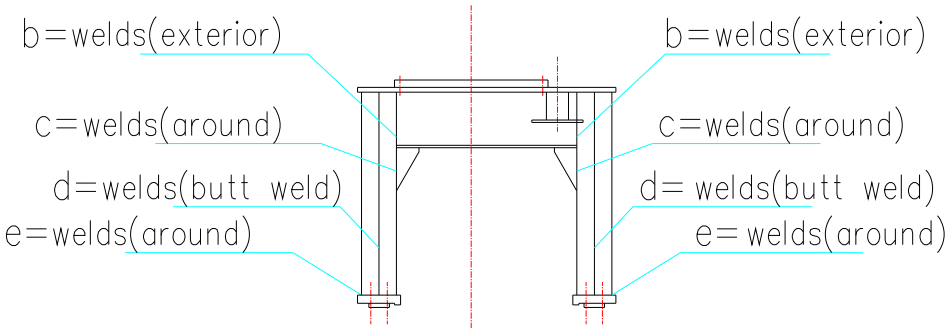
ZPMC	金属结构检查记录	页码 :
项目: *** RTG	检查部位 车架 (从动)	图号 : (8)



ITEM	TEXT			ACCEPT	REJECT	COMMENTS	BY	DATE
	VT	MT	UT					
a (R)	○							
b (R)	○							
c (R)	○							
d (R)	○							
e (R)	○							
a (L)	○							
b (L)	○							
c (L)	○							
d (L)	○							
e (L)	○							

ZPMC	金属结构检查记录	页码 :
项目: *** RTG	检查部位 车架 (主动)	图号 : (9)





ITEM	TEXT			ACCEPT	REJECT	COMMENTS	BY	DATE
	VT	MT	UT					
a (R)	○							
b (R)	○							
c (R)	○							
d (R)	○							
e (R)	○							
a (L)	○							
b (L)	○							
c (L)	○							
d (L)	○							
e (L)	○							

USER MANUAL

(mechanical)

CICT

FORTY (40)　RUBBER TYRED GANTRY CRANES



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